



RX INTERVIEWS

**Venezuelan Volatility and the
Sovereign Restructuring Rodeo**

**RESTRUCTURINGINTERVIEWS.COM
NEW YORK, NEW YORK**



NOTE BEFORE BEGINNING

If one's to dabble in distress, then the line from Lewis Carroll's *The Hunting of the Snark* ought to be committed to memory or at least should be printed out in a place that can't be missed: "They threatened its life with a railway-share".

The 1840s Railway Mania that Carroll references was marked by speculators falling under the spell of railway shares – for whatever reason, maybe it was the ceaseless upward price action alone, who knows. But it was also probably at least partially because it seemed axiomatic that railways, in some capacity, were the future and that to buy into railways, the specifics of the railways one bought into be damned, was a no-brainer. It was the *macro narrative*, not the *micro specifics*, that mattered.

You'd be forgiven for thinking that those drawn to distress – preternaturally pessimistic, dour in disposition – would be immune to all this. And perhaps that's true generally but with the one caveat that *when* the siren song of sovereign debt calls out, once or twice a decade, lots of buy side analysts can't resist getting drawn in.

Which is all sensible enough. Because, look, as much as I've harped on the "micro" in the hundreds of thousands of words I've written (e.g., the specific language in docs that permit, or not, certain transaction structures) I've also at times tried to discuss the importance of the macro backdrop since that is such a determiner of both what opportunity set one looks to *and* aggregate fund performance across most credit funds (no matter how "opportunistic" one pretends to be).

Because there are only so many idiosyncratic situations out there at any one time (e.g., LMEs that one can participate in). Thus, almost all funds ride the wave of HY spread compression or expansion to various extents and then fight tooth and nail, when idiosyncratic situations arrive, to carve out a high return since it'll be *those* situations that are most additive to their overall return in excess of their benchmark (as much as funds might moan that they shouldn't be compared to *any* benchmark).

Anyway, this is partly the reason that the siren song of sovereign debt is so hard to ignore: sure, sure, if there were a financial crisis and HY spreads blew out then sovereign debt of dubious value would trade wider all else equal – but, outside that, it's going to be largely uncorrelated to the HY universe and, as important, is going to be a large enough and (most often) liquid enough pool that trades can be put on at whatever size one wants whenever one wants (within reason).

This latter point is worth underlining since outside of huge, sprawling capital structures (e.g., Altice France) even if one *thinks* they've found a fantastic opportunity one can be limited, either by the amount of debt outstanding or the amount of debt that can be practically acquired, in being able to do the trade in sufficient size to move the needle much for the fund overall.

But, as I've said elsewhere, sovereign debt restructurings are a world unto themselves, and if one thinks there are no rules to the road when it comes to LMEs (untrue, although recent rulings have made things murkier) there are *legitimately* few hard and fast rules when it comes to sovereign restructurings despite the reams of papers, frameworks, initiatives, and associated gobbledygook produced by multilateral institutions (that are adhered to when convenient, disregarded in spirit or outright ignored when not).



Sure, lots of sovereign debt (e.g., VENZ bonds!) are issued [under New York state law](#), and New York courts will hear arguments from cantankerous creditors who believe they've been aggrieved. But while these have the look and feel of traditional disputes – and opinions can be rendered that have the same superficial look and feel as we've come accustomed to – the remedies are, well, sometimes vague and difficult to enforce against an uncooperative or ill-defined sovereign (at least until Elliott buys their own [discombobulation](#) machine).

This applies to even – in the grand scheme of things – small issues. For example, in 2023 a group of institutional funds that held around \$10b of VENZ and PDVSA debt banded together to form the Venezuela Creditor Committee (VCC) and sought, in part, to have New York courts bump out the maturities on their debt (read: agree to a tolling arrangement). In other words, something more or less akin to an amend-and-extend.

But since the first Trump administration, back in 2019, recognized the leader of the Venezuelan National Assembly as Venezuela's political leader, not Maduro, this meant that when the VCC brought their litigation there was, uh, confusion as to the appropriate counterparty.

This lasted for months until, in Nov 2023, [an agreement was reached](#) between the VCC and the opposition-backed National Authority to kick the maturities to 2028 – which, if one squints, solves the problem but in another sense is the definition of a paper solution (after all, while on paper the opposition might now be the quasi-issuer of this debt, in practice they are, uh, not able to action anything – and if that changes in the future, where what's true on paper *and* in practice are in alignment, their equanimity in front of New York courts might as well).

Although I would hasten to add (as a broader point we'll return to) that the U.S. Treasury has a habit in some sovereign restructurings of pulling a stunt not dissimilar to what we've seen from judges in some of our recent LME rulings in which a [thin veneer of legalese](#) is brushed over their opinion on what ought to be (often without much thought to pragmatism or the setting of precedence – not dissimilar to when another UST, the U.S. Trustee, pokes their nose into chapter 11s out of “principle” and then unleashes chaos). So, in the case of Venezuela, *if* a restructuring is to come, one would imagine a Secretary Bessent sized thumb firmly on the scale.

Anyway, what we saw in the first few weeks of Jan 2026 – as everyone was discombobulated and disoriented to different degrees – was a sudden resurgence of interest in Venezuela and PDVSA (but I repeat myself) debt. And much as with the Railway Mania two centuries ago, the *macro narrative* in the immediate aftermath of the Venezuela operation (Maduro out, must be favorable) superseded all consideration of the *micro specifics* (the eventual recoveries these scraps of defaulted paper will have and *when* the recoveries will practically flow).

One of the innumerable reasons I haven't touched much on sovereign restructurings is that they're, sure, drawn out and always a bit bespoke – but they're also, as one would expect given the number of countries at any one time that are distressed, rare.

Which means that unless one works within a mile radius of the *Arc de Triomphe* the lessons of past sovereign restructurings – especially those that have had the U.S. squarely in the mix – can be forgotten in the heat of the moment, and those in opportunistic seats can get a little over their skis applying the intuitions they have about corporate situations to sovereign situations...



Note: When it comes to getting over one's skis, that includes myself. Sovereign restructurings are not an area of focus, and the below should all be treated as food for thought and read out of interest. Also, as a word to the wise, don't spend too much time here as no one will ever want to talk sovereign restructurings in an interview and generally it's best not to show too much interest or talk at length about things you won't be working on anyway (it can lead to the impression that you misunderstand what the day-to-day job involves, etc.).

BRIEF VENZ AND PDVSA NOTES FOR THE WEEK ENDED JAN 09, 2026

As saber-rattling vis-à-vis Venezuela became more intense through Q4 25, both VENZ and PDVSA bonds began to grind higher in sympathy with each other until, of course, a violent upward lurch occurred as the Truth-based saber-rattling turned into real-world kinetic action.

When the dust settled, and the week closed after the capture of Maduro, VENZ debt was up 9-11pts and PDVSA debt was up 7-10pts with the lower coupon bonds outperforming. Of course, volumes were pretty high – as few seemed shy about taking a position even though the capture didn't represent a clean resolution that illuminated a clear path forward – at a little under \$2b.

VENZ 2027s and PDVSA 2024s, -1y Px Performance



It takes two to tango, and what was observed through the week was that legacy holders – those that had been in the mix in some cases for years waiting for some catalyst – saw this as an opportunity to depart the complex while (as per GS and JPM trading desks) more opportunistic players that weren't previously involved in the situation rushed in.

Look, no one knows the future (truer now than ever before, it seems). But until this all happened if you rung up someone (almost anyone!) and asked them what they thought the recoveries on VENZ debt would be they'd shrug and then say they've penciled in 30-35 like everyone else on *claim* value (which implies different trading levels depending on what one believes total claims will be – in other words, depending on the treatment of interest on interest which there are disagreements about).

But it'd be stressed on the call that this all amounts to fantasies about what could be and even if one is (somehow) confident in what the recoveries will be, one needs to (somehow) be confident in the timeline those recoveries will be realized on and what happens in the interim (lest the IRR drifts ever lower as the timeline to realization marches ever further into the distance).

Anyway, based on the price action we saw through the first week sans-Maduro, there are two broad takeaways one should have. First, that somehow the actions of the administration will redound to somewhat higher recoveries than the consensus view was beforehand. Second, that in order for the price levels across the VENZ and PDVSA complexes to be sensical, a resolution will be needed in somewhat short fashion (read: this doesn't kick the can *another* eight years or whatever).



In other words, it's hard to look at the price action we saw and not conclude that the market (at least some corners of it) thought that the administration would both strong-arm a solution to the overlevered capstack of a sovereign somewhat quickly (read: low single-digit years) and in a manner that boosts recoveries of creditors all else equal (read: the price action isn't purely signaling a bump up in the timeline to resolution). With the latter (lest the sovereign be left still too levered) requiring a pretty immediate *and* significant turnaround in the economic fundamentals of Venezuela based on the high levels of debt to begin with and the amount of past due interest that's accrued.

Perhaps a better way to frame all this is that in the immediate aftermath of the capture of Maduro, there was an impression that this was the beginning of true regime *change* and the administration would ensure that what came after this change would be a regime aligned with U.S. interests.

And the surest way to do that would be to make the populace prosperous (read: if the average individual's standard of living doubles a few years after a regime change then whoever receives the credit for precipitating that change will probably stand in good odor – most of the time, at least).

In the case of Venezuela with a current GDP of ~\$80b – suppressed after decades of economic suffocation – it wouldn't take much to see GDP reinflate rather quickly: mix together sanctions relief, encouragement of foreign investment, remigration of those that fled over the last few decades, and a rejuvenation of the oil sector and that should do the trick. The first steps on the road to economic recovery aren't the difficult ones (those come later as true potential GDP comes closer into view).

Anyway, the market seems to have believed that as these first few economic revitalization steps are taken – which could happen without much issue over the next 12-16 months – the contours of a full-scale sovereign restructuring, that'd encompass PDVSA, could be set.

And because when the contours are finalized and negotiations begin in earnest the GDP of Venezuela will have reinflated somewhat from its previously almost completely deflated state this would mean that significant recoveries for holders *could* be achieved because while the post-reorg level of debt would be high based on \$80-100b of GDP, it'd be more modest, manageable, and reasonable based on \$200b or whatever

To back up for a minute, before the kerfuffle in Caracas VENZ bonds traded around 29 on par (or 17 based on the total claim if we count all accrued interest), and PDVSA bonds traded at 25 on par (or 15 based on the total claim if we count all accrued interest). So, with a sufficient ramp up in oil production, foreign investment resurgence, sanctions relief, capital control relinquishment, and a (somewhat) stabilized political system, sure, it's not impossible to imagine a world in which there are ~45-50ish recoveries (based on par – so at a slight premium to where we're trading now, and a much higher level than in Q3 2025 before anyone thought twice about a VENZ solution being in the offing soon).

It'll all come down to the haircut – both on principal and past due interest – that is viewed as acceptable and which doesn't overburden Venezuela going forward. But it's not inconceivable that with 50-60% haircuts on both principal and past due interest, \$30-40b of new official money, and GDP that bounces to \$100b (possibly up to \$200b) in the short-term, we could see debt-to-GDP in the ~50-60% context which should be agreeable. (Remember GDP was \$350b in the early 2000s.)



Now, there might not be exact procedural parallels between corporate and sovereign restructurings but, of course, debt is debt, leverage is leverage, and thus no matter the type of issuer – sovereign or corporate – it all boils down to the same set of ideas that inform the end state and creditor IRRs.

Nuance aside, in both a corporate and sovereign context (outside some anomalous situations) what precipitates the need to restructure is an unsustainable capital structure – often one that the issuer themselves might not believe is unsustainable, but that the market has decided is (as we’ve previously discussed, often it’s a bit of a feather that broke the camel’s back situation where an issuer looks resilient in the face of bad headlines but then it all becomes a bit too much and their access to capital markets is suddenly foreclosed and their debt begins to trade down significantly).

Anyway, in a corporate context when we think about recoveries we, of course, think about EV as the representation of the value pie that will then be divided amongst holders. In the sovereign context it’s somewhat similar except the representation of value can be thought of as captured by GDP instead (which isn’t as fair of a representation but since multi-lateral institutions, who often pony up new-money post-reorg, care about debt-to-GDP it’s a representation that practically matters).

In both cases, at a high level, what we’re trying to *back into* is the right level of debt post-reorg that makes the issuer viable – a level that provides equitable value to pre-reorg holders that are not getting back what they originally bargained for but that also doesn’t burden the issuer such that they’ll perhaps land in the same spot in a few years (read: having to do yet another restructuring).

And, of course, if one thought that for all the superficial rigidity of the in-court process – how mechanized and formulistic it all looks – the process of determining the right value, and the right mix of consideration for holders, seems pretty subjective then it *really* is when dealing with sovereign issuers.

In both cases there’s the patina of formalism wrapped around a process that, at heart, is a kind of large-scale bargain that can be swayed by a myriad of non-objective factors (e.g., the willingness of large holders to be a nuisance; the intransigence of the issuer despite their current predicament; the involvement of quasi-third-parties like the UST, whether that’s the U.S. Trustee in the corporate context or the U.S. Treasury in the sovereign context; etc.).

Likewise, as we discussed at length in the corporate context, most often when a corporate files not only do they seem to not have a sustainable capital structure – at least based on what their EV is now, and what the market believes their EV will be in the future – but they often have de minimis cash (since capital markets were closed to them due to their unsustainable capital structure, etc.).

Thus, while it’s nice to file, have holders take a significant haircut, and delever – and that can make the post-reorg capital structure appear more sustainable on paper at emergence – there’s often the need for a significant cash infusion (whether to fund in-court costs or to have cash on the balance sheet to, like, operate after emergence while the ship is steadied). And, of course, this can be done through fresh debt or an equity rights offering as we’ve talked at length about before.

So, when we think about the post-reorg capital structure of a debtor, it’s both the new debt that’s consideration for the pre-reorg debt (if that’s part of the consideration makeup) *and* the new debt raised to fund in-court costs and/or put cash on the balance sheet at emergence (if that’s done).



Likewise, in a sovereign context, things tend to happen gradually then suddenly. In the case of Venz, they've been more or less frozen out of traditional capital markets for almost a decade since, well, they stopped paying interest (which is one of those things that tends to make people skittish about lending to you – well, that and all the sanctions, I suppose).

Thus, when we think about how to back our way into what we believe will be a sustainable capital structure we need to think about the possible *aggregate* debt levels post-reorg relative to GDP (read: debt-to-GDP). And the aggregate debt will be comprised of both the potential new debt provided to pre-reorg creditors as consideration for their holdings *and* then new debt raised to support the sovereign post-reorg (e.g., most often from multi-lateral institutions who, as previously mentioned, really care about post-reorg debt-to-GDP levels).

But, of course, this all becomes a bit circular – because, at base, the amount of haircut that current holders will “need” to take will be informed by both the amount of new-money believed to be “required”, and how GDP is expected to evolve in the near-term.

Or, put another way, the amount of new-money that can be raised will be limited by what's believed to be an acceptable debt-to-GDP ratio (one that is “sustainable”) and the amount of post-reorg debt provided to current holders as consideration for their holdings. It all depends at which end of the concentric circle one begins at.

And much like a company in a volatile sector that's fallen on hard times, it's often the case that when a sovereign restructures everyone acknowledges it's at (or near) the nadir of its GDP – that this level of GDP is far below its potential and it'll rebound quickly. Thus, while 30-60% debt-to-GDP might be the ideal, safe level to aim for, is that based on near-term GDP or what consensus forecasts are for GDP 24, 36, or 48 months out?

In other words, can the sovereign “emerge” a bit more levered than one would hope for at time zero based on the assumption it should look appropriately levered in a few years as its economy reinflates? Is it a disservice to base recoveries on near-term GDP when that'll make debt-to-GDP look low in short succession?

Of course, nowhere is this dilemma – over what GDP level is reasonable to expect – more obvious and stark than in Venezuela where there's been years-long entrenchment of Chavismo with the sovereign cut from capital markets over the last eight years (since its default) and its GDP collapsing to what appears to be near the minimal viable level (from ~\$350b to ~\$80b in a few decades).

Note: One way to paper over this dilemma – to an extent – is to have a larger nominal haircut but have some of the consideration mix come in the form of Value Recovery Instruments (VRIs) linked to either GDP or commodity-related revenue (read: providing creditors some participation in the “upside” or “turnaround” of the sovereign, in the case of Venz perhaps through VRIs linked to the performance of state-owned energy companies or oil prices outright). I'm personally (for what little it's worth) skeptical of all this in the Venz context and see VRIs or other forms of contingent instruments as potentially something that can be a modest tool in the toolbox at most (something that can bridge a *narrow* gap between both sides, if needed). Because with the kind of debt load we're talking about here, and creditor appetite (or lack thereof) for more equity-like instruments comprising a significant part of their consideration, I don't see this as a viable, real-world solution to start from.



Here's a nice summation from GS' EM desk in late-2025 that sketched out what was more or less consensus at the time (with not much changing in the market's mind after the capture of Maduro in terms of the variables except, for some, penciling in a bit higher GDP and a bit faster timeline...).



Venezuela & PDVSA

Weighing Regime Change | Assessing Recovery Scenarios

1/ Venezuela and PDVSA complex up 110-120% year-to-date with Venezuela complex trading at average price of 29.8c (17.3c on claim) and PDVSA at 25.7c (15.9c on claim)

2/ EM Credit Team view on recovery is, assuming: 1) three-year process (Iraq), 2) 50-60% principal and PDI haircuts, 3) new 2045 bond with step-up coupons starting at 4.5% going to 6.5% of instrument life 4) amortizing starting in mid-2030s, 5) Brent at \$60-65/bbl, 6) production at 1.5-2mmmbbl/d, 7) new money needs of \$40-60bn, 8) debt stack of around \$165-175bn;

3/ Then at exit yields of 12-15%, arrive at theoretical recovery values in the 25c to 30c *on claim* context

4/ Venezuela would still exit with high debt to GDP metrics, but grow into capital structure over medium-term | assume GDP around \$85-100bn today

5/ Market (obviously) increasingly pricing in probability of political catalyst | main scenarios for market participants are: 1) kinetic regime change, 2) negotiated regime change, 3) no regime change, but deal with U.S., 4) no regime change, no deal

Anyway, the final variable to add to the mix that's worth a brief discussion – because it's high vol impacts pricing on the screens – is time (read: discount rate). While pricing *is* informed by factors that are more traditionally thought of as informing recoveries (e.g., how big the pie will be, what types of slices everyone is getting, who will be taking what slices, etc.) it's also informed by the timeline on which the anticipated restructuring will happen – which is, as much as anything can be, more of a raw geo-political calculation (since it's more or less informed by when the current regime sufficiently changes or when a new government deemed legitimate enters into office and *then* how much they prioritize, perhaps with the cajoling of the UST and State, the beginning of a restructuring process).

This is a point that I spent quite a bit of time harping on in the Bond Math Guide and the Case Study Guide. Because even in a corporate context timelines are apt to change and even if one is right on the “fundamentals” (e.g., the anticipated recoveries) if one is wrong on *when* those recoveries will be realized (e.g., when a company will file, how long it'll take to emerge thereafter, etc.) then that can skew returns to such a degree that it changes whether it was a worthwhile trade to enter into or not.

While it's impossible to disentangle all the contributors to upward price action when you're dealing with a unique situation – and I would think the extraction of the leader of a country in the dead of night thanks to discombobulation devices would qualify – there are without doubt some shops that decided to take a kick at the VENZ / PDVSA can in the immediate aftermath because they thought, “Well, if at T-1, before the extraction, the price of this bond was [X] then all else equal the price at T+1, right after the extraction, should be higher because the timeline on which a recovery can be realized *has* to be at least *somewhat* shorter than it was before. Maybe GDP will increase because of more U.S. directed reforms, etc. in the short-term and thus boost recoveries, maybe not. But we don't need to engage in some scenario analysis on that right now – if we receive the same recovery but sooner than we were expecting before, then that recovery is per se more valuable now. QED.”



The above quote more or less matches the chatter – at least coming out of the distressed desks that were flipping VENZ/PDVSA debt at a rapid clip – in the 24-72 hours after the capture. However, it seemed to me at the time (and more so as the week came to a close) to be at odds with both our most recent experience with more UST-involved sovereign restructurings and also the President’s own, uh, disposition toward creditor treatment in restructurings (see: the old yarns in *The Vulture Investors* by Rosenberg – published in 2000 and that I’ve referenced before – which feature a few Trump default stories alongside some cameos from those that would also go on to have a larger public profile than they had at the time like the then young-ish Ken Moelis).

To this end, late on Friday (Jan 9) [an executive order](#) was issued that [declared a national emergency](#) to “...safeguard Venezuelan oil revenue held in U.S. Treasury accounts for attachment or judicial process, ensuring these funds are preserved to advance U.S. foreign policy objectives”.

Which amounts to a pretty clear message to creditors that Venezuelan oil revenue – which before Maduro’s capture was, of course, at arms-lengths from creditors – will continue to be at arms-length from creditors because that revenue is explicitly tied to the ambiguous advancement of U.S. natsec and foreign policy interests.

In other words, the typical tools of attachment and enforcement weren’t operative before and may not be operative now or in the future because the order “...blocks any attachment, judgement, lien, execution, garnishment, or judicial process against Foreign Government Deposit Funds.” With Foreign Government Deposit Funds defined as including “...the Venezuelan oil revenues and diluent sales held in U.S. Treasury accounts.”

So, who do these funds belong to? Well, the order continues that it “...affirms the funds are sovereign property of Venezuela held in U.S. custody for governmental and diplomatic purposes, not subject to private claims”. Which most would read (although there are disagreements) as requiring that until the U.S. Treasury deems a valid government to be in place, no matter how much funds accrue, over whatever length of time, these funds will not be up for grabs – and when they are, it’ll be up to the “valid” government vis-à-vis how they’re used (or not) as part of a restructuring transaction. (Or at a minimum the UST will have, in the interim, significant say in how these funds are used by Venez.)

In the immediate aftermath of the Maduro capture, there was a narrative that it was going to somehow be an essential component of the Donroe Doctrine to take an active interest in turning Venezuela around and cleaning up Venezuela’s legacy debt. For reasons that remain elusive, the narrative was (is?) that this latter point would be a foreign policy win for the administration.

While perhaps more skeptical than most, I’m not sure that the most pressing matter on the Donroe docket is significant time and energy being placed to support a fundamental rejiggering of Venezuela’s economy and a restructuring of VENZ and PDVSA debt.

And even if it were to be that a more fundamental rejiggering and restructuring was sought, would the likelihood be that the administration would ensure that it’s on favorable terms to creditors or would the administration look around and find it more favorable for a bunch of opportunistic funds to take a bit bigger hit on their debt so that Venezuela can emerge with a reasonable debt load and the administration can look equanimous for keeping the prying creditors at bay and do that instead?



The foreign policy win for the administration is surely not to leave a bunch of holders of long since defaulted VENZ and PDVSA paper with nice IRRs, but the sovereign itself burdened by a still significant debt load that leads to resentment and bitterness that festers over time as the new government feels stymied due to the “unjustified” level of debt they were strong-armed into maintaining (with the salt in the wound that the debt was layered on by an old government with, presumably, minimal linkages to the new government).

Given this, the current situation might not be a repeat of Iraq, but from the EO it does look an awful lot like it rhymes. Because in 2003 the Bush administration signed [EO 13303](#) (Protecting the Development Fund for Iraq and Certain other Property in Which Iraq Has an Interest) that, no different than the Trump administration’s EO, shielded certain Iraqi funds and revenues from attachment and, by extension, removed one of the most important ways for creditors to flex muscle in a sovereign restructuring.

In other words, in both cases one of the most important sources of ongoing cashflow – of value – to creditors was deemed by both administrations better used to support the sovereigns in the future than used in some sense to repay the debts of the past accrued by regimes that have been ousted, replaced, or reconfigured (pick your euphemism). Again: an obvious thing to do from a foreign policy perspective, but one that leaves creditors a bit in the lurch.

Now, things can always change – and with this administration perhaps a few times in a few different directions. But of interest – and in support of the possibility that we’ll perhaps see an Iraq-approach taken, not a new [Third Way](#) designed *de novo* – is that in the President’s roundtable discussion with oil executives, it seemed like the President at least partially adopted a “what’s done is done” attitude toward legacy claims.

Because when Ryan Lance – the CEO of ConocoPhillips – brought up ConocoPhillips’ old \$12b claim against Venezuela due to losses incurred from a past nationalization of their assets, the President joked that it’d be a “good write-off” and that companies like ConocoPhillips would “...start with an even plate” in Venezuela.

To be fair, this is somewhat orthogonal to more classic VENZ and PDVSA claims that’ll be at issue in an eventual restructuring. But it speaks to a mindset – that has been shared across administrations when it comes to sovereign restructurings in countries where the U.S. has natsec interests – that it’s better to see things start from a *relatively* clean slate (or an “even plate”, in this case) than to strong-arm a few extra points of recovery out of a sovereign which could potentially breed resentment and bitterness that’ll rear its head in the future.

With all that said, if we take the admin at face value, then while this might be regime change it’s a revamped style from what we became accustomed to in the early-aughts with the definition of change in the admin’s mind perhaps less “removal” and more “recalibration”. Thus, insofar as Rodriguez, et al. can engage with the admin in a more constructive fashion to make the changes the admin would like to see – the removal of the ELN and other groups active in Colombia, the stop of drug trafficking, an operational restructuring of the oil sector, and the end of Venezuela “cozying up”, in the words of Sec Rubio, to adversaries in search of a foothold in the Western hemisphere – then the thornier, messier, more time-intensive work of a proper sovereign restructuring that places Venezuela on a pathway toward a more durable economic future can be shelved for a bit.



Which all means that if the rationale for some opportunistic funds getting into the mix post-capture was less based on a change in recoveries and more on *when* those recoveries would flow (read: much sooner than previously thought), then it's not that this rationale has been disproved but it's at least a bit more doubtful than it was when MH-47 Chinooks were flying low over Caracas...

BRIEF VENZ AND PDVSA NOTES FOR THE WEEK ENDED JAN 30, 2026

With the close of this week what we saw (although opinions differ, of course) is more aligned with what I expected at first blush: that this is less true regime change than regime recalibration, and that while the capture of Maduro might accelerate the timeline on which a restructuring occurs, and might precipitate reforms that lead to enhanced GDP, there's no real restructuring-related time pressure felt by the administration and, when push comes to shove, if there's any thought that more favorable creditor treatment could undermine the natsec and foreign policy aims of the administration than such favorable treatment won't likely occur.

And this has all begun to be reflected in the price action we've seen after the first week of tumult: bonds this week closed around a point higher (give or take a few bps across the complex) with VENZ debt outperforming PDVSA a touch (as modest concerns rise that maybe whatever restructuring occurs will see Republic debt take priority over PDVSA debt – I have my doubts on that, but it's understandable that there's a bit higher of a risk premium attached to PDVSA debt, so fair enough).

While uncertainties still abound, this week saw a crystallization of the admin's strategy going forward – one that can perhaps be summarized as managing Venz as it exists now to best serve current natsec and foreign policy aims, with more structural issues on the back burner for the time being.

For example, this week saw the National Assembly's passage of a new hydrocarbons bill which then led to OFAC issuing a new general license that authorizes oil-related transactions between U.S. persons and VENZ/PDVSA so long as contracts are governed by U.S. law and payments are directed to the Foreign Government Deposit Funds outlined in the previously discussed Jan 9 executive order.

Of interest, and it shouldn't be a surprise, is that the license excludes any transactions that involve Cuba, China, Iran, North Korea, and Russia in an attempt to halt Venezuela's capacity to implicitly support U.S. geo-political adversaries (or, from the other side of the ledger, have these adversaries develop a stronger foothold in Venezuela through closer economic ties).

While not a surprise, it does raise questions as to how China, who holds around \$12-13b of Venz debt, will be treated in a future restructuring transaction based off of this (read: does this signal a potential where not all creditors will be treated *pari* in a future restructuring – there's no clear answer, and I have my doubts a bifurcation would be worth the inevitable delays and complications, but worth keeping in the back of one's mind).

More of note – and what did cause more downside price action before things reset on Friday – was Sec. Rubio's Senate testimony that shed light on the current flow of funds. As of this week, \$500m from initial Venz crude sales have been placed in Qatari accounts in Venz's name – done explicitly to prevent creditor attachment or seizure or, in his own words, to avoid revenues being “...picked apart by creditors”.



Rubio went on to acknowledge that “...we will eventually have to take care of creditors” but didn’t elaborate much beyond that (not to enter into a semantics squabble, but note the use of *have* instead of *will* which suggests, based on the tone taken throughout, that pre-existing creditors are thought of as more of an obstacle to be dealt with rather than a favored party to be prioritized moving forward).

Further, to revisit what “change” means in the context of this regime change, it’s beginning to look more like Rodriguez is pursuing an accommodative stance on administration priorities (e.g., the new hydrocarbons law, release of U.S. prisoners, etc.) with the belief that if this is done true regime change (via free and fair elections) can be kicked down the road (perhaps through planting the idea that such change runs the risk of a less accommodative government entering into office, etc.).

Which all means, for creditors, that the situation continues to become more nuanced and the similarities to Iraq begin to become clearer. On the one hand, the steps that look like they’ll be taken (although it’s still early days) should enhance GDP, and all else equal there’s no doubt a restructuring *should* still occur on a quicker timeline than if Maduro was still in power.

However, if a restructuring isn’t a priority of anyone that could precipitate it right now, then we could still be looking at a timeline that stretches years into the future (read: only beginning after significant economic liberalizations occur *and* a political transition occurs). In other words, the notion that recoveries, all else equal, will be higher than they appeared to be a month ago remains true (hard for that not to be the case with how deflated GDP was pre-capture). But, on the other hand, and as I’ve harped on, the timeline matters – and while Rubio acknowledges that “eventually” creditors will need to be dealt with, in the admin’s priority list the restructuring of VENZ/PDVSA debt is quite clearly subordinate to natsec and foreign policy aims...

Week Ended Jan 30 Price Action

Americas EM Credit Monitor

	Last			Last week			Last month			Last quarter		
	Price	G-Spread	Yield	Price	G-Spread	Yield	Price	G-Spread	Yield	Price	G-Spread	Yield
Distressed				Price			Price			Price		
VENZ 8 1/4 10/13/24	37.10	n/a	n/a	0.84			8.55			11.45		
VENZ 11.95 08/05/31	44.23	n/a	n/a	0.73			11.66			15.37		
PDVSA 6 05/16/24	30.59	n/a	n/a	(6.29)			6.90			8.68		
PDVSA 9 3/4 05/17/35	36.68	n/a	n/a	0.23			10.51			12.46		

Week Ended Jan 09 Price Action

Americas EM Credit Monitor

	Last			Last week			Last month			Last quarter		
	Price	G-Spread	Yield	Price	G-Spread	Yield	Price	G-Spread	Yield	Price	G-Spread	Yield
Distressed				Price			Price			Price		
VENZ 8 1/4 10/13/24	37.87	n/a	n/a	9.32			8.29			16.73		
VENZ 11.95 08/05/31	43.65	n/a	n/a	11.08			10.69			18.99		
PDVSA 6 05/16/24	31.20	n/a	n/a	7.51			5.62			14.06		
PDVSA 9 3/4 05/17/35	36.37	n/a	n/a	10.20			8.62			16.96		



APPENDIX

In case it's of interest, attached are some good pieces on Venez from over the years as potential recovery estimates and restructuring frameworks have been sketched out (plus, some other stuff). Note that Moatti and Muci's framework is the most frequently cited that I've seen, despite its age. But, of course, until rubber meets road – whenever that is – these all skew to the more philosophical end of things rather than the practical...

- Barclays – Bond Prices after Maduro (2026)
- GS – A Look at Recovery Estimates for Venezuela (2023)
- [Moatti and Muci – an Economic Framework for Venezuela's Debt Restructuring \(2019\)](#)
- GS – Lessons from 40 Years of Sovereign Defaults and Distress (2020)
- GS – Americas First (2026)
- [Lazard – Policy Brief: The 2020-2025 Sovereign Debt Crisis \(2025\)](#)
- [Lazard – Policy Brief: Domestic Debt Restructuring. An Exercise in Laser Surgery \(2023\)](#)
- [Gelpern – The Strained Marriage of Public Debts and Private Contracts \(2017\)](#)
- [Hinrichsen – Tracing Iraqi Sovereign Debt Through Defaults and Restructuring \(2019\)](#)

Venezuela/PDVSA Sovereign Strategy

Bond prices after Maduro

On Monday, VENZ and PDVSA bonds rallied approximately 30-35%, implying a current recovery value of \$0.37 (versus par). PDVSA trades at a c.11% discount while past due interest continues to screen as cheap.

Jason Keene

+1 212 526 7275

jason.keene@barclays.com

BCI, US

Key takeaways from Monday's price action

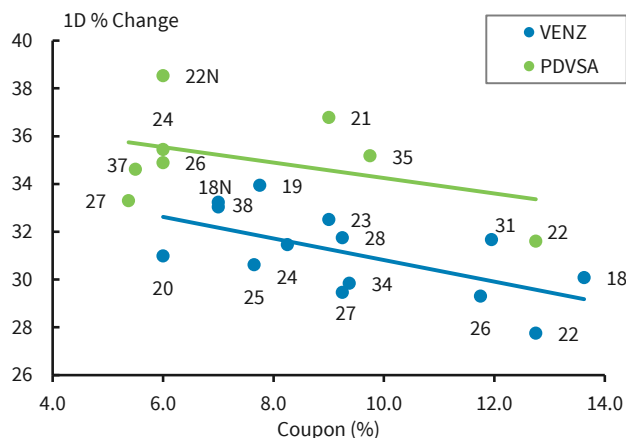
- **The market has priced a recovery value of \$0.37:** The total market value of unsecured debt reached \$20bn, implying a current recovery value of \$0.37 on the dollar (from \$1.00 par) and \$0.21 (from \$1.00 in total claim).
- **PDVSA discount of c.11%:** Adjusting for past due interest (PDI), PDVSA bonds trade at a c.11% discount to VENZ bonds.
- **Past due interest continues to screen as cheap:** Across both PDVSA and VENZ bond complexes, one unit of PDI is trading at c.40% the value of one unit of principal.

This document is intended for institutional investors and is not subject to all of the independence and disclosure standards applicable to debt research reports prepared for retail investors under U.S. FINRA Rule 2242. Barclays trades the securities covered in this report for its own account and on a discretionary basis on behalf of certain clients. Such trading interests may be contrary to the recommendations offered in this report.

Please see analyst certifications and important disclosures beginning on page 6.

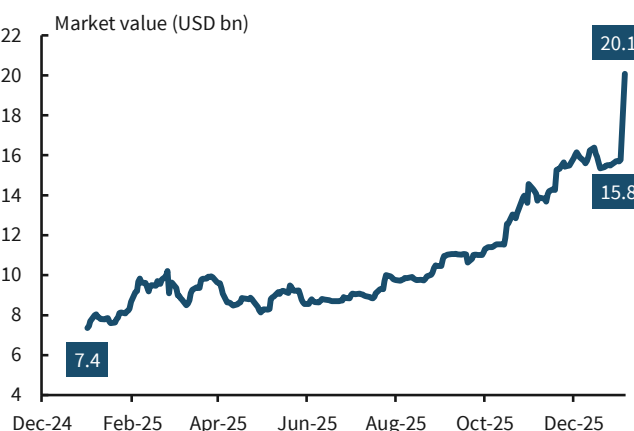
Monday's price action: On Monday, VENZ and PDVSA bonds rallied approximately \$7.50-9.50 (or approximately 30-35%, on average). On a percentage basis, PDVSA outperformed VENZ and lower-coupon bonds outperformed higher-coupon bonds (Figure 1). The total market value of VENZ and PDVSA unsecured debt has risen above \$20bn (Figure 2). Unsecured bondholders' par claim is \$56.5bn and their total claim including PDI is \$98.4bn. This implies a current recovery value of \$0.36 (from \$1.00 par) and \$0.21 (from \$1.00 total claim).

FIGURE 1. VENZ and PDVSA performance (1D %)



Source: Bloomberg, Barclays Research

FIGURE 2. Market value of PDVSA/VENZ unsecured debt (USD bn)

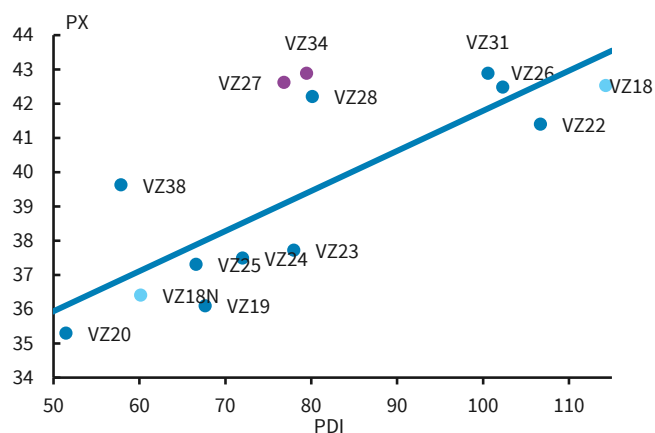


Source: Bloomberg, Barclays Research

What's priced in on a relative value basis? In line with previous exercises, we regress both the VENZ (Figure 3) and PDVSA (Figure 4) curves separately, controlling for PDI in both cases. In the VENZ curve, we also control for stronger CAC protections (VENZ 9.25% 2027s and VENZ 9.375% 2034s) and off-index exposure (VENZ 13.625% 2018s and VENZ 7% 2018s). The two most-important conclusions are as follows:

- PDVSA bonds trade at an 11% discount to VENZ bonds, on average (Figure 5)
- Past due interest continues to screen as cheap, with one unit of PDI trading at c.40% the value of unit of principal (Figure 6)

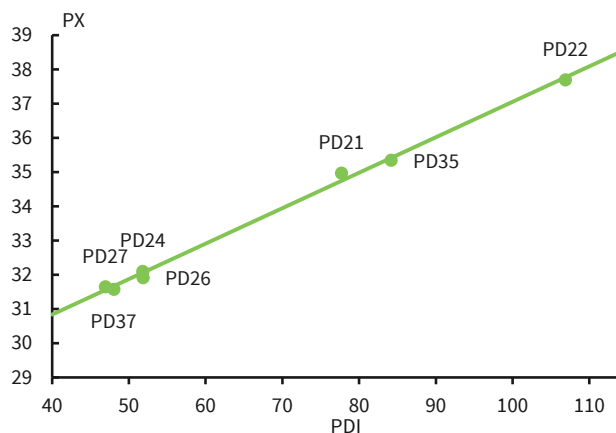
FIGURE 3. VENZ bond prices and past due interest



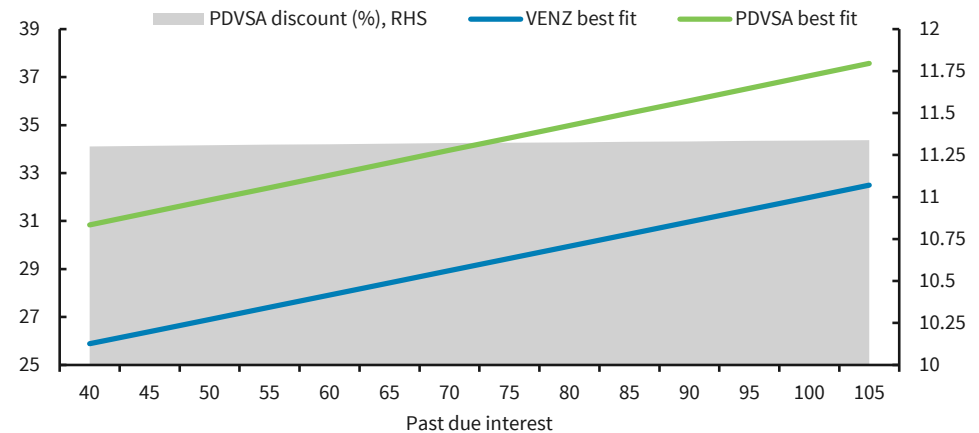
Light blue data points represent off-index bonds. Purple data points represent bonds with stronger collective action clauses (CACs).

Source: Bloomberg, Barclays Research

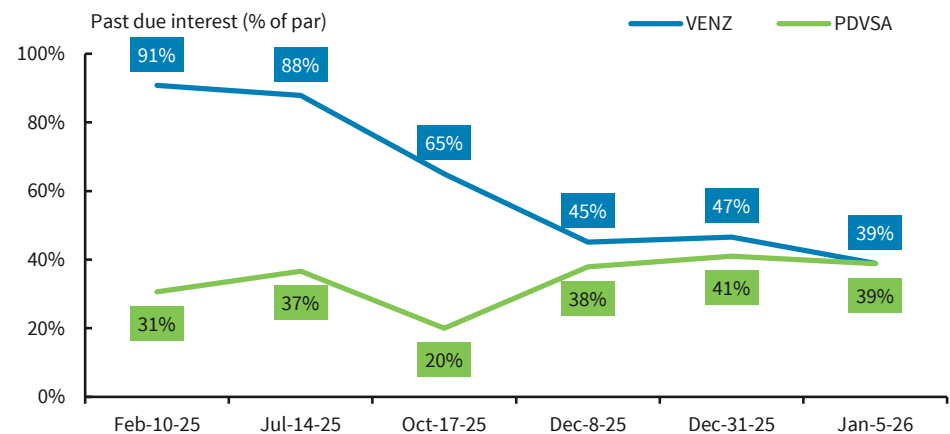
FIGURE 4. PDVSA bond prices and past due interest



Source: Bloomberg, Barclays Research

FIGURE 5. PDVSA bonds trade at a c.11% discount to VENZ bonds


Source: Bloomberg, Barclays Research

FIGURE 6. The market-implied value of past due interest has converged at c.40%


Source: Bloomberg, Barclays Research

Explaining the model: Ahead of the market open, we published a table with potential upside estimates based on a series of potential recovery values (\$0.35/\$0.40/\$0.45). We have refined our model and included it here. We make several simplifying assumptions:

- All VENZ and PDVSA unsecured bonds are treated as *pari passu* with one another
- Past due interest (PDI) is treated as *pari passu* with par claims

To start, we assume that recovery values of \$0.35/\$0.40/\$0.45 are measured from par (not from total claim). This allows us to come up with a market value for new, restructured bonds that would accrue to VENZ and PDVSA bondholders in an eventual restructuring. These recovery values allow us to back into the market value of the new, restructured bonds — labeled in the table below as *Total Recovery (\$bn)*. Then, we calculate recovery values as a percentage of total claim. See [Figure 7](#) for more details.

FIGURE 7. Converting recovery values from % of par to % of total claim, including past due interest

Recovery (% of par)	35.0	40.0	45.0
Recovery (% of total claim)	20.1	23.0	25.8
VENZ recovery (\$bn)	10.9	12	14
PDVSA recovery (\$bn)	8.9	10	11
Total recovery (\$bn)	19.8	22.6	25.4

Source: Bloomberg, Barclays Research

Why is this an important step? Because in this stylized model, we are assuming that past due interest will be treated as *pari passu* with par claims. We take the total recovery value (the market value of the new restructured debt) and divide by the total bonded debt claim. Then, we distribute new bonds as a percentage of each bond's total claim (par plus PDI). Finally, we convert each bond's total claim back into face value terms. The math in the subsequent columns is as follows:

$$\frac{\text{Bond claim (\$mn)}}{\text{Face value (\$mn)}} * \text{Recovery (\% of all bonded claims)}$$

Finally, the last six columns in [Figure 10](#) illustrate how far current pricing is from model-implied recovery values under the various scenarios.

FIGURE 8. Summary of VENZ upside/downside under various recovery value assumptions

VENZ						
	35c	40c	45c	35c T+2	40c T+2	45c T+2
Low	-7.8	-3.3	1.2	-11.3	-7.3	-3.3
High	0.9	7.1	13.2	-3.8	1.6	7.1
Avg	-3.6	1.7	6.9	-7.6	-2.9	1.7

For a bond-by-bond breakdown, see [Figure 10](#).
Source: Bloomberg, Barclays Research

FIGURE 9. Summary of PDVSA upside/downside under various recovery value assumptions

PDVSA						
	35c	40c	45c	35c T+2	40c T+2	45c T+2
Low	-2.2	2.1	6.3	-5.4	-1.6	2.1
High	3.9	9.8	15.7	-0.7	4.6	9.9
Avg	0.0	4.6	9.3	-3.7	0.5	4.7

For a bond-by-bond breakdown, see [Figure 10](#).
Source: Bloomberg, Barclays Research

FIGURE 10. VENZ and PDVSA unsecured bonds and upside/downside under various recovery value assumptions

Ticker	Price	1D	1D %	Issue Date	CAC	PDI	Market Value (\$bn)	Face Value (\$bn)	Total Claim (\$bn)	35c	40c	45c	35c T+2	40c T+2	45c T+2	35c	40c	45c	35c T+2	40c T+2	45c T+2
VENZ 13.63% 18	42.5	9.5	28.6	Aug-98	-	114	0.32	0.75	1.61	43.1	49.2	55.4	38.3	43.8	49.3	0.5	6.7	12.8	-4.2	1.3	6.8
VENZ 13.63% 18	42.2	9.8	30.2	Sep-01	-	114	0.13	0.30	0.64	43.1	49.2	55.4	38.3	43.8	49.3	0.9	7.1	13.2	-3.8	1.6	7.1
VENZ 7% 18	36.4	9.1	33.1	Dec-03	85%	60	0.36	1.00	1.60	32.2	36.8	41.4	28.6	32.7	36.8	-4.2	0.4	5.0	-7.8	-3.7	0.4
VENZ 7.75% 19	36.1	9.2	34.0	Oct-09	75%	68	0.90	2.50	4.18	33.7	38.5	43.3	30.0	34.3	38.6	-2.4	2.4	7.2	-6.1	-1.8	2.4
VENZ 6% 20	35.4	8.5	31.4	Dec-05	75%	51	0.53	1.50	2.27	30.4	34.8	39.1	27.1	31.0	34.8	-5.0	-0.6	3.7	-8.3	-4.4	-0.6
VENZ 12.75% 22	41.4	9.0	27.8	Aug-10	75%	107	1.24	3.00	6.20	41.5	47.5	53.4	37.0	42.2	47.5	0.1	6.1	12.0	-4.4	0.8	6.1
VENZ 9% 23	37.7	9.3	32.5	May-08	75%	78	0.75	2.00	3.56	35.8	40.9	46.0	31.8	36.4	40.9	-2.0	3.1	8.3	-5.9	-1.3	3.2
VENZ 8.25% 24	37.5	9.0	31.5	Oct-09	75%	72	0.94	2.50	4.29	34.6	39.5	44.4	30.8	35.2	39.6	-2.9	2.0	6.9	-6.7	-2.3	2.1
VENZ 7.65% 25	37.3	8.7	30.6	Apr-05	75%	67	0.60	1.60	2.67	33.5	38.3	43.0	29.8	34.1	38.3	-3.8	1.0	5.7	-7.5	-3.3	1.0
VENZ 11.75% 26	42.5	9.6	29.3	Oct-11	75%	102	1.27	3.00	6.07	40.7	46.5	52.3	36.2	41.4	46.5	-1.8	4.0	9.8	-6.3	-1.1	4.0
VENZ 9.25% 27	42.6	9.7	29.4	Sep-97	-	77	1.70	4.00	7.07	35.5	40.6	45.7	31.6	36.1	40.7	-7.1	-2.0	3.1	-11.0	-6.5	-1.9
VENZ 9.25% 28	42.2	10.2	31.7	May-08	75%	80	0.84	2.00	3.60	36.2	41.4	46.5	32.2	36.8	41.4	-6.0	-0.8	4.3	-10.0	-5.4	-0.8
VENZ 11.95% 31	42.9	10.3	31.7	Aug-11	75%	101	1.80	4.20	8.42	40.3	46.1	51.8	35.9	41.0	46.1	-2.6	3.2	8.9	-7.0	-1.9	3.2
VENZ 9.38% 34	42.8	9.8	29.7	Jan-04	85%	79	0.64	1.50	2.69	36.1	41.2	46.4	32.1	36.7	41.3	-6.8	-1.6	3.5	-10.7	-6.2	-1.6
VENZ 7% 38	39.6	9.8	33.0	Nov-07	75%	58	0.49	1.25	1.97	31.7	36.3	40.8	28.2	32.3	36.3	-7.8	-3.3	1.2	-11.3	-7.3	-3.3
PDVSA 9% 21	35.0	9.4	36.8	Nov-11	-	78	0.84	2.39	4.25	35.7	40.8	45.9	31.8	36.3	40.9	0.7	5.8	10.9	-3.2	1.4	5.9
PDVSA 12.75% 22	37.7	9.1	31.6	Feb-11	-	107	1.13	3.00	6.21	41.6	47.5	53.5	37.0	42.3	47.6	3.9	9.8	15.7	-0.7	4.6	9.9
PDVSA 6% 22*	28.6	-	38.5	Oct-14	-	49	0.86	3.00	4.47	30.0	34.2	38.5	26.7	30.5	34.3	1.3	5.6	9.9	-2.0	1.9	5.7
PDVSA 6% 24	32.1	8.4	35.5	May-14	-	52	1.61	5.00	7.59	30.5	34.9	39.2	27.2	31.0	34.9	-1.6	2.8	7.1	-5.0	-1.1	2.8
PDVSA 6% 26	31.9	8.3	34.9	Nov-13	-	52	1.44	4.50	6.83	30.5	34.9	39.2	27.2	31.0	34.9	-1.4	2.9	7.3	-4.8	-0.9	3.0
PDVSA 5.38% 27	31.7	7.9	33.4	Apr-07	-	47	0.95	3.00	4.41	29.5	33.7	38.0	26.3	30.0	33.8	-2.2	2.1	6.3	-5.4	-1.6	2.1
PDVSA 9.75% 35	35.4	9.2	35.2	May-12	-	84	1.06	3.00	5.53	37.0	42.3	47.6	32.9	37.6	42.4	1.7	6.9	12.2	-2.4	2.3	7.0
PDVSA 5.5% 37	31.6	8.2	34.8	Apr-07	-	48	0.47	1.50	2.22	29.7	34.0	38.2	26.5	30.3	34.0	-1.9	2.4	6.6	-5.1	-1.4	2.4
VENZ	40.3	9.5	30.9				12.5	31.1	56.9	36.8	42.0	47.3	32.7	37.4	42.1	-3.6	1.7	6.9	-7.6	-2.9	1.7
PDVSA (ex-2020s)	32.0	7.6	35.0				8.4	25.4	41.5	32.9	37.5	42.2	29.2	33.4	37.6	0.0	4.6	9.3	-3.7	0.5	4.7
Total	37.0						20.9	56.5	98.4	35.0	40.0	45.0	31.1	35.6	40.0	-2.0	3.0	8.0	-5.8	-1.4	3.1
% of GDP							25.2	68.2	118.9												

Source: Bloomberg, Barclays Research

Analyst(s) Certification(s):

I, Jason Keene, hereby certify (1) that the views expressed in this research report accurately reflect my personal views about any or all of the subject securities or issuers referred to in this research report and (2) no part of my compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this research report.

Important Disclosures:

Barclays Research is produced by the Investment Bank of Barclays Bank PLC and its affiliates (collectively and each individually, "Barclays").

All authors contributing to this research report are Research Analysts unless otherwise indicated. The publication date at the top of the report reflects the local time where the report was produced and may differ from the release date provided in GMT.

Availability of Disclosures:

For current important disclosures regarding any issuers which are the subject of this research report please refer to <https://publicresearch.barclays.com> or alternatively send a written request to: Barclays Research Compliance, 745 Seventh Avenue, 13th Floor, New York, NY 10019 or call +1-212-526-1072.

Barclays Capital Inc. and/or one of its affiliates does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that Barclays may have a conflict of interest that could affect the objectivity of this report. Barclays Capital Inc. and/or one of its affiliates regularly trades, generally deals as principal and generally provides liquidity (as market maker or otherwise) in the debt securities that are the subject of this research report (and related derivatives thereof). Barclays trading desks may have either a long and / or short position in such securities, other financial instruments and / or derivatives, which may pose a conflict with the interests of investing customers. Where permitted and subject to appropriate information barrier restrictions, Barclays fixed income research analysts regularly interact with its trading desk personnel regarding current market conditions and prices. Barclays fixed income research analysts receive compensation based on various factors including, but not limited to, the quality of their work, the overall performance of the firm (including the profitability of the Investment Banking Department), the profitability and revenues of the Markets business and the potential interest of the firm's investing clients in research with respect to the asset class covered by the analyst. To the extent that any historical pricing information was obtained from Barclays trading desks, the firm makes no representation that it is accurate or complete. All levels, prices and spreads are historical and do not necessarily represent current market levels, prices or spreads, some or all of which may have changed since the publication of this document. Barclays Research Department produces various types of research including, but not limited to, fundamental analysis, equity-linked analysis, quantitative analysis, and trade ideas. Recommendations and trade ideas contained in one type of Barclays Research may differ from those contained in other types of Barclays Research, whether as a result of differing time horizons, methodologies, or otherwise.

In order to access Barclays Statement regarding Research Dissemination Policies and Procedures, please refer to <https://publicresearch.barclays.com/S/RD.htm>. In order to access Barclays Research Conflict Management Policy Statement, please refer to: <https://publicresearch.barclays.com/S/CM.htm>.

Disclosure(s) regarding Information Sources

Bloomberg® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg") and the Bloomberg Indices are trademarks of Bloomberg. Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Bloomberg does not approve or endorse this material, or guarantee the accuracy or completeness of any information herein, or make any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, Bloomberg shall have no liability or responsibility for injury or damages arising in connection therewith.

All pricing information is indicative only. Unless otherwise indicated, prices are sourced from LSEG Data & Analytics and reflect the closing price in the relevant trading market, which may not be the last available price at the time of publication.

Types of investment recommendations produced by Barclays FICC Research:

In addition to any ratings assigned under Barclays' formal rating systems, this publication may contain investment recommendations in the form of trade ideas, thematic screens, scorecards or portfolio recommendations that have been produced by analysts in FICC Research. Any such investment recommendations produced by non-Credit Research teams shall remain open until they are subsequently amended, rebalanced or closed in a future research report. Any such investment recommendations produced by the Credit Research teams are valid at current market conditions and may not be otherwise relied upon.

Disclosure of other investment recommendations produced by Barclays FICC Research:

Barclays FICC Research may have published other investment recommendations in respect of the same securities/instruments recommended in this research report during the preceding 12 months. To view all investment recommendations published by Barclays FICC Research in the preceding 12 months please refer to <https://live.barcap.com/go/research/Recommendations>.

Barclays does not assign ratings to asset backed securities. Barclays Capital Inc. and/or one of its affiliates may have acted as an underwriter for public offerings of any asset backed securities that are otherwise recommended in trade ideas contained within its securitised research reports.

Legal entities involved in producing Barclays Research:

Barclays Bank PLC (Barclays, UK)

Barclays Capital Inc. (BCI, US)

Barclays Bank Ireland PLC, Frankfurt Branch (BBI, Frankfurt)

Barclays Bank Ireland PLC, Paris Branch (BBI, Paris)

Barclays Bank Ireland PLC, Milan Branch (BBI, Milan)

Barclays Securities Japan Limited (BSJL, Japan)

Barclays Bank PLC, Hong Kong Branch (Barclays Bank, Hong Kong)

Barclays Bank Mexico, S.A. (BBMX, Mexico)

Barclays Capital Casa de Bolsa, S.A. de C.V. (BCCB, Mexico)

Barclays Securities (India) Private Limited (BSIPL, India)

Barclays Bank PLC, Singapore Branch (Barclays Bank, Singapore)

Barclays Bank PLC, DIFC Branch (Barclays Bank, DIFC)

Disclaimer:

This publication has been produced by Barclays Research Department in the Investment Bank of Barclays Bank PLC and/or one or more of its affiliates (collectively and each individually, "Barclays").

It has been prepared for institutional investors and not for retail investors. It has been distributed by one or more Barclays affiliated legal entities listed below or by an independent and non-affiliated third-party entity (as may be communicated to you by such third-party entity in its communications with you). It is provided for information purposes only, and Barclays makes no express or implied warranties, and expressly disclaims all warranties of merchantability or fitness for a particular purpose or use with respect to any data included in this publication. To the extent that this publication states on the front page that it is intended for institutional investors and is not subject to all of the independence and disclosure standards applicable to debt research reports prepared for retail investors under U.S. FINRA Rule 2242, it is an "institutional debt research report" and distribution to retail investors is strictly prohibited. Barclays also distributes such institutional debt research reports to various issuers, media, regulatory and academic organisations for their own internal informational news gathering, regulatory or academic purposes and not for the purpose of making investment decisions regarding any debt securities. Media organisations are prohibited from re-publishing any opinion or recommendation concerning a debt issuer or debt security contained in any Barclays institutional debt research report. Any such recipients that do not want to continue receiving Barclays institutional debt research reports should contact debtresearch@barclays.com. Unless clients have agreed to receive "institutional debt research reports" as required by US FINRA Rule 2242, they will not receive any such reports that may be co-authored by non-debt research analysts. Eligible clients may get access to such cross asset reports by contacting debtresearch@barclays.com. Barclays will not treat unauthorized recipients of this report as its clients and accepts no liability for use by them of the contents which may not be suitable for their personal use. Prices shown are indicative and Barclays is not offering to buy or sell or soliciting offers to buy or sell any financial instrument.

Without limiting any of the foregoing and to the extent permitted by law, in no event shall Barclays, nor any affiliate, nor any of their respective officers, directors, partners, or employees have any liability for (a) any special, punitive, indirect, or consequential damages; or (b) any lost profits, lost revenue, loss of anticipated savings or loss of opportunity or other financial loss, even if notified of the possibility of such damages, arising from any use of this publication or its contents.

Other than disclosures relating to Barclays, the information contained in this publication has been obtained from sources that Barclays Research believes to be reliable, but Barclays does not represent or warrant that it is accurate or complete. Appearances by Third-Party Speakers: Any views or opinions expressed by third-party speakers during this event are solely those of the speaker and do not represent the views or opinions of Barclays. Barclays is not responsible for, and makes no warranties whatsoever as to, the information or opinions contained in any written, electronic, audio or video presentations by any third-party speakers at the event ("Third-Party Content"). Any such Third-Party Content has not been adopted or endorsed by Barclays and does not represent the views or opinions of Barclays. Third-Party Content is provided for information purposes only and has not been independently verified by Barclays for its accuracy or completeness.

The views in this publication are solely and exclusively those of the authoring analyst(s) and are subject to change, and Barclays Research has no obligation to update its opinions or the information in this publication. Unless otherwise disclosed herein, the analysts who authored this report have not received any compensation from the subject companies in the past 12 months. If this publication contains recommendations, they are general recommendations that were prepared independently of any other interests, including those of Barclays and/or its affiliates, and/or the subject companies. This publication does not contain personal investment recommendations or investment advice or take into account the individual financial circumstances or investment objectives of the clients who receive it. Barclays is not a fiduciary to any recipient of this publication. The securities and other investments discussed herein may not be suitable for all investors and may not be available for purchase in all jurisdictions. The United States imposed sanctions on certain Chinese companies (<https://home.treasury.gov/policy-issues/financial-sanctions/sanctions-programs-and-country-information/chinese-military-companies-sanctions>), which may restrict U.S. persons from purchasing securities issued by those companies. Investors must independently evaluate the merits and risks of the investments discussed herein, including any sanctions restrictions that may apply, consult any independent advisors they believe necessary, and exercise independent judgment with regard to any investment decision. The value of and income from any investment may fluctuate from day to day as a result of changes in relevant economic markets (including changes in market liquidity). The information herein is not intended to predict actual results, which may differ substantially from those reflected. Past performance is not necessarily indicative of future results. The information provided does not constitute a financial benchmark and should not be used as a submission or contribution of input data for the purposes of determining a financial benchmark.

This publication is not investment company sales literature as defined by Section 270.24(b) of the US Investment Company Act of 1940, nor is it intended to constitute an offer, promotion or recommendation of, and should not be viewed as marketing (including, without limitation, for the purposes of the UK Alternative Investment Fund Managers Regulations 2013 (SI 2013/1773) or AIFMD (Directive 2011/61)) or pre-marketing (including, without limitation, for the purposes of Directive (EU) 2019/1160) of the securities, products or issuers that are the subject of this report.

Third Party Distribution: Any views expressed in this communication are solely those of Barclays and have not been adopted or endorsed by any third party distributor.

United Kingdom: This document is being distributed (1) only by or with the approval of an authorised person (Barclays Bank PLC) or (2) to, and is directed at (a) persons in the United Kingdom having professional experience in matters relating to investments and who fall within the definition of "investment professionals" in Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (the "Order"); or (b) high net worth companies, unincorporated associations and partnerships and trustees of high value trusts as described in Article 49(2) of the Order; or (c) other persons to whom it may otherwise lawfully be communicated (all such persons being "Relevant Persons"). Any investment or investment activity to which this communication relates is only available to and will only be engaged in with Relevant Persons. Any other persons who receive this communication should not rely on or act upon it. Barclays Bank PLC is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority and is a member of the London Stock Exchange.

European Economic Area (“EEA”): This material is being distributed to any “Authorised User” located in a Restricted EEA Country by Barclays Bank Ireland PLC. The Restricted EEA Countries are Austria, Bulgaria, Estonia, Finland, Hungary, Iceland, Liechtenstein, Lithuania, Luxembourg, Malta, Portugal, Romania, Slovakia and Slovenia. For any other “Authorised User” located in a country of the European Economic Area, this material is being distributed by Barclays Bank PLC. Barclays Bank Ireland PLC is a bank authorised by the Central Bank of Ireland whose registered office is at 1 Molesworth Street, Dublin 2, Ireland. Barclays Bank PLC is not registered in France with the Autorité des marchés financiers or the Autorité de contrôle prudentiel. Authorised User means each individual associated with the Client who is notified by the Client to Barclays and authorised to use the Research Services. The Restricted EEA Countries will be amended if required.

Finland: Notwithstanding Finland’s status as a Restricted EEA Country, Research Services may also be provided by Barclays Bank PLC where permitted by the terms of its cross-border license.

Americas: The Investment Bank of Barclays Bank PLC undertakes U.S. securities business in the name of its wholly owned subsidiary Barclays Capital Inc., a FINRA and SIPC member. Barclays Capital Inc., a U.S. registered broker/dealer, is distributing this material in the United States and, in connection therewith accepts responsibility for its contents. Any U.S. person wishing to effect a transaction in any security discussed herein should do so only by contacting a representative of Barclays Capital Inc. in the U.S. at 745 Seventh Avenue, New York, New York 10019.

Non-U.S. persons should contact and execute transactions through a Barclays Bank PLC branch or affiliate in their home jurisdiction unless local regulations permit otherwise.

This material is distributed in Canada by Barclays Capital Canada Inc., a registered investment dealer, a Dealer Member of Canadian Investment Regulatory Organization (www.ciro.ca), and a Member of the Canadian Investor Protection Fund (CIPF).

This material is distributed in Mexico by Barclays Bank Mexico, S.A. and/or Barclays Capital Casa de Bolsa, S.A. de C.V. This material is distributed in the Cayman Islands and in the Bahamas by Barclays Capital Inc., which it is not licensed or registered to conduct and does not conduct business in, from or within those jurisdictions and has not filed this material with any regulatory body in those jurisdictions.

Japan: This material is being distributed to institutional investors in Japan by Barclays Securities Japan Limited. Barclays Securities Japan Limited is a joint-stock company incorporated in Japan with registered office of 6-10-1 Roppongi, Minato-ku, Tokyo 106-6131, Japan. It is a subsidiary of Barclays Bank PLC and a registered financial instruments firm regulated by the Financial Services Agency of Japan. Registered Number: Kanto Zaimukyokucho (kinsho) No. 143.

Asia Pacific (excluding Japan): Barclays Bank PLC, Hong Kong Branch is distributing this material in Hong Kong as an authorised institution regulated by the Hong Kong Monetary Authority. Registered Office: 41/F, Cheung Kong Center, 2 Queen's Road Central, Hong Kong.

All Indian securities-related research and other equity research produced by Barclays’ Investment Bank are distributed in India by Barclays Securities (India) Private Limited (BSIPL). BSIPL is a company incorporated under the Companies Act, 1956 having CIN U67120MH2006PTC161063. BSIPL is registered and regulated by the Securities and Exchange Board of India (SEBI) as a Research Analyst: INH000001519; Portfolio Manager INP000002585; Stock Broker INZ000269539 (member of NSE and BSE); Depository Participant with the National Securities & Depositories Limited (NSDL): DP ID: IN-DP-NSDL-478-2020; Investment Adviser: INA000000391. BSIPL is also registered as a Mutual Fund Advisor having AMFI ARN No. 53308. The registered office of BSIPL is at Nirlon Knowledge Park, 9th floor, Block B-6, Off. Western Express Highway, Goregaon (East), Mumbai – 400063, India. Telephone No: +91 22 61754000 Fax number: +91 22 61754099. Any other reports produced by Barclays’ Investment Bank are distributed in India by Barclays Bank PLC, India Branch, an associate of BSIPL in India that is registered with Reserve Bank of India (RBI) as a Banking Company under the provisions of The Banking Regulation Act, 1949 (Regn No BOM43) and registered with SEBI as Merchant Banker (Regn No INM000002129) and also as Banker to the Issue (Regn No INBI00000950). Barclays Investments and Loans (India) Limited, registered with RBI as Non Banking Financial Company (Regn No RBI CoR-07-00258), and Barclays Wealth Trustees (India) Private Limited, registered with Registrar of Companies (CIN U93000MH2008PTC188438), are associates of BSIPL in India that are not authorised to distribute any reports produced by Barclays’ Investment Bank.

This material is distributed in Singapore by the Singapore Branch of Barclays Bank PLC, a bank licensed in Singapore by the Monetary Authority of Singapore. For matters in connection with this material, recipients in Singapore may contact the Singapore branch of Barclays Bank PLC, whose registered address is 10 Marina Boulevard, #23-01 Marina Bay Financial Centre Tower 2, Singapore 018983.

This material, where distributed to persons in Australia, is produced or provided by Barclays Bank PLC.

This communication is directed at persons who are a “Wholesale Client” as defined by the Australian Corporations Act 2001.

Please note that the Australian Securities and Investments Commission (ASIC) has provided certain exemptions to Barclays Bank PLC (BBPLC) under paragraph 911A(2)(l) of the Corporations Act 2001 from the requirement to hold an Australian financial services licence (AFSL) in respect of financial services provided to Australian Wholesale Clients, on the basis that BBPLC is authorised by the Prudential Regulation Authority of the United Kingdom (PRA) and regulated by the Financial Conduct Authority (FCA) of the United Kingdom and the PRA under United Kingdom laws. The United Kingdom has laws which differ from Australian laws. To the extent that this communication involves the provision of financial services by BBPLC to Australian Wholesale Clients, BBPLC relies on the relevant exemption from the requirement to hold an AFSL. Accordingly, BBPLC does not hold an AFSL.

This communication may be distributed to you by either: (i) Barclays Bank PLC directly, (ii) Barrenjoey Markets Pty Limited (ACN 636 976 059, “Barrenjoey”), the holder of Australian Financial Services Licence (AFSL) 521800, a non-affiliated third party distributor, where clearly identified to you by Barrenjoey. Barrenjoey is not an agent of Barclays Bank PLC or (iii) such other non-affiliated third-party distributor(s) as may be clearly identified to you. Such non-affiliated third-party distributor(s) are not agents of Barclays Bank PLC.

This material, where distributed in New Zealand, is produced or provided by Barclays Bank PLC. Barclays Bank PLC is not registered, filed with or approved by any New Zealand regulatory authority. This material is not provided under or in accordance with the Financial Markets Conduct Act of 2013 (“FMCA”), and is not a disclosure document or “financial advice” under the FMCA. This material is distributed to you by either: (i) Barclays Bank PLC directly or (ii) Barrenjoey Markets Pty Limited (“Barrenjoey”), a non-affiliated third party distributor, where clearly identified to you by Barrenjoey. Barrenjoey is not an agent of Barclays Bank PLC. This material may only be distributed to “wholesale investors” that meet the “investment business”, “investment activity”, “large”, or “government agency” criteria specified in Schedule 1 of the FMCA.

Middle East: Nothing herein should be considered investment advice as defined in the Israeli Regulation of Investment Advisory, Investment Marketing and Portfolio Management Law, 1995 (“Advisory Law”). This document is being made to eligible clients (as defined under the Advisory Law) only. Barclays Israeli branch previously held an investment marketing license with the Israel Securities Authority but it cancelled such license on 30/11/2014

as it solely provides its services to eligible clients pursuant to available exemptions under the Advisory Law, therefore a license with the Israel Securities Authority is not required. Accordingly, Barclays does not maintain an insurance coverage pursuant to the Advisory Law.

This material is distributed in the United Arab Emirates (including the Dubai International Financial Centre) and Qatar by Barclays Bank PLC. Barclays Bank PLC in the Dubai International Financial Centre (Registered No. 0060) is regulated by the Dubai Financial Services Authority (DFSA). Principal place of business in the Dubai International Financial Centre: The Gate Village, Building 4, Level 4, PO Box 506504, Dubai, United Arab Emirates. Barclays Bank PLC-DIFC Branch, may only undertake the financial services activities that fall within the scope of its existing DFSA licence. Related financial products or services are only available to Professional Clients, as defined by the Dubai Financial Services Authority. Barclays Bank PLC in the UAE is regulated by the Central Bank of the UAE and is licensed to conduct business activities as a branch of a commercial bank incorporated outside the UAE in Dubai (Licence No.: 13/1844/2008, Registered Office: Building No. 6, Burj Dubai Business Hub, Sheikh Zayed Road, Dubai City) and Abu Dhabi (Licence No.: 13/952/2008, Registered Office: Al Jazira Towers, Hamdan Street, PO Box 2734, Abu Dhabi). This material does not constitute or form part of any offer to issue or sell, or any solicitation of any offer to subscribe for or purchase, any securities or investment products in the UAE (including the Dubai International Financial Centre) and accordingly should not be construed as such. Furthermore, this information is being made available on the basis that the recipient acknowledges and understands that the entities and securities to which it may relate have not been approved, licensed by or registered with the UAE Central Bank, the Dubai Financial Services Authority or any other relevant licensing authority or governmental agency in the UAE. The content of this report has not been approved by or filed with the UAE Central Bank or Dubai Financial Services Authority. Barclays Bank PLC in the Qatar Financial Centre (Registered No. 00018) is authorised by the Qatar Financial Centre Regulatory Authority (QFCRA). Barclays Bank PLC-QFC Branch may only undertake the regulated activities that fall within the scope of its existing QFCRA licence. Principal place of business in Qatar: Qatar Financial Centre, Office 1002, 10th Floor, QFC Tower, Diplomatic Area, West Bay, PO Box 15891, Doha, Qatar. Related financial products or services are only available to Business Customers as defined by the Qatar Financial Centre Regulatory Authority.

Russia: This material is not intended for investors who are not Qualified Investors according to the laws of the Russian Federation as it might contain information about or description of the features of financial instruments not admitted for public offering and/or circulation in the Russian Federation and thus not eligible for non-Qualified Investors. If you are not a Qualified Investor according to the laws of the Russian Federation, please dispose of any copy of this material in your possession.

Sustainable Investing Related Research: There is currently no globally accepted framework or definition (legal, regulatory or otherwise) of, nor market consensus as to what constitutes a 'sustainable', 'ESG', 'green', 'climate-friendly' or an equivalent company, investment, strategy or consideration or what precise attributes are required to be eligible to be categorised by such terms. This means there are different ways to evaluate a company or an investment and so different values may be placed on certain sustainability credentials as well as adverse ESG-related impacts of companies and ESG controversies. The evolving nature of sustainable investing considerations, models and methodologies means it can be challenging to definitively and universally classify a company or investment under a sustainable investing label and there may be areas where such companies and investments could improve or where adverse ESG-related impacts or ESG controversies exist. The evolving nature of sustainable finance related regulations and the development of jurisdiction-specific regulatory criteria also means that there is likely to be a degree of divergence as to the interpretation of such terms in the market. We expect industry guidance, market practice, and regulations in this field to continue to evolve.

Any information, data, image, or other content including from a third party source contained, referred to herein or used for whatsoever purpose by Barclays or a third party ("Information"), in relation to any actual or potential 'ESG', 'sustainable' or equivalent objective, issue, factor or consideration is not intended to be relied upon for ESG or sustainability classification, regulatory regime or industry initiative purposes ("ESG Regimes"), unless otherwise stated. Nothing in these materials, including any images included therein, is intended to convey, suggest or indicate that Barclays considers or represents any product, service, person or body mentioned in these materials as meeting or qualifying for any ESG or sustainability classification, label or similar standards that may exist under ESG Regimes. Barclays has not conducted any assessment of compliance with ESG Regimes. Parties are reminded to make their own assessments for these purposes.

IRS Circular 230 Prepared Materials Disclaimer: Barclays does not provide tax advice and nothing contained herein should be construed to be tax advice. Please be advised that any discussion of U.S. tax matters contained herein (including any attachments) (i) is not intended or written to be used, and cannot be used, by you for the purpose of avoiding U.S. tax-related penalties; and (ii) was written to support the promotion or marketing of the transactions or other matters addressed herein. Accordingly, you should seek advice based on your particular circumstances from an independent tax advisor.

© Copyright Barclays Bank PLC (2026). All rights reserved. No part of this publication may be reproduced or redistributed in any manner without the prior written permission of Barclays. Barclays Bank PLC is registered in England No. 1026167. Registered office 1 Churchill Place, London, E14 5HP. Additional information regarding this publication will be furnished upon request.

BRCF2242

LatAm Credit Notes

A Look At Recovery Estimates for Venezuela

- On October 18, the US Treasury suspended certain sanctions measures on Venezuela's oil and gas sector operations. This effectively removed the ban on secondary trading of certain Venezuela bonds and pre-2017 bonds issued by PdVSA. If the suspension of sanctions turns out to be permanent, this could be an opportunity for Venezuela to restructure its debt. In this piece, we look into Venezuela's fundamentals to assess the type of possible restructuring outcomes under different scenarios.
- To assess the restructuring outcome of a country's defaulted USD bonds, one needs to take into account its current and future fundamentals. In the case of Venezuela, these have changed substantially over the past 10 years: oil production has fallen by ~70%, and GDP per capita by ~80%. This suggests that any recovery value estimates of Venezuela's debt would have changed significantly since the initial default in 2017, and future recovery estimates will be highly dependent on the trajectory of Venezuela's oil production and, by extension, economic recovery.
- In addition to the uncertainty around Venezuela's economic outlook, the overall level of debt outstanding remains unclear given limited data availability. From the bonds and multilateral debt outstanding, we can infer that Venezuela and its government-related entities have ~\$99bn outstanding (including accrued interest). In terms of bilateral debt, suppliers, ongoing litigation and 'debt for oil' contracts, various sources suggest that the debt outstanding could amount to an additional ~\$48bn, although this may be as high as ~\$78bn. This would place the total external debt of Venezuela and its government-related entities (primarily PdVSA) in the range of ~\$148bn-\$177bn.
- To assess the potential restructuring outcome based on the available data we use a cash flow analysis and our own top-down recovery value model. Using these frameworks, our cash flow analysis points to a recovery estimate of ~19-25 cents, assuming a 15% exit yield. Our top-down NPV haircut model, which does not include an exit yield assumption, puts the recovery rates at the lower end of that range, at ~13-20 cents on the USD (depending on the level of debt), but with more upside if oil production recovers.

Nathan Fabius

+1(212)855-0805 |
nathan.fabius@gs.com
Goldman Sachs & Co. LLC

Sara Grut

+44(20)7774-8622 | sara.grut@gs.com
Goldman Sachs International

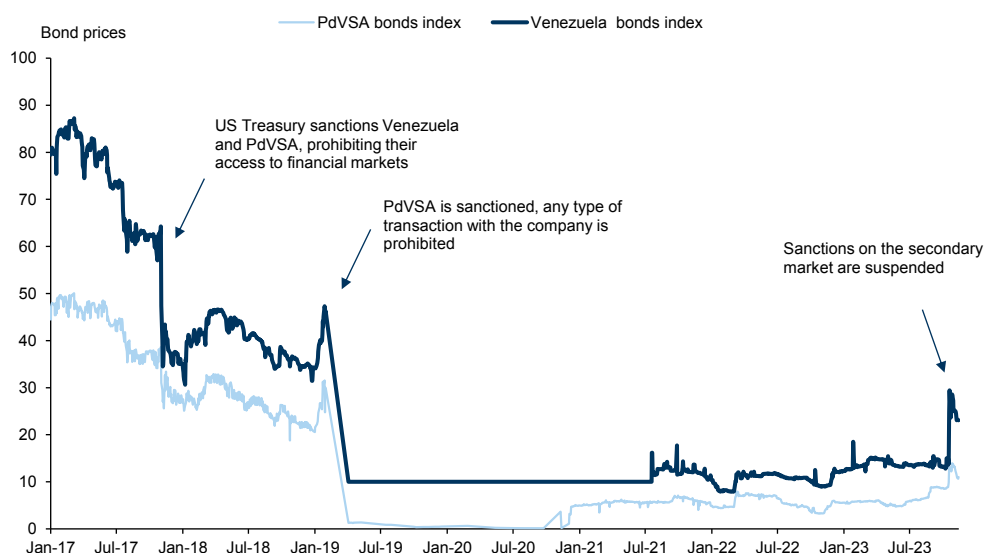
Temporary Sanction Relief

On October 18, the US Treasury lifted sanctions on the trading of Venezuela's sovereign bonds and securities issued by Petróleos de Venezuela, S.A. (PdVSA) before August 25, 2017, after Venezuela's opposition parties under the *Unitary Platform* umbrella group and the Maduro administration signed an electoral roadmap and guarantee agreement. As part of the roadmap and electoral guarantees agreed with the opposition, the Maduro administration committed to hold general elections in 2H2024. The US Department of State made public the expectation and understanding that Venezuela would take the following steps before the end of November: (1) define a specific timeline and process for the expedited reinstatement of all presidential candidates and political parties, and (2) begin the release of all wrongfully detained US nationals and Venezuelan political prisoners. Failure to abide by the terms of this arrangement would lead the United States to reverse the removal of sanctions.

The sanctions relief covers authorization of most transactions involving Venezuela's oil & gas and gold sectors, as well as removing the ban on secondary trading of certain securities. The ban on trading Venezuelan sovereign bonds and certain PdVSA securities in the primary markets remains in place. Venezuela and PdVSA's USD bonds performed well on the news of the sanction relief (Exhibit 1). Venezuela and PdVSA have a combined \$64bn in USD bonds outstanding, most of which have been in default since 2017, with the sanctions on secondary debt weighing on the prospect of a restructuring.

If the suspension of sanctions turns out to be permanent, opening the way for a sustainable economic recovery, it is possible that Venezuela would be able to restructure its debt. In this piece, we look into Venezuela's fundamentals, in particular its debt dynamics, to assess the type of restructuring outcomes that are possible under different scenarios.

We emphasize that the uncertainty around our estimates is high, given the uncertainty around Venezuela's underlying macroeconomic data, but also around the suspension of sanctions: the US Treasury stated that it is *"prepared to amend or revoke authorizations at any time, should representatives of Maduro fail to follow through on their commitments."* A six-month general license temporarily authorizing transactions involving the oil & gas sector will be renewed only if Venezuela *"meets its commitments under the electoral roadmap as well as other commitments with respect to those who are wrongfully detained"*.

Exhibit 1: PdVSA's and Venezuelan sovereign bonds have rallied since the US Treasury's announcement

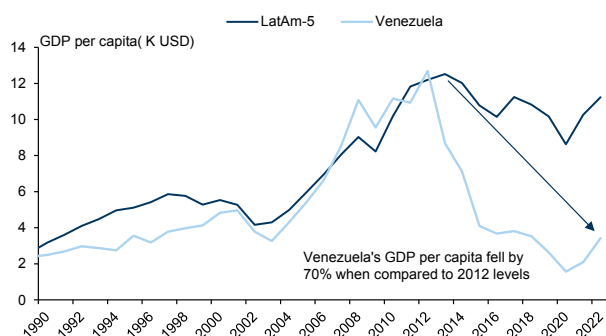
*Excludes the PdVSA 2020 secured bond

Source: Bloomberg, Goldman Sachs Global Investment Research

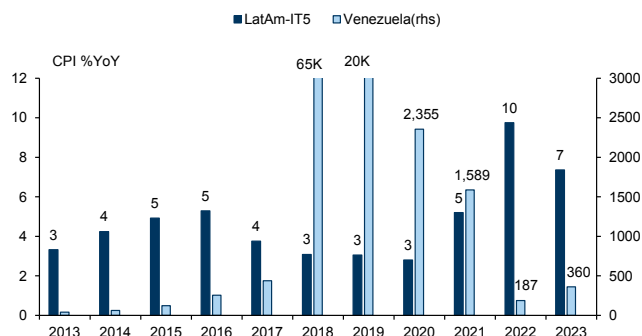
Venezuela's Macro Environment

Over the past 10 years, Venezuela has experienced a significant drop in GDP and hyperinflation

To analyze the restructuring outcome of a country's defaulted USD bonds, one needs to take into account its current and future fundamentals. In the case of Venezuela, these have changed substantially over the past 10 years: GDP per capita has fallen to less than 20% of 2012 levels ([Exhibit 2](#)), alongside severe hyperinflation ([Exhibit 3](#)) that has eroded the purchasing power of Venezuela's currency, the Bolivar. The erosion of Venezuela's economy came on the back of a mix of heterodox policies, and was exacerbated by the imposition of sanctions on the sovereign and on its main state-owned oil company.

Exhibit 2: Venezuela's GDP per capita has collapsed in absolute and relative terms

Source: IMF, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 3: Beyond hyperinflation but far from price stability

Latam Inflation Targeting 5: Brazil, Chile, Colombia, Mexico and Peru

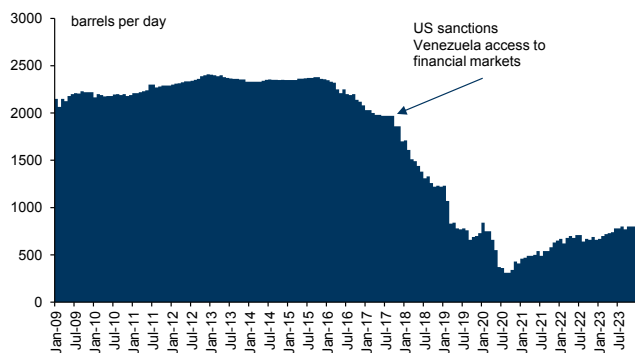
Source: IMF, Bloomberg

Venezuela's oil production has declined as a result of a default and sanctions ...

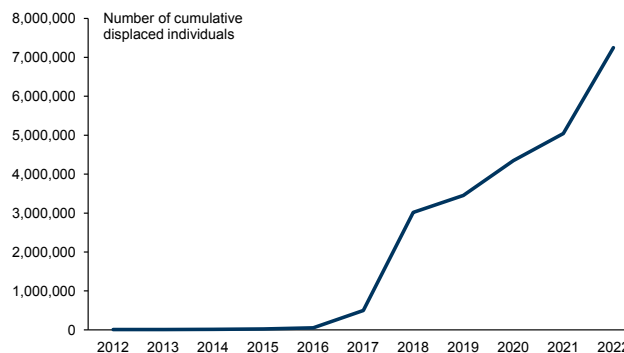
The fall in oil production is a key reason for the decline in Venezuela's GDP. According to the latest official data from the early 2010s, the oil sector accounted for 25-30% of total GDP and 96% of all exports.

As can be seen in [Exhibit 4](#), oil production has fallen from ~2.5 million barrels per day in 2016, to around ~0.8 million barrels per day currently. This decline initially followed the default of Venezuela's USD bonds, which limited Venezuela's access to capital markets and hence the capital expenditure needed for oil production. Current estimates place the investment needed to allow oil production to return to 2.5 million barrels per day in eight years at between \$78bn and \$120bn ([Hernandez and La Rosa Reyes, 2021](#)), which is significant compared with Venezuela's GDP, which the IMF estimates is ~\$92bn as of 2023.

The collapse of Venezuela's economy also led to large-scale emigration, with the IMF estimating that 7 million individuals have left the country since 2016 ([Exhibit 4](#)), which amounts to ~23% of the country's total population. This is a significant drop in the size and skills of its labor force, which is likely to weigh on Venezuela's economic recovery going forward.

Exhibit 4: Sanctions and the collapse of oil production in Venezuela

Source: Bloomberg, OPEC, Goldman Sachs Global Investment Research

Exhibit 5: Over 7 million Venezuelans have emigrated

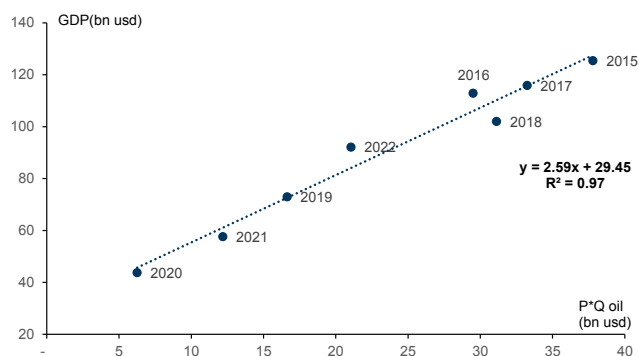
Source: UNHCR

GDP developments are highly dependent on oil production

As can be inferred from the above, economic recovery for Venezuela is highly dependent on oil production and capital expenditure in the oil & gas sector, among other things. This in turn creates a key challenge in assessing Venezuela's restructuring outcomes, as its ability to pay off its debt would increase with higher levels of oil production, which requires a substantial amount of capital in the form of FDI. Some estimates consider that the country could recover its 2016 oil production levels in 5 years.

We take a look at how a recovery in GDP might look if oil production recovers, by regressing the IMF's GDP estimates for Venezuela on oil production, using OPEC's official monthly series and the barrel price for Mars blend, which we find to have a close historical relation in price levels to Venezuela's oil barrel, the Merey blend. Here, it is worth noting that, from 2018 onwards, the IMF estimates Venezuela's GDP level, as the underlying data were unavailable for collection.

Generally, we find a strong correlation between Venezuela's GDP level and the total value of its oil production, with a multiplier of around ~2.6 from the oil sector to the overall economy. As a result of this exercise, we can estimate GDP base levels on oil price levels and expected production: Exhibit 7 shows that if oil production were to increase from ~0.8mbd to ~2.2mbd, with Brent at ~\$80/bbl, which is consistent with our commodity team's forecasts, the size of Venezuela's USD GDP would almost double.

Exhibit 6: IMF's GDP estimates: combining oil production and prices

Source: IMF, Bloomberg

Exhibit 7: Venezuela's GDP based on oil production and prices

GDP Level estimates (bn USD)						
Oil production '000 / Price USD	50	60	70	80	90	100
800	64	71	78	85	92	98
1000	73	81	90	98	107	115
1200	81	92	102	112	122	132
1400	90	102	114	126	138	150
1600	98	112	126	139	153	167
1800	107	122	138	153	168	184
2000	115	132	150	167	184	201
2200	124	143	161	180	199	218

Source: Goldman Sachs Global Investment Research

Debt Stock: A Complex Starting Point

Beyond GDP, the key variable needed to assess a country's debt sustainability is the size of its debt. For Venezuela, the total (government and government-related entities) foreign debt is difficult to track due to the lack of publicly available and reliable information. Below, we describe what we currently know about Venezuela's foreign outstanding debt (bonds, multilateral), and where we have to make assumptions (bilateral debt, suppliers, ongoing litigation cases, debt for oil).

What we know ...

All in all, we know that Venezuela owes ~\$99bn to bondholders and multilateral institutions, between bonds, loans and interest in arrears.

1) ~\$64bn outstanding in international bonds and ~\$31bn in interest arrears

Starting with the bonds, Venezuela and its state-owned entities have a total of 26 defaulted bonds. Of these, 16 are sovereign bonds, 9 are PdVSA bonds, and 1 was issued by public utility company Elecar. These bonds amount to ~\$64bn in defaulted notional, and as most of them are in default since late 2017, they have incurred close to ~\$31bn interest in arrears.

Collective agreement clauses (CACs) are an important element of USD bond restructurings, whereby the same terms of debt restructurings apply to all bondholders. This limits the risk of creditor holdouts, which can slow the restructuring process, as was the case in Argentina's 2001 debt restructuring. Looking across the prospectus of the outstanding USD bonds in Venezuela, we find that most of the sovereign bonds have CACs (Exhibit 9), unlike the bonds of Elecar and PdVSA, which would need to be restructured in accordance with each bond indenture and the specific bankruptcy law in each of their jurisdictions.

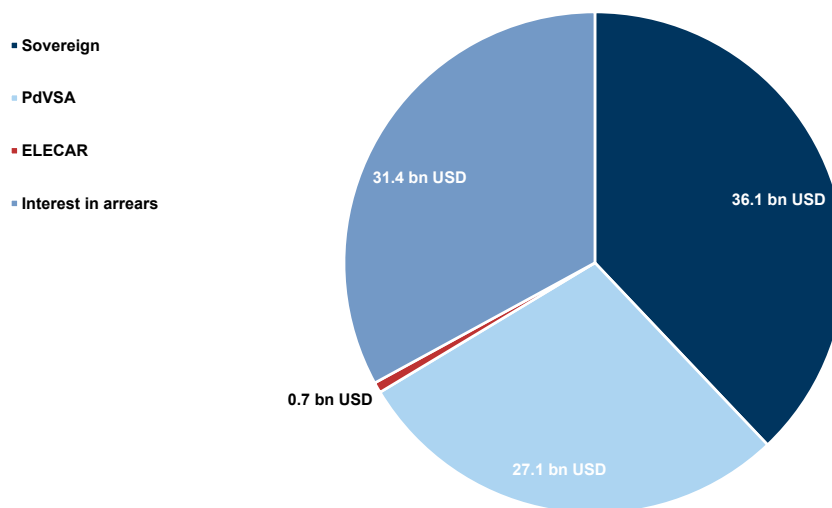
Some bonds have special characteristics and could face legal challenges when the restructuring takes place: for example, the PdVSA 2020 8.5% is a secured bond, which

has as collateral part of PdVSA's shares in Citgo Holding which partially own Citgo Petroleum Corporation, and as a result trades at much higher Dollar prices than the non-secured bonds; however, it was issued without approval of the national assembly at the time. The Venezuela 2036 5% bond is another example of bonds issued without approval; it is not registered in Euroclear and therefore would have different treatment in negotiations. The PdVSA 6% 2022 bond could also face court challenges, as data suggest that it may not have been issued at par in an arm's length transaction. Finally, we have excluded the legacy Brady oil warrants, which have \$600mn outstanding, as these are not technically bonds.

2) ~\$4.3bn in Multilateral Debt

In terms of multilateral debt, according to the latest available data, Venezuela owes the Inter-American Development Bank a total of \$2bn in loans and interest in arrears, and owes \$2.3bn to the CAF Development Bank of Latin America and the Caribbean.

Exhibit 8: Venezuela's defaulted debt (sovereign and quasi) is \$64bn in notional plus \$31bn in interest in arrears



We do not include in the total the outstanding \$60mn in Brady oil warrants

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 9: A diverse set of bonds

Issuer Name	Coupon	Maturity	Currency	Mid Price	Default Date	Amt Out(bn)	CAC
ELECAR	8.5	10-Apr-18	USD	8	10-Nov-17	0.65	NA
PDVSA	5.375	12-Apr-27	USD	12	12-Nov-17	3.00	NA
	6	28-Oct-22	USD	8	28-May-18	3.00	NA
	5.5	12-Apr-37	USD	11	12-Nov-17	1.50	NA
	8.5	27-Oct-20	USD	78	26-Nov-19	1.68	NA
	6	16-May-24	USD	12	16-Dec-17	5.00	NA
	12.75	17-Feb-22	USD	16	17-Mar-18	3.00	NA
	9.75	17-May-35	USD	14	17-Dec-17	3.00	NA
	6	15-Nov-26	USD	12	15-Dec-17	4.50	NA
	9	17-Nov-21	USD	13	17-Dec-17	2.39	NA
Venezuela Government	11.95	5-Aug-31	USD	19	5-Mar-18	4.20	75%
	12.75	23-Aug-22	USD	16	23-Mar-18	3.00	75%
	9.25	7-May-28	USD	17	7-Dec-17	2.00	75%
	7.75	13-Oct-19	USD	13	13-Nov-17	2.50	75%
	11.75	21-Oct-26	USD	19	21-Nov-17	3.00	75%
	8.25	13-Oct-24	USD	16	13-Nov-17	2.50	75%
	9	7-May-23	USD	16	7-Dec-17	2.00	75%
	7	31-Mar-38	USD	15	30-Apr-18	1.25	75%
	6	9-Dec-20	USD	13	9-Jan-18	1.50	75%
	7.65	21-Apr-25	USD	16	21-Nov-17	1.60	75%
	6.5	29-Dec-36	USD	NA	9-Jan-18	5.00	75%
	7	1-Dec-18	USD	13	1-Jan-18	1.00	85%
	9.375	13-Jan-34	USD	17	13-Feb-18	1.50	75%/85%*
	9.25	15-Sep-27	USD	19	15-Apr-18	4.00	NA
	13.625	15-Aug-18	USD	14	15-Mar-18	0.75	NA
	13.625	15-Aug-18	USD	14	15-Mar-18	0.30	NA

Source: Bloomberg, Goldman Sachs Global Investment Research

... and what we know we don't know

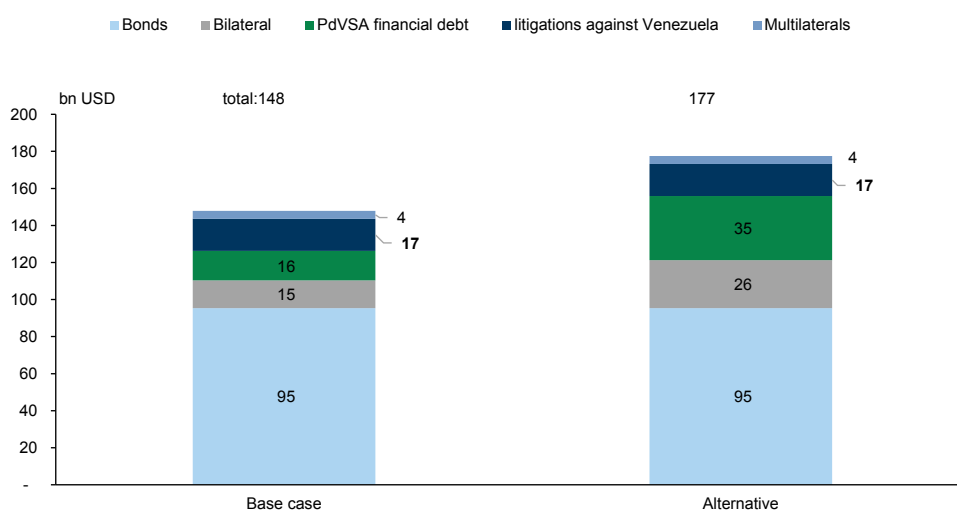
In addition to the above-listed debt, there are several liabilities that Venezuela and PdVSA are known to hold, but where the precise amount is unknown. Below we provide estimates of how large the debt outstanding might be, according to the latest available data. We estimate this debt (excluding bonds and multilateral debt) to be ~\$48bn, with alternative estimates from Moatti and Muci putting it as high at ~\$78bn.

- **Venezuela – Arbitrations and litigations (~\$17bn).** During the 2000s, Venezuela's government seized a number of local assets owned by international companies. As a result, multiple arbitration and litigation cases were brought against the country at the World Bank ICSID arbitration court. Assessing these on an individual case by case basis falls beyond the scope of this analysis, but many of them ruled in favor of the companies and adjudicated a monetary award. We have identified 9 cases that have been ruled on and that are not related to defaulted bonds, which amount to ~\$12.75bn in claims that would form part of any sovereign negotiation, as well as ~\$4.5bn in accrued interest to be paid. Furthermore, in one of the recent rulings a US-based judge ruled that PdVSA and the sovereign were one and the same, meaning that Venezuela's debts are also owed by PdVSA, which could create further complexities in negotiations by creditors of Venezuela and PdVSA.
- **Venezuela – Bilateral debt (~\$16bn).** Venezuela has negotiated bilateral loans with other governments, including China, Russia and Brazil. There have been several bilateral agreements and loans between Russia and Venezuela over the years but, according to the latest data, the outstanding loans amount to \$3bn, although could

reach \$6bn. The loans between Venezuela and China are reported to amount to \$10bn, which is consistent with the recorded oil flows between the two countries, although some estimates place this number closer to \$20bn. With Brazil, the current debt in arrears appears to be closer to \$2bn, according to the current available information.

- **PdVSA – Arbitration and litigations (~\$5bn).** PdVSA issued a series of promissory notes after the entity lost market access in 2015-16. The company has subsequently defaulted on these notes, which have found their way to litigation cases in the US, such as the Red Tree Investments case, which was awarded earlier this year. Estimating the total amount of outstanding promissory notes is a difficult task as these are not listed and therefore there is no public data on them. Published estimates range between \$5bn and \$20bn (see Buchheit & Gulati, 2017; Moatti & Muci, 2019; Lerrick, 2018; PdVSA, 2016). However, due to the limited amount of litigation and claims on this front, we tend to estimate these claims in the lower bound of the range at \$5bn.
- **PdVSA – Debt to oil companies (\$10bn).** PdVSA also owes money to other oil companies, among them ENI, Repsol, the successor to Rosneft – Roszarubezhneft, and Chevron, with which the company has debts that amount to at least \$8.6bn according to the latest available data. Most of these debts are likely to be paid with oil production in agreements known as ‘debt for oil’ contracts. As there could be other agreements that we are not aware of, we estimate this type of debt at \$10bn in our base case.

Exhibit 10: Limited information, but high debt levels



Source: Goldman Sachs Global Investment Research, Bloomberg, Haver, Reuters

Bottom-Up NPV Haircut Estimates — Cash Flow

To analyze the potential restructuring outcome, we build a comprehensive model assuming a potential economic recovery in Venezuela following a debt restructuring. This model is based on our proprietary estimates and relies heavily on Moatti and Muci's 2019 paper, "[An Economic Framework for Venezuela's Debt Restructuring](#)". Our flows analysis assumes that Venezuela will receive IMF and other multilateral organizations' support to get the economy back on track in the first 4 years. Assuming that a full reopening of the economy and an economic recovery is under way, we assume the country can finance itself after 5 years, and will generate a surplus of dollars after 6 years. This surplus could be used to repay its creditors.

To model the balance of payments, we divide the economy between two broad sectors: oil and non-oil sectors. An important assumption that we make here is that the authorities return to more orthodox economic policies, which would allow for an economic recovery, and the release of funding from multilateral organizations. The results of the final assumptions are presented in [Exhibit 11](#).

Our Balance of Payments Assumptions

Non-oil sector

- **Non-oil imports:** For non-oil **imports of goods**, we estimate a multiplier effect of 2.8x to the rest of the non-oil sector, which seems consistent with the 2010 level, when there was substantial government involvement in the economy. We expect this multiplier to increase to 4.6x the level seen in the pre-Chavez period up to 1998. We present the numbers for imports in CIF terms, estimate the **freight and insurance** according to the averages from 1997 to 2014 for all categories. For **service imports** we assume that these return to their historical average of 2.9% of non-oil GDP, consistent with Moatti and Muci's assumptions.
- **Non-oil exports:** For non-oil exports, which include Venezuela's gold production (for which estimates vary greatly, at between 28 tons and 80 tons per year), we opt for the middle of the range, at 50 tons. For total goods exports, we assume a level of 5% of non-oil GDP, which tends to converge to 8%, the historical average in the 2000s. Non-oil service exports start at 0.3% of non-oil GDP and converges over 10 years to the historical average of 1.2%, adjusted by world inflation.
- **Remittances and FDI:** Finally, as we have previously shown, millions of Venezuelans have left the country over the last decade, as they have settled in and joined the workforce in other countries, the volume of **remittances** to Venezuela has steadily increased, reaching 4.2bn USD per year according to estimates. We assume that this level of inflows continues for some years after the reopening, but then steadily decreases as the local economy normalizes and Venezuelan's return to the country. On **FDI for the non-oil sector**, we assume that FDI starts at the 2019 level of 1% and steadily increases to the regions average of 2.6%.

Oil sector

To assess Venezuela's future potential cash flow, we model the oil sector as a single company in order to simplify the analysis, though we acknowledge that in practice this is unlikely, due to the interplay between PdVSA, the government and other joint ventures. We would expect PdVSA to keep its current joint venture model with private investors, due to its lack of access to capital markets, and as a result private participation in the sector would increase.

We assume that oil production will recover back to **3 million barrels** per day, which is slightly below the previous peak of 3.3 million barrels per day in 1998. This recovery in the oil sector is followed by a similar recovery in the non-oil sector, but we expect the share of the energy industry to remain constant at 30% of total GDP, which is consistent with the past economic participation of this sector in Venezuela.

We follow Moatti and Muci's assumptions that the oil industry will increase its productivity from current decline rates of 35% to 21% in 3 years, as capital expenditure increases and operations normalize. Operating expenditure (opex) starts at \$12/bbl and decreases to \$11/bbl in 6 years, while capital expenditure (capex) for existing production starts at \$60/bbl and declines to \$40/bbl over the same period. We assume that the country would initially need to import 90% of the required materials used in capex and opex, declining to 80% over a decade as the industrial sector in Venezuela recovers. The total amount of capex invested in the first 8 years is \$105bn, consistent with other estimates shown above.

We assume that domestic consumption of oil grows in line with GDP from the level of 2022, where data is available for both oil production and exports. This is slightly higher than Moatti and Muci, but we assume that as income increases and the economy normalizes during the first years of the recovery, there would be a higher elasticity from fuel consumption to growth than the 0.6x observed in the prevailing literature. In relation to income tax and oil royalties, we keep the current rates of 50% and 30% constant. We assume that in the first years the oil sector will fully re-invest its after-tax earnings in order to finance the required capex, and finance the rest with FDI. As net income increases and FDI decreases, companies start to send their profit to their headquarters, as can be seen in the income balance.

The IMF and multilateral organizations

To get Venezuela's economy back on track will need the support of the IMF and other multilateral organizations, such as the CAF and IDB, which have historically been present in the country. The IMF's EFF program with Argentina, approved in 2022, reached \$44bn over 30 months, and could be comparable to what Venezuela will need.

We assume that the country would need ~\$50bn in the first 4 years of the reopening to finance its imports while it builds up its international reserves. We acknowledge that this is a significant assumption, especially given Venezuela's relatively small GDP compared with Argentina, which would mean it has less quotas than Argentina. Finally, we assume that the country targets a level of international reserves consistent with covering 12 months of imports and reaching 20% of GDP by 2033.

Exhibit 11: BoP analysis, future flows dependent on oil production recovery

US\$ mm	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Current Account (excl. interest payments)	0	-16130	-11381	-5429	8	-722	-78	2951	8616	15462	23567
Total Exports (G+S)	17628	24378	32512	39527	45945	52107	58129	62369	70165	78819	82876
Goods Exports (FOB)	17174	23642	31639	38552	44867	50925	56847	60990	68695	77267	81237
Oil	13391	19688	27390	33872	39588	44888	49777	52513	58276	64141	64287
Non-Oil	3783	3954	4249	4681	5279	6037	7070	8477	10419	13126	16950
Service Exports	454	736	873	975	1078	1181	1282	1379	1470	1552	1638
Total Imports (G+S)	-22191	-44708	-47673	-48358	-48999	-49703	-51074	-52143	-52901	-53357	-49128
Goods Imports (CIF)	-23144	-44424	-47745	-48814	-49846	-50936	-52633	-54031	-55118	-55889	-52352
Oil	-9934	-13839	-14470	-14499	-14681	-15140	-16449	-17723	-18961	-20163	-16994
Non-Oil	-11440	-30585	-33275	-34315	-35165	-35796	-36184	-36309	-36158	-35726	-35358
Freight & Insurance	-1771	-3680	-3955	-4044	-4129	-4219	-4360	-4476	-4566	-4630	-4337
Service Imports	2724	3396	4028	4500	4976	5453	5918	6363	6783	7162	7561
Income Balance (excl interest)	364	0	0	0	0	-5881	-9613	-9506	-10657	-11808	-11808
Foreign Direct Investment	0	0	0	0	0	-5881	-9613	-9506	-10657	-11808	-11808
Other Investment	364	0	0	0	0	0	0	0	0	0	0
Current Transfers	4200	4200	3780	3402	3062	2756	2480	2232	2009	1808	1627
Remittances	4200	4200	3780	3402	3062	2756	2480	2232	2009	1808	1627
Financing inflows (excl debt repayment)		16,130	11,381	5,429	-8	723	2,084	2,646	3,302	3,966	4,508
Foreign Direct Investment		2,025	5,482	9,228	3,248	3,492	4,084	4,646	5,302	5,966	6,508
Oil Related		622	3575	6833	322	0	0	0	0	0	0
Other		1,403	1,907	2,395	2,925	3,492	4,084	4,646	5,302	5,966	6,508
Portfolio Flows		0	0	0	0	0	0	0	0	0	0
IMF program		20000	15000	10000	5000	0	0	0	0	0	0
Change in Reserves (negative is increase)		-5895	-9101	-13799	-8256	-2769	-2000	-2000	-2000	-2000	-2000
Gross Financing Needs (excl. debt service)	0	0	0	0	0	0	-2,006	-5,597	-11,918	-19,428	-28,076
Memorandum Items											
Nominal GDP (USDbn)	95	119	140	155	170	184	196	206	217	228	232
Merely Oil Prices (USD/bbl)	71	76	77	75	72	70	68	64	64	64	64
Average Oil Production (mbpd)	0.76	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.00
International Reserves (USDbn)	0	6	15	29	37	40	42	44	46	48	50
Months of Imports in Reserves (multiple)	-	2X	4X	7X	9X	10X	10X	10X	10X	11X	12X

Source: Bloomberg, Goldman Sachs Global Investment Research, IMF, Harvard

Implied Recovery Values

In order to provide a range instead of a specific target for recovery values, we model the worst and best case scenario with the available information. That is, assuming Venezuela's debt at either the lower or the higher end of the historical range.

For both scenarios we assume that the IMF will make its lending programs conditional on the result of negotiations with creditors to be consistent with its Debt Sustainability Framework (DSF). The DSF establishes that a low income country such as Venezuela should have an **external debt to GDP ratio that does not exceed 30% of GDP**. For GDP, we assume the modelled 2025 GDP of ~\$140bn.

In both scenarios we assume that the Venezuela 2036 bond, which has not been registered under NY law and was issued without approval from the National Assembly, will be repudiated and the PdVSA 2022 bond, consistent with current market pricing, will be paid at the discounted issuance level. We also assume that multilateral debt will be renegotiated at favorable terms for the country and its terms will be extended, while bilateral debt that is mainly paid in kind will be repaid over time.

Conventional bonds

- **We assume maturity extension, and no coupons in the first five years.** For both

cases, consistent with our BoP scenario we model an exit bond with 15 years to maturity with a face coupon of 8%. The bond capitalizes the interest during the first 5 years as Venezuela will not be generating enough dollars to make any type of external payments. In years 6-8 included, the bond pays interest only, while throughout the years 9-14 it amortizes 10% of the initial face value. This type of new debt is not too different from previous restructurings, as seen in Argentina and Ecuador's 2020 debt restructurings.

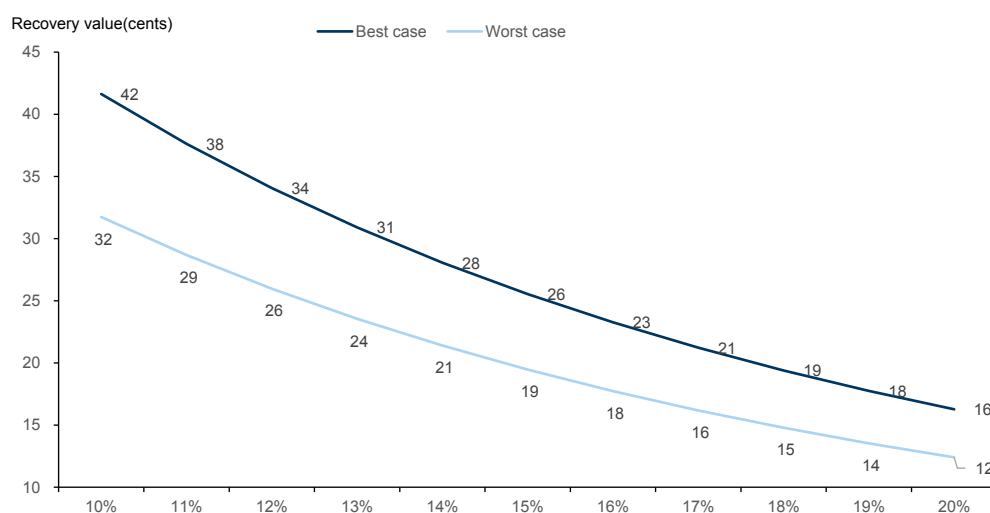
- **Upside scenario points to a recovery value of 25 cents, at a 15% exit yield.** For the **best case scenario** we use the lower end of our total debt estimate of \$148bn explained above in [Exhibit 10](#). We assume that Citgo's auction will reach the higher level of the valuation range estimate at \$20bn. This amount would be sufficient to cover the current litigation and almost all the PdVSA 8.5% 2020 bond that has Citgo shares as collateral.

The rest of the bondholders would receive a haircut of 64% on both their accrued interest and principal; this would amount to \$537 per \$1,000 of original face value. We expect the exit yield of the new issued bond to be close to 15%, which is at the higher end of the range of historical exit yields that we observe in the literature. This would result in a present value of **\$255 per \$1000 of original face value** (excluding oil warrants). However, as shown in [Exhibit 12](#), this estimate is highly dependent on the exit yield assumption.

- **Downside scenario points to a recovery value of 19 cents.** In our worst case scenario, we start with a total debt level of \$177bn, and see the Citgo auction value at the lower end of the range, at \$13bn. As a result of these conditions, the amount of debt coming to the negotiation with creditors is higher. The PdVSA 8.5% 2020 bond is only partially repaid by the auction and around 66% of its notional and accrued interest is still outstanding.

To achieve a total external debt of less than 30% of GDP, bondholders would receive a higher haircut, in this scenario, of 73% on both their accrued interest and their principal, which would amount to \$409 per \$1,000 of the original face value. We expect the exit yield of the new issued bond to be close to 20%, which would result in a present value of **\$194 per \$1000 of original face value**.

As can be seen in [Exhibit 12](#), in both the worst and best case scenarios, exit yields for any proposed bond would have a meaningful impact on recovery values. This would not only be subject to the outcome of negotiations between Venezuela and its creditors but also dependent on market conditions at the time of the future exchange between the defaulted bonds and the newly issued one.

Exhibit 12: Exit yields will matter for recovery values in all cases

Source: Goldman Sachs Global Investment Research

Oil warrants

An additional source of value for investors could be the issuance of oil warrants together with the resulting bond. These warrants would likely be linked to oil production levels and the export price for Venezuelan oil, with their payment capped at a maximum amount that will be disbursed to the holder per period.

In the late 1980s Venezuela issued oil warrants under the Brady Plan; these traded at 3 cents before defaulting in 2018. We assume that, depending on the scenario and conditions, these new issued warrants could add between 5 cents and 10 cents in recovery value, but would face lower liquidity than the bonds. At this moment, we do not model this instrument due to the high level of uncertainty surrounding its structure, but we decided to include it here as a possible source of value.

Top-Down NPV Haircut Estimates — Model Approach

The above analysis is highly contingent on assumptions ranging from cash flow to GDP and exit yield estimates. An alternative approach is to use our top-down haircut model, a cross-sectional model that estimates the historical NPV haircuts of sovereigns since the 1980s, based on external debt (% GDP), public external debt (% total external debt), log(GDP) and an FX regime dummy. In other words, the model is designed to show what a country's NPV haircut might be by looking at previous restructurings and taking into account the country's macroeconomic fundamentals at the time.

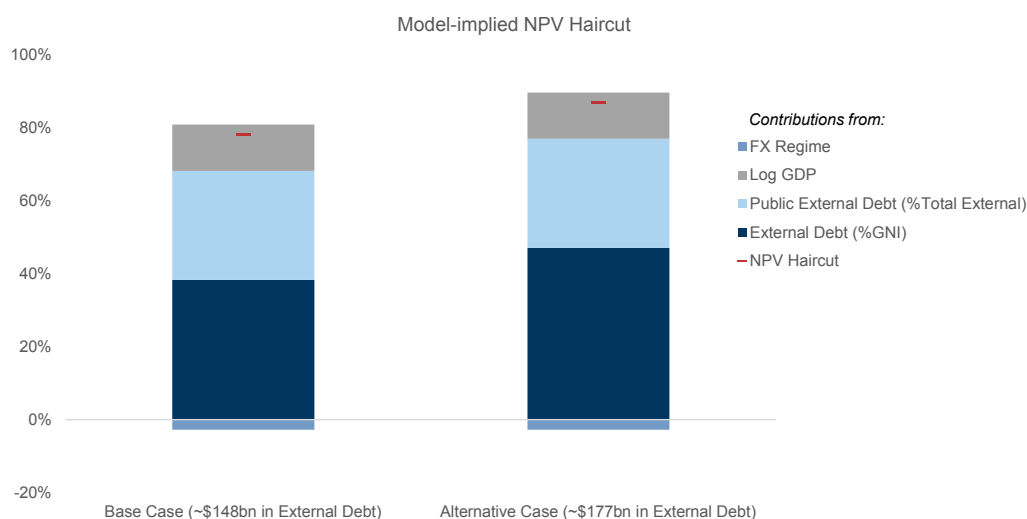
Our top-down NPV haircut model points to a 13-20% recovery value under current fundamentals ...

For Venezuela's macro fundamentals, we use the IMF's 2023 GDP level estimate of

\$92bn and the above estimates of total external debt, which assume Venezuela's total external debt at \$148bn, all of which is public or publicly guaranteed. In our analysis, we also include the alternative case where Venezuela's total external debt is higher, at ~\$177bn. Finally, we assume that the country is de facto dollarized due to its very high levels of inflation (as described above).

Putting those inputs into our model in [Exhibit 13](#) results in an NPV haircut of ~80% of Venezuela's USD bonds (i.e., a recovery value of 20%), which is largely due to the high level of Venezuela's debt relative to GDP. In the alternative case, the model points to a higher NPV haircut of ~87%.

Exhibit 13: Our top-down NPV haircut model points to an NPV haircut of ~80% for Venezuela's USD bonds under current fundamentals



Source: Ilzetzki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), World Bank, IMF, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

... but more upside if oil production increases

However, the uncertainty around this estimate is high, not only because of the sizeable standard errors that come with these types of model estimates, but also as it relates to Venezuela's economic outlook, which can shift quickly depending on its oil production levels. In [Exhibit 14](#), we assume a total external debt of ~\$148bn, and show how the model's NPV haircut estimate would shift depending on Venezuela's oil production and on oil prices. More specifically, we draw from our estimates of how Venezuela's USD GDP has typically correlated to its oil production and oil prices (as described above), and use this as an input for the model.

This model gives some indication of how the recovery value estimates may change, depending on Venezuela's economic recovery. Of course, as mentioned above in the cash flow analysis, even if oil production increases, it will take time for the economy to generate a surplus of dollars, due to the high level of capital expenditures needed initially to get oil production up and running, a factor which is not taken into account in this model.

Exhibit 14: Venezuela's recovery value is highly dependent on oil production and oil prices**\$ GDP Level Estimates**

Oil production '000/Price \$/bbl	50	60	70	80	90	100
800	64	71	78	85	92	98
1000	73	81	90	98	107	115
1200	81	92	102	112	122	132
1400	90	102	114	126	138	150
1600	98	112	126	139	153	167
1800	107	122	138	153	168	184
2000	115	132	150	167	184	201
2200	124	143	161	180	199	218

Total External Debt (% GDP)

Oil production '000/Price \$/bbl	50	60	70	80	90	100
800	226	204	186	171	158	147
1000	199	178	161	147	136	126
1200	178	158	143	129	119	109
1400	161	143	128	115	105	97
1600	147	129	115	104	95	87
1800	136	119	105	95	86	79
2000	126	109	97	87	79	72
2200	117	102	90	80	73	67

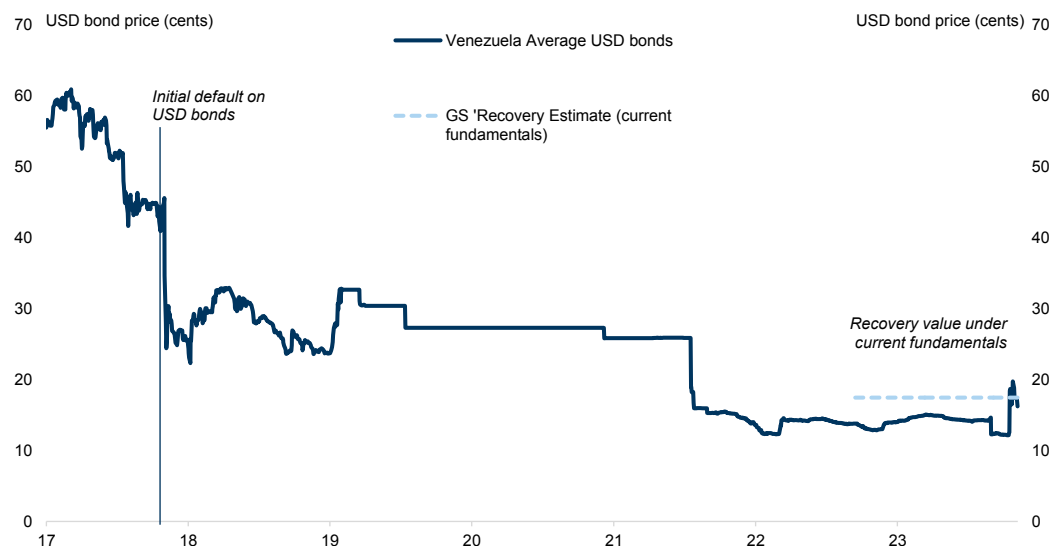
GS Top-Down Recovery Value Estimates

Oil production '000/Price \$/bbl	50	60	70	80	90	100
800	6%	11%	15%	18%	21%	24% <- Current Environment
1000	12%	17%	21%	24%	26%	29%
1200	17%	21%	25%	28%	30%	32%
1400	21%	25%	28%	31%	33%	35%
1600	24%	28%	31%	33%	35%	37%
1800	26%	30%	33%	35%	37%	39%
2000	29%	32%	35%	37%	39%	40%
2200	31%	34%	37%	39%	40%	41%

Source: Ilzetki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), World Bank, IMF, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

The market is priced roughly in line with our estimates of Venezuela's current fundamentals

When comparing our recovery value estimates to the current market pricing of Venezuela's USD bonds, we find that the market is priced roughly in line with our estimates of a 13%-20% recovery, with an average USD bond price of ~16 cents on the USD.

Exhibit 15: The market is currently pricing a recovery value that is slightly below 20%

Source: Bloomberg, Goldman Sachs Global Investment Research

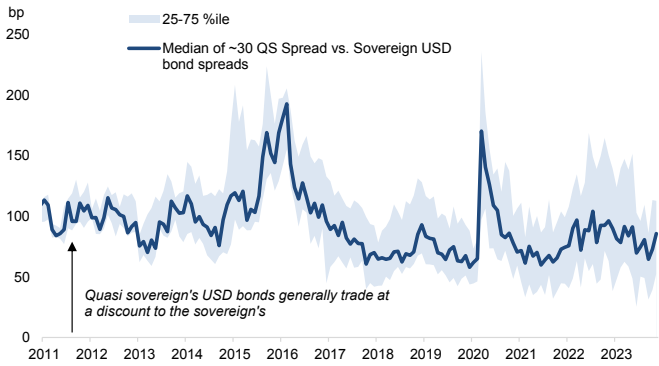
Treatment of Quasi-Sovereigns

A key question for bondholders is whether PdVSA's debt restructuring will be treated differently from Venezuela's. Here, we would note that it is rare for sovereigns and oil-exporting quasi-sovereigns to default at the same time, providing limited historical examples to draw from. For example, Russia's 1998 sovereign default did not coincide with a default of its oil-exporting company Gazprom, nor did Ukraine's 2015 default result in a default of its gas-exporting company Naftogas.

More broadly, we would note that quasi-sovereigns tend to trade at a discount to the sovereign's USD bonds ([Exhibit 16](#)), just as PdVSA's USD bonds have often traded below Venezuela's USD bonds ([Exhibit 17](#)). This is interesting given that these quasi-sovereigns are either 100% or majority government-owned, and suggests the market places a lower hurdle for quasi-sovereign defaults, and possibly a lower recovery rate.

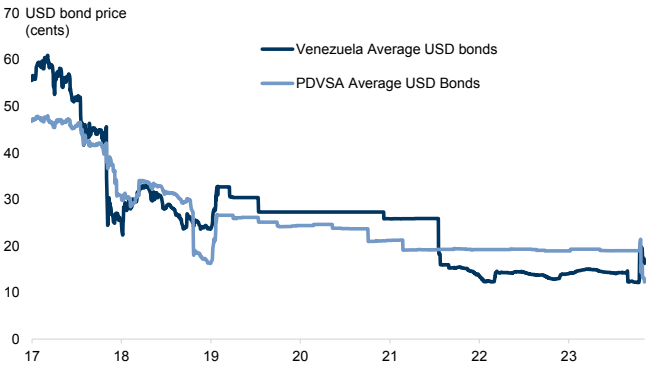
In the case of Venezuela and PdVSA, however, we would note that the very high correlation between Venezuela's GDP and oil output makes it difficult to disentangle the two. Here, the challenge will be how to deal with the above-mentioned CACs, and any potential holdouts to the debt-restructuring negotiations, which would likely put upward pressure on PdVSA's recovery rate.

Exhibit 16: Quasi-sovereign debt tends to trade at a discount to the sovereign



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 17: PdVSA's USD bonds have often traded below Venezuela's USD bonds



Source: Bloomberg, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

I, Nathan Fabius, hereby certify that all of the views expressed in this report accurately reflect my personal views about the subject company or companies and its or their securities. I also certify that no part of my compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

I, Sara Grut, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with subject company, public appearances and trading securities held by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 951-A, Rational House, Appasaheb Marathe Marg, Prabhadevi, Mumbai 400 025, India, Corporate Identity Number U7140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited Investor Grievance E-mail: india-client-support@gs.com. Compliance Officer: Anil Rajput [Tel: + 91 22 6616 9000 | Email: anil.m.rajput@gs.com]. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are

available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, The Investment Trusts Association, Japan, and Japan Investment Advisers Association. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brasil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: GSI, authorised by the PRA and regulated by the FCA and the PRA, disseminates research in the following jurisdictions within the European Economic Area: the Grand Duchy of Luxembourg, Italy, the Kingdom of Belgium, the Kingdom of Denmark, the Kingdom of Norway, the Republic of Finland and the Republic of Ireland; GSI - Succursale de Paris (Paris branch) which is authorised by the French Autorité de contrôle prudentiel et de résolution ("ACPR") and regulated by the Autorité de contrôle prudentiel et de résolution and the Autorité des marchés financiers ("AMF") disseminates research in France; GSI - Sucursal en España (Madrid branch) authorized in Spain by the Comisión Nacional del Mercado de Valores disseminates research in the Kingdom of Spain; GSI - Sweden Bankfilial (Stockholm branch) is authorized by the SFSA as a "third country branch" in accordance with Chapter 4, Section 4 of the Swedish Securities and Market Act (Sw. lag (2007:528) om värdepappersmarknaden) disseminates research in the Kingdom of Sweden; Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research in the Federal Republic of Germany and those jurisdictions within the European Economic Area where GSI is not authorised to disseminate research and additionally, GSBE, Copenhagen Branch filial af GSBE, Tyskland, supervised by the Danish Financial Authority disseminates research in the Kingdom of Denmark; GSBE - Sucursal en España (Madrid branch) subject (to a limited extent) to local supervision by the Bank of Spain disseminates research in the Kingdom of Spain; GSBE - Succursale Italia (Milan branch) to the relevant applicable extent, subject to local supervision by the Bank of Italy (Banca d'Italia) and the Italian Companies and Exchange Commission (Commissione Nazionale per le Società e la Borsa "Consob") disseminates research in Italy; GSBE - Succursale de Paris (Paris branch), supervised by the AMF and by the ACPR disseminates research in France; and GSBE - Sweden Bankfilial (Stockholm branch), to a limited extent, subject to local supervision by the Swedish Financial Supervisory Authority (Finansinspektionen) disseminates research in the Kingdom of Sweden.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of

individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.fiadocumentation.org/fia/regulatory-disclosures_1/fia-uniform-futures-and-options-on-futures-risk-disclosures-booklet-pdf-version-2018. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2023 Goldman Sachs.

No part of this material may be (i) copied, photocopied or duplicated in any form by any means or (ii) redistributed without the prior written consent of The Goldman Sachs Group, Inc.

Global Economics Paper

EM credit after the Covid-19 shock — Lessons from 40 years of sovereign defaults and distress

- The 2020 Covid-19 crisis has already seen five sovereign debt events, more than the three defaults that occurred during the Global Financial Crisis. As the world awaits a potential vaccine in 2021, a key question for EM credit investors is whether the worst is now behind us, or whether the lagged effect from the scarring of national debt trajectories in EM is still to come.
- To address this question, we look across episodes of sovereign defaults and debt restructuring of external commercial borrowing since the 1980s to identify the fundamental macro factors that have typically led sovereigns to default. Given the small number of debt restructurings in any given year (and over time), we extend our analysis to include episodes of sovereign credit distress over the past decade in order to provide a more complete picture of the EM sovereign credit outlook. These are episodes where credit spreads are sustained at extreme wide levels (typically above the 90th percentile of the EM HY universe) for an extended period. In addition, we model the factors that typically matter for haircuts (i.e., losses given default). Taken together, this provides a framework for expected investor losses in the asset class.
- Our results suggest that the probability of default has risen across the EM HY sovereign universe, and is likely to continue to increase in 2021 under the IMF's forecasts that debt levels will remain high after the sharp increase this year. We therefore see risks of more default or restructuring events in 2021. That said, we may already be past the peak of broad-based distressed pricing of EM HY sovereign credit. And with EM HY credit spreads still at the higher end of their historical range, we find the risk premium for the asset class provides attractive compensation for the underlying fundamental risk.
- Although most EM HY sovereigns are still below the probability levels that have previously been associated with defaults, a number of countries screen as having probabilities of distress in excess of the conditional median. This includes recent defaulters (such as Ecuador, Argentina and Lebanon), but also identifies a high probability of prolonged debt distress in Iraq, Sri Lanka, Angola and Gabon. These sovereigns also have a relatively high probability of default, albeit below their historical medians in the three latter. Finally, our results point to a few sovereigns with a high probability of debt distress but still relatively low probability of default; these include Belarus, Mozambique, Mongolia, El Salvador and Pakistan.

Sara Grut

+44(20)7774-8622 | sara.grut@gs.com
Goldman Sachs International

Kamakshya Trivedi

+44(20)7051-4005 | kamakshya.trivedi@gs.com
Goldman Sachs International

Teresa Alves

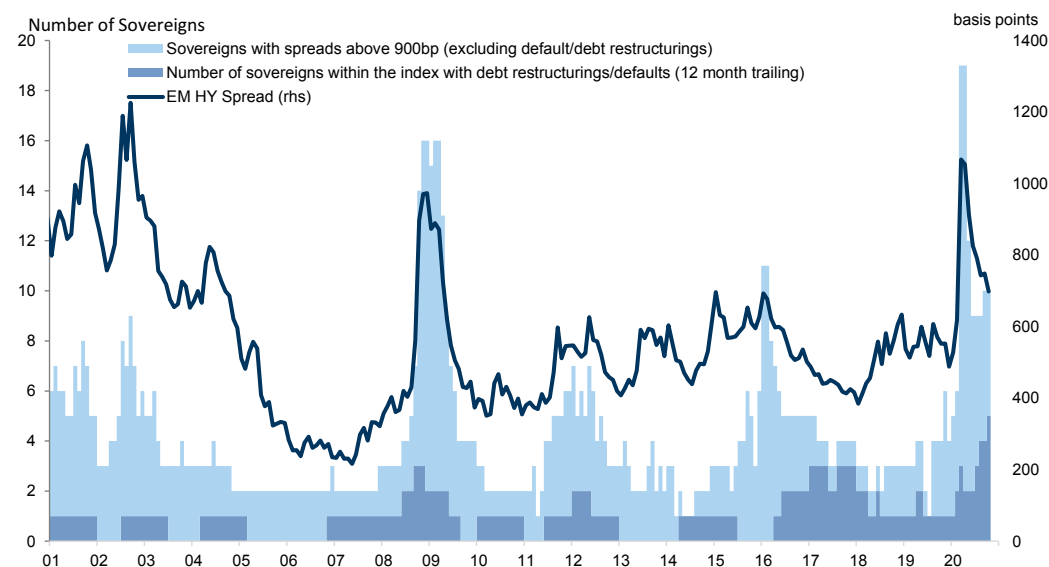
+44(20)7051-7566 | teresa.alves@gs.com
Goldman Sachs International

Is the worst behind us?

The shock to global growth from Covid-19 has hit some of the most vulnerable countries the hardest, leaving credit spreads at distressed levels for a record number of lower-rated (high-yield) sovereigns ([Exhibit 1](#)). While the [official sector](#) was quick to respond to the shock with necessary aid, some of the most fragile sovereigns were pushed into debt restructurings. After [Argentina's default in 2019](#) and [Lebanon's default in 2020Q1](#) (debt restructuring is ongoing), Ecuador followed suit with a default and subsequent [debt restructuring](#), [Suriname](#) restructured part of its debt in July, [Belize](#) restructured part of its debt in August, and, after [announcing intentions to restructure its debt](#), [Zambia](#) missed a coupon payment in October.

So far, 2020 has been associated with five events of sovereign debt distress (defaults/distressed debt exchanges), more than during the Global Financial Crisis (by comparison, there were three sovereign defaults during the Global Financial Crisis). That said, this is a relatively benign outcome considering that some of those sovereign risks were well flagged in 2019 and given the acute nature of the shock. As the world awaits a potential vaccine in 2021, the question now is whether the worst is behind us, or whether the lagged effect from the scarring of Covid-19 is still to come.

Exhibit 1: The Covid-19 shock has left a number of EM HY sovereigns with spreads at distressed levels



Source: Cruces and Trebesch (2014), Catao and Mano (2015), Moodys, Bloomberg, Datastream, Goldman Sachs Global Investment Research

In this *Global Economics Paper*, we address the question with a deep dive into the history and stylised facts of EM sovereign defaults. More specifically, we look across episodes of sovereign defaults and debt restructurings of external commercial borrowing since the 1980s to identify the fundamental macro factors that have typically led sovereigns to default. In line with the literature on this topic, our analysis points to the importance of FX reserve developments, along with the current account and short-term external debt. We also find evidence that it is not necessarily the level of

debt (public or external), but rather the change in debt that matters for defaults.

The small number of debt restructurings in any given year (and over time) is an important limitation to any attempt to quantify these relationships. To provide a more complete picture of the EM sovereign credit outlook, we extend our analysis to include episodes of sovereign credit distress over the past decade — where credit spreads are sustained at extreme wide levels (typically above the 90th percentile of the EM HY universe) for an extended period. For this sample, we find that variables such as dollarisation of deposits and FX valuations have been additional good indicators.

Finally, we complement this with an analysis of the macro factors that matter in determining the size of haircuts (i.e., losses given default). In particular, external debt and the share of external debt that is public are helpful in predicting the relative size of NPV haircuts over time. Together with our models for default and distress, this provides an overall framework for quantifying expected investor losses in EM HY sovereign credit.

The historical playbook points to a number of sovereigns where the probability of default and debt distress has risen and will likely continue to rise as a result of the Covid-19 shock. Nevertheless, most sovereigns are still below levels which screen as alarming, and we find that EM HY credit spreads are currently compensating for the higher credit risks (measured by 2020 fundamentals), leaving the risk premium for the asset class at the higher end of its historical range.

The higher risk premium likely reflects the fragile equilibrium for at least a handful of EM sovereigns. For this group, a combination of official sector support and a swift market recovery has staved off more severe debt distress. But there is a clear vulnerability in the event that the growth collapse of 2020 is repeated, for example if a vaccine is not efficacious and/or if policy support is not forthcoming.

Forty years of defaults

We start by looking at a brief history of sovereign defaults, focusing in particular on defaults since the 1980s, where detailed data are more readily available for a larger subset of countries. [Exhibit 2](#) summarises the number of sovereign debt restructurings per year on external commercial debt going back to 1978. While episodes of defaults and debt restructurings have indeed increased this year, sovereign defaults in the last 20 years have been relatively benign compared with the 1980s, which saw a wave of sovereigns default on their bank loans.

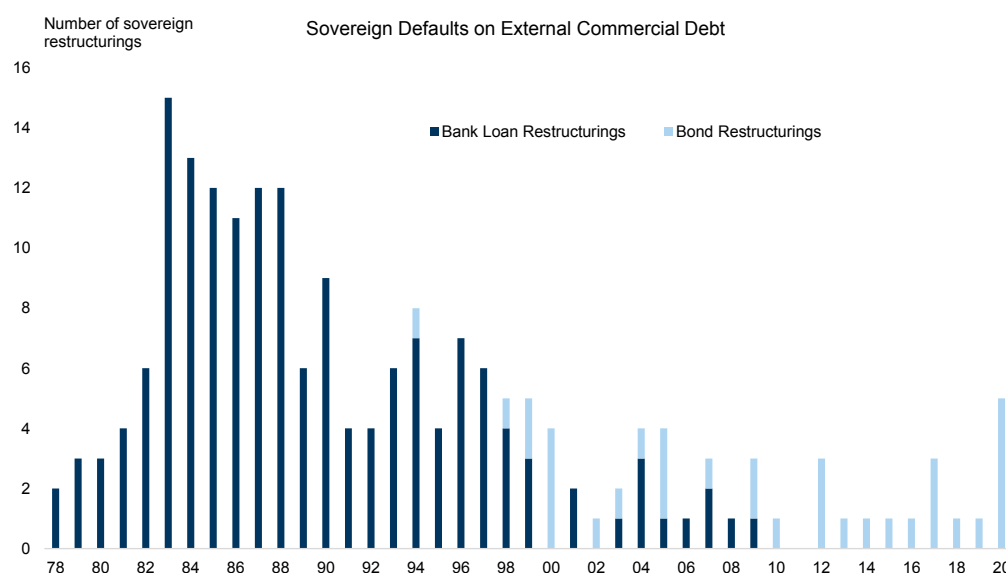
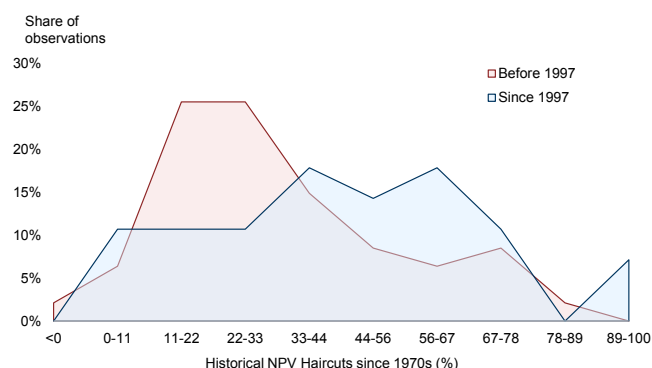
Exhibit 2: Episodes of sovereign defaults have fallen after a wave of bank debt restructurings in the 1980s

Exhibit shows number of successful restructurings per year, except for 2020 which includes Lebanon where the debt restructuring is ongoing

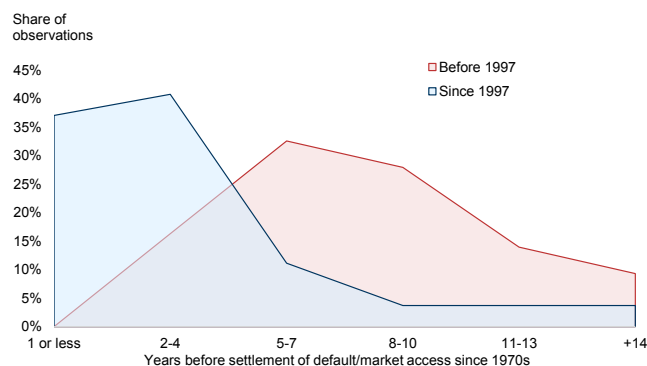
Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, Moody's, Goldman Sachs Global Investment Research

There may be a number of reasons for the relatively benign default picture. For one, while fundamentals in the emerging world still lag the developing world, they have improved over the period, with more anchored inflation expectations, rising GDP per capita and deeper local debt markets. Indeed, our own [sovereign ESG metrics](#) are consistent with this finding even over the past 10 years, and show a particular improvement in social and governance indicators for EM overall.

Another factor could be the increasing participation of emerging market sovereigns in the international bond market (aided by the transition from bank loans to [Brady bonds](#)), which may have allowed for a better sharing and understanding of sovereign risk. For example, the 1980s were marked by repeated debt defaults in a number of countries. These repeated default episodes meant that while NPV haircuts were relatively low at ~10-30% ([Exhibit 3](#)), it typically took a sovereign ~5-10 years to re-access the market once it had defaulted on its first loan ([Exhibit 4](#)). Conversely, debt restructurings over the past ~20 years have been settled faster (within ~1-4 years) but with a much wider range of NPV haircuts (between 30%-60%).

Exhibit 3: The last 20 years have seen a wider range of NPV haircuts ...

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, Moody's, Goldman Sachs Global Investment Research

Exhibit 4: ... and much faster debt restructurings

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, Moody's, Goldman Sachs Global Investment Research

While the difference between bank loans and Eurobonds is important in understanding the development of sovereign defaults over time, we still find that macro fundamentals can help explain what might lead a sovereign to default on its debt.

Literature Review

The literature that analyses sovereign defaults is extensive and tends to focus on three main areas: (i) the macroeconomic fundamentals associated with sovereign debt restructurings, (ii) the debt restructuring process, including the haircut, and (iii) the aftermath and return to capital markets.

Macroeconomic fundamentals and defaults

As expected, the debt stock is a key focus in the literature, with Manasse and Roubini (2005) suggesting the debt-to-GDP ratio is a crucial predictor of sovereign defaults. This idea has been explored in Reinhart, Rogoff and Savastano (2003), who find that the external debt-to-GDP threshold for default could be as low as 15% depending on a country's "debt intolerance," which is largely a function of inflation (particularly above 40%) and previous default history. Moody's (2020) study indicates that rather than the debt-to-GDP ratio, measures of debt affordability have been more correlated with defaults historically and that, for the last 21 defaults since 2010, in the year before default, the share of foreign-currency debt was ~70% of total debt on average. Additionally, of these same 21 defaults, most were caused by either political/institutional weaknesses (36%) or the burden of high debt (33%), with lower proportions attributed to chronic economic stagnation (17%) and banking crises (14%). Looking at a wider sample, Sturzenegger and Zettelmeyer (2006) suggest defaults are triggered by terms of trade deterioration, increases in international borrowing costs, recessions in main lender countries or contagion from a crisis in a key economy. Finally, macroeconomic and FX weakness and volatility are also mentioned in the literature (i.e., Catao and Kapur, 2006 or Manasse and Roubini, 2005), as well as illiquidity episodes, often measured through the short-term debt to FX reserves ratio (Manasse and Roubini, 2005).

Haircut size determinants

Cruces and Trebesch (2013) find that the average haircut is ~37%, although there is a significant variance around this estimate. Catao and Mano (2015) suggest that almost 90% of the actual haircut can be

predicted by the default's length (i.e, time a country spends in default) and the sovereign-specific fundamentals. Asonuma et al (2017) conclude that haircuts on shorter-term debt tend to be larger, reflecting the fact that short-term bond prices converge from above to the prices of long-term bonds in the run-up to the restructuring.

The default aftermath

The literature suggests that defaults not only impact a sovereign's credit market access but also have spillovers to the broader economy. In terms of market access, the literature touches on a few different channels. The reputational channel, put forth by Eaton and Gersovitz (1981) and extended by Aguiar and Gopinath (2006), suggests defaulters are punished by creditors. Authors such as Cole and Kehoe (1998) and Sandleris (2008) also conclude that higher haircuts imply lower trust in the government, which has adverse consequences for future spreads.

In terms of magnitude, the earlier literature suggests that creditors have "short memories," with credit spreads affected for only two years post-default (Borensztein and Panizza, 2009), and market access regained within one or two years for most defaulters (Gelos et al., 2004 and Richmond and Dias, 2009). Panizza, Sturzenegger and Zettelmeyer (2009) also find no significant increase in post-default spreads if ratings are controlled for. They further find that downgrades tend to reverse after two years, making the effect on spreads through the rating temporary. In contrast, Cruces and Trebesch (2013) suggest that haircut size is the main determinant of the market re-access conditions that a sovereign will face. More specifically, they find that a one standard deviation increase in the haircut (22 percentage points) is associated with spreads that are 150bp higher in the first year post-restructuring, and 70bp wider in the fourth and fifth years after, pointing to a trade-off: larger debt relief in the default may imply worse borrowing conditions in the long run. They also find that exclusion from capital markets increases with the size of the haircut, with NPV haircuts above 60% implying a high likelihood of capital market exclusion for over 10 years.

Additionally, Catao and Mano (2015), through the calculation of a Default Premium that accounts for as much as 60% of spreads in the initial five years post-default, find two crucial determinants of market re-access conditions: first, the proportion of years in default in the sample and, second, the number of years since the last default. Other papers by Eichengreen and Portes (1986), Reinhart, Rogoff and Savastano (2003), and Esteves (2007) have reached similar conclusions, suggesting that "serial defaulters" pay a premium, not due to how often they default, but because they spend a higher percentage of their time in default.

On the effects on the overall economy, output losses are estimated to be up to 5% a year for up to 10 years by De Paoli, Hoggarth and Saporta (2009). Papers by Cole and Kehoe (1998) and Sandleris (2008) suggest that there might be reputational spillovers to FDI and credit access for the private sector, as well as declines in trade flows post-default, as suggested by Rose (2005). Additionally, Gennaioli, Martin and Rossi (2010) find that public defaults are followed by lowered aggregate financial activity.

Sovereign debt restructurings

To identify the fundamental factors that may lead a sovereign to restructure its external commercial debt, we draw from Cruces and Trebesch (2014) data on sovereign defaults since the 1980s (a widely used dataset in the literature) and complement this with more recent data from Moody's. We define distressed debt exchanges and defaults as restructurings which have left creditors worse off, and focus on all sovereign debt restructurings of commercial external debt in the last ~40 years which were not donor funded. More specifically, we include two types of debt events in our analysis: 1) defaults, defined as the failure to make a principal or interest payment on time, past the grace period, and 2) distressed debt events, which is when a restructuring takes place within the grace period. As we include all restructurings on external bond and bank loans, our analysis extends beyond the universe of sovereigns covered by the benchmark EMBI Global Diversified Index (which tracks EM USD sovereign and quasi-sovereign debt).

For data going back to the 1980s, when serial defaulters were more common, we focus on the initial default that removed market access for a given country. Here, we use data by Catão and Mano (2015), whose analysis focuses on years of exclusion from the market following the initial sovereign default. For example, between July 1983 and December 1991, Nigeria restructured its external commercial debt seven times, resulting in ~9 years without market access. For this, we define 1983 as the year of initial default for Nigeria, and assume that the relationship between the fundamental developments and defaults in the interim period were endogenous to one another.

Stylised Facts: Debt restructurings have been associated with rising debt levels and high inflation

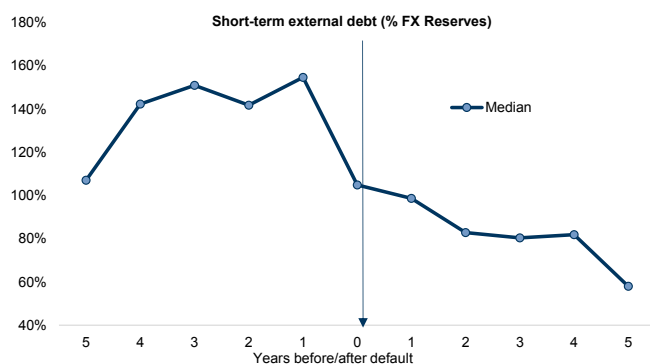
As described in the literature review, there is no strong consensus on which factors matter the most for a sovereign default, likely reflecting the wide range of circumstances leading to sovereign distress. Nevertheless, looking at the episodes of sovereign defaults over time, **we find some common trends in the median sovereign's macro variables in the years leading up to a default:**

- 1. Rising short-term external debt relative to reserves:** We find that sovereigns have typically seen their short-term external debt rise relative to FX reserves, and that this tends to be above 100% in the year before a default ([Exhibit 5](#)).
- 2. Rising central and external government debt:** Moreover, we do not find strong evidence that sovereign defaulters tend to have higher public debt relative to GDP or public external debt relative to GDP, but do find some evidence that these tend to have been rising in the years leading up to a default ([Exhibit 6](#) and [Exhibit 7](#)). This finding is similar to Reinhart, Rogoff and Savastano (2003), who argue that the external debt-to-GDP threshold for default is dependent on a sovereign's "debt intolerance" and could be as low as 15%.
- 3. Highly managed FX regimes:** We find some evidence that a country's FX regime is correlated with defaults. We show this in [Exhibit 8](#), which uses data by Ilzetzki,

Reinhart and Rogoff (2016) to classify countries' FX regimes. Here, FX regimes are scaled from 1 to 6, whereby 1 is the most managed currency, and 5-6 is the least managed. The exhibit shows that countries with more managed FX regimes tend to have a higher frequency of defaults. However, it also shows that this tends to change slightly in the years prior to a default, where countries that otherwise have a highly managed currency (regimes 1 or 2) are re-classified as having "freely-falling" currencies (regime 5). This category refers to countries with chronically collapsing currencies that accompany very high inflation (>40%). While the causality between the apparent shift in FX regimes and default episodes is less clear, these sudden FX moves suggest that the sovereign is adjusting for external imbalances, mainly through inflation.

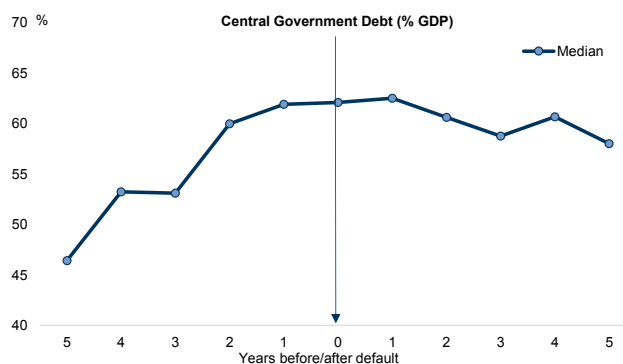
- 4. High inflation:** Related to this, a sovereign's inflation history is a key component of its "debt intolerance" level, according to Reinhart, Rogoff and Savastano (2003), who suggest that EMs that have never defaulted have never had inflation above 40%, compared with serial defaulters, which surpass this threshold ~25% of the time. We find evidence of this in our sample and show it in [Exhibit 9](#) with a threshold of 15% and [Exhibit 10](#) with a threshold of 40%. Indeed, the median non-defaulter has never recorded inflation above 40% in our sample, with only ~10% of inflation prints since 1978 above 15%. Conversely, sovereigns that have defaulted once and, to a greater extent, those that have defaulted more than once, exhibit higher inflation: the median country defaulting more than once has had inflation above 40% for around a quarter of time (similar to the literature's findings) and above 15% for roughly 40% of the time. It is worth noting that there are exceptions to this rule. For example, Angola and Belarus have had inflation above the 40% threshold for 50% and 44% of the time respectively, but without any debt events in our sample.

Exhibit 5: Short-term external debt has typically been rising relative to FX reserves prior to a default

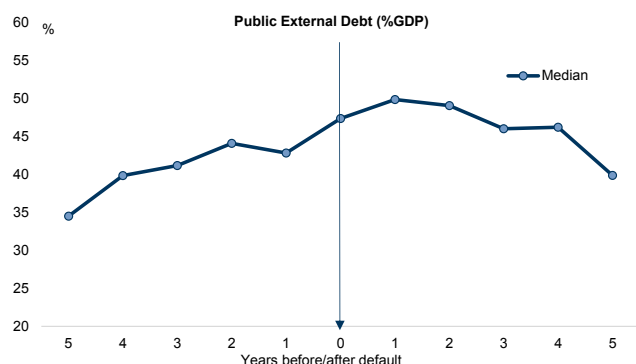


Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

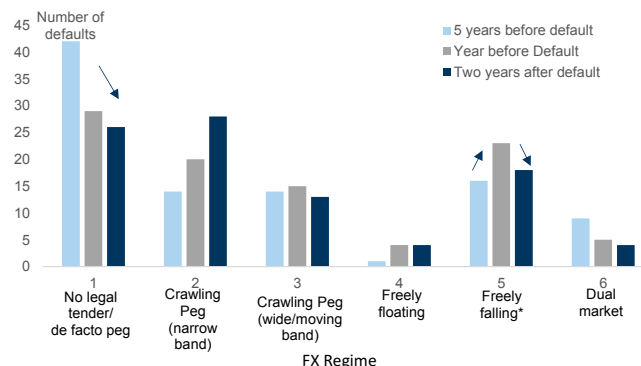
Exhibit 6: Similarly, central government debt has tended to rise relative to GDP in the years prior to a default ...



Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

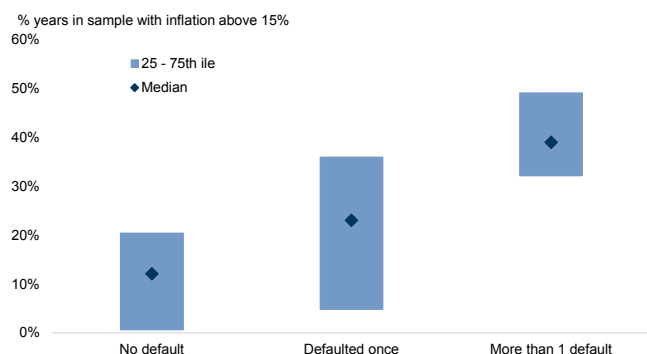
Exhibit 7: alongside rising public external debt

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

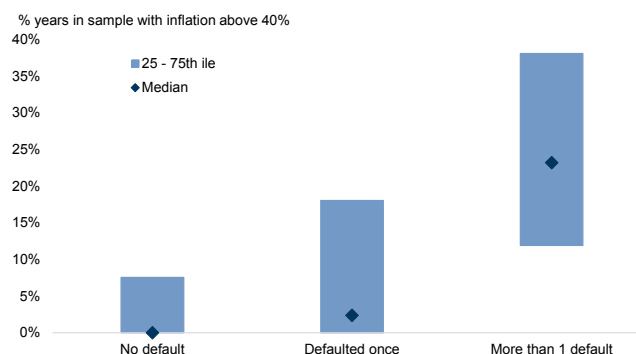
Exhibit 8: FX regimes tend to become more flexible in the year before a default

*chronically collapsing currencies that accompany very high inflation (>40%)

Source: Ilzetzki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), Moody's, Goldman Sachs Global Investment Research

Exhibit 9: Countries that have defaulted tend to have higher inflation

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

Exhibit 10: ... and countries that have defaulted more than once tend to have several years with inflation above 40%yoy

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

Modeling defaults: We find that gross FX reserve developments matter significantly for the probability of default

To further quantify the impact of these macro fundamentals on the default probability over time, we use a Binary Logit model, whereby the dependent variable is 1 in the year of default, and zero otherwise. This is an unbalanced panel, running from 1980 to 2020, across ~80 countries. To improve the model fit, we exclude wealthier sovereigns which do not have a history of default in our sample (captured as sovereigns with a GDP per capita above ~\$22,000).

The results are shown in [Exhibit 11](#), which estimates the probability of sovereign default using various macroeconomic variables that have been discussed in the literature, with broadly similar signs and statistical significance. For example, we find that **higher FX reserves significantly reduces the probability of a sovereign default, alongside a higher current account and lower short-term external debt relative to GDP**.

Moreover, we find that when a country has recently defaulted, there tends to be a low

probability of a renewed default within the next 15 years (with the probability falling over time), although this variable is less significant across different model specifications. We also find that **changes in stock variables, such as the central government debt, total external debt or changes in public external debt, tend to matter** for the probability of default.

Given the rarity of sovereign defaults, and the challenge of getting reliable macro data over time, we encounter a significant trade-off between adding additional variables to the analysis, without reducing the number of sovereign defaults in the sample. For example, we find that institutional quality (measured through the Bayesian Corruption Index) matters for the probability of default, but reduces the observations with sovereign defaults to 33, and we encounter a similar trade-off with other macro variables. Interestingly, we do not find evidence that higher inflation, lower primary balances or over-valued currencies (measured through a REER-gap) have led to a higher probability of default over time, when controlling for other variables.

Exhibit 11: Modeling the probability of defaults and debt restructurings

Dependent Variable: Default/Restructuring Episode

Explanatory variables (1 Year prior to default)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
FX Reserves (% GDP)	-8.6 (2.79)***		-8.6 (2.63)***	-8.4 (2.61)***	-7.4 (1.51)***	-8.8 (2.95)***	-12.6 (3.48)***	-2.9 (0.35)***	-8.5 (2.79)***
Current Account (%GDP) - 3YMA	-4.5 (2.53)*	-3.4 (2.18)''	-4.6 (2.37)*	-4.9 (2.39)**	-4.7 (2.54)*	-4.9 (2.58)*	-4.0 (3.08)'	-5.3 (2.57)**	-5.1 (2.61)*
Short-term External Debt (%GDP) -3YMA	2.7 (1.11)**	2.2 (1.17)*	2.2 (1.09)**	2.4 (1.06)**	3.2 (1.21)***	2.6 (1.14)**	2.9 (1.32)**	2.0 (1.27)''	2.7 (1.11)**
Years since last default LOG (ysd+1) set to constant at 15yrs	-0.2 (0.15)''	-0.3 (0.14)**	-0.1 (0.14)	-0.1 (-0.14)	-0.4 (0.16)***	-0.2 (0.15)''	-0.3 (0.17)''	-0.3 (0.16)**	-0.2 (0.15)''
Central Government Debt (%GDP) - YoY Change in 3YMA	2.5 (1.46)*	3.0 (1.43)**			2.5 (1.47)*	2.2 (1.46)''	0.3 (1.76)	1.4 (2.59)	2.4 (1.47)*
Total External Debt (%GDP) (%GDP) - YoY Change in 3YMA			2.1 (0.91)**						
Public External Debt (%GDP) - YoY Change in 3YMA				2.6 (1.13)**					
Bayesian Corruption Index					0.1 (0.03)***				
High Inflation Dummy Inflation above 15%yoy						0.4 (0.36)			
Real GDP growth (%) 3YMA							-17.7 (5.21)***		
Primary Balance (% GDP) 3YMA								4.7 (4.81)	
REER gap to 5YMA									0.00 (0.01)
Constant	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
McFadden R-squared	8%	4%	7%	7%	11%	8%	13%	8%	8%
Defaults	40	40	45	45	33	39	34	34	40
Total Observations	1891	1860	2062	2050	1679	1869	1703	1605	1878

Standard errors in parenthesis. P-value: ***<1%, **<5%, *<10%, ''<15%, '<20%

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Samuel Standaert (2015), Moody's, Goldman Sachs Global Investment Research

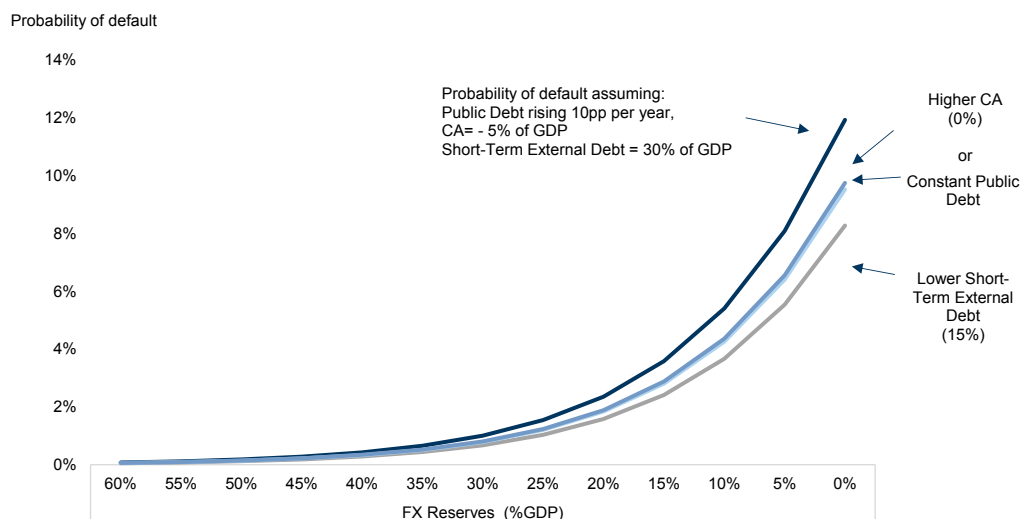
While most macro variables get the correct expected sign, it is important to note that

due to the nature of the data and the exercise, the R-squared is still relatively low, and suggests that the output should be read with some care. For the purpose of our exercise, which is to predict episodes of debt restructurings on a forward-looking basis, we choose Model 1 for the rest of this paper, where the R-squared is slightly higher, and where we are confident that the macro variables will be available on relatively high frequency in the future.

Exhibit 12 shows how the various coefficients from Model 1 work together to estimate the probability of debt distress. In particular, the exhibit highlights the importance of developments in FX reserves, whereby lower FX reserves significantly increase the probability of a default. For example, a country that has seen its public debt rise by ~10pp a year, and has a 5% current account deficit and short-term external debt at 30% of GDP, will have a ~12% probability of default if FX reserves are 10% of GDP, and a ~0% probability of default if FX reserves are 60% of GDP. Other variables, such as the current account or the public debt to GDP, have a more marginal impact on the model-implied probability, whereas developments in the short-term external debt can also have a significant impact on the output.

As such, Model 1 poses a challenge in instances where data on gross FX reserves have proven less reliable (such as Lebanon), or where countries are in monetary unions (such as Senegal and Cote d'Ivoire). In these instances, we refer to Model 2 instead.

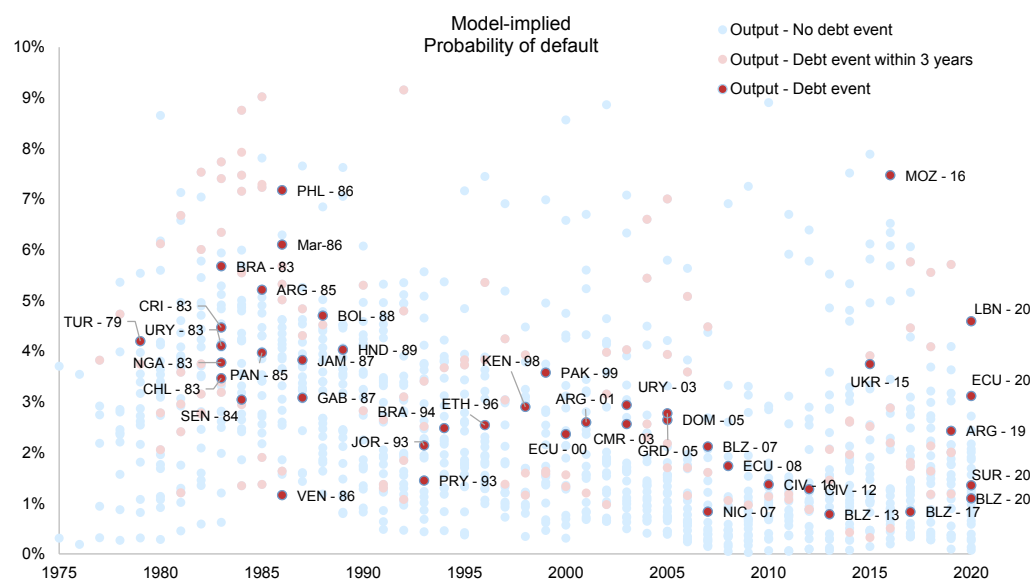
Exhibit 12: The level of FX reserves matters significantly for the model-implied probability of default



Source: Goldman Sachs Global Investment Research

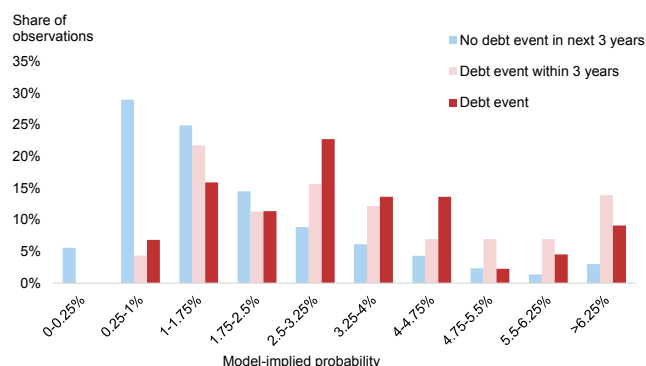
Model output: The implied probability has typically been rising ahead of a debt event

Exhibit 13 shows the model-implied output over time, highlighting the countries which have experienced debt events in that year (red dot), or within the next 3 years (pink dot). The exhibit also shows the output for sovereigns which did not default over the period. In relation to the low R-squared discussed above, the model-implied probabilities are generally low, at around ~2% for the median.

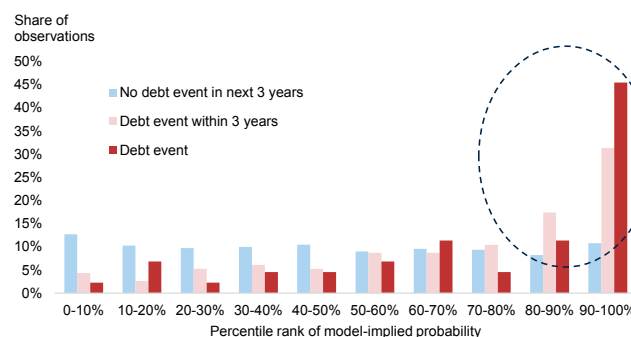
Exhibit 13: The model-implied probabilities for defaults tend to be low

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

While the absolute levels of default probabilities are low, we do find that there is some information in the model-implied output which can help distinguish defaulters from non-defaulters. For example, we find that the median model-implied probability of default tends to be ~2pp higher for sovereigns with debt events, at ~3% (Exhibit 14). Moreover, we find that there is information in the historical ranking of the model-implied probabilities, where the sovereigns that end up defaulting also tend to have a probability which is at the higher end of its historical range (80-100th percentile; Exhibit 15).

Exhibit 14: Probabilities have been higher for countries which have subsequently restructured ...

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

Exhibit 15: ... and the percentile rank of the model-implied probabilities tends to be at the higher end of the historical range

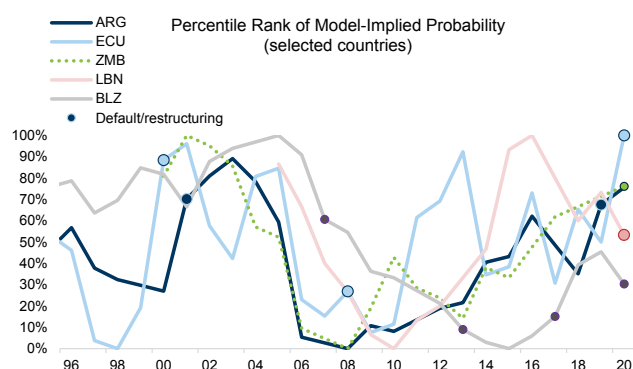
Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

Using the percentile ranks of the model-implied probabilities also shows that countries that have recently defaulted (or announced debt restructurings) have also seen a rising probability of default towards the higher end of the historical range (Exhibit 16). Nevertheless, the model did not perfectly predict the defaults even for this subsample. For example, while Ecuador defaulted on its debt in 2008-09, the probability of default

was at the lower end of the historical range at the time. This likely reflects the nature of the default, which followed political developments and a change in government, which would not be captured by macro factors in the model. Moreover, the model has not captured the series of restructurings in Belize in recent years. While the percentile rank was high for the 2006-07 episode, it was relatively low in 2012-13, and had only been rising slightly prior to the 2017 debt restructuring. In this instance, the weaker model-implied fit was a function of otherwise positive gross FX reserve developments prior to the debt events.

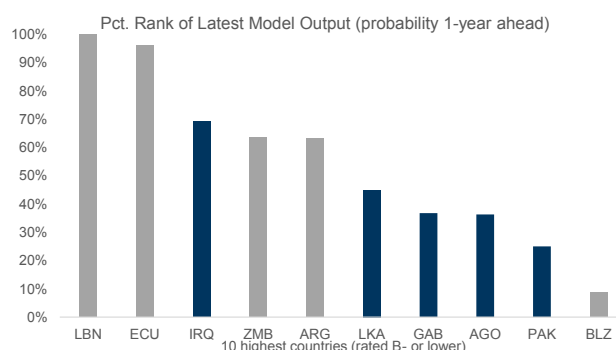
Looking at the broader subset of sovereigns which are currently rated B- or lower (i.e., where rating agencies already assign a higher probability of debt distress), we find that the probabilities (based on percentile ranks) are highest for Iraq, Sri Lanka, Gabon and Angola, although the last three are still below the median ([Exhibit 17](#)).¹

Exhibit 16: Probabilities were rising for countries which recently defaulted



Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

Exhibit 17: Outside of countries that have recently experienced debt distress, the model-implied probability is highest for Iraq, Sri Lanka, Gabon and Angola



The exhibit shows the percentile rank of the model-implied probability of default relative to the sovereign's history (i.e., ~1990-2020 for most countries in the sample), where a higher value equates a higher probability of default.

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Bloomberg, Moody's, Goldman Sachs Global Investment Research

As discussed above, however, these results should be read with some care due to the small number of sovereign defaults over time. To gain a more complete picture of sovereign debt vulnerability, we widen the lens to look at episodes of sovereign debt distress over the last ~10 years in the following section.

Sovereign distress

Stylised facts: Sovereign distress is usually (but not always) associated with debt restructurings

We complement our default analysis by looking into distressed credits to get a more

¹ The latest output uses the latest available data across sovereigns. For countries where we use IMF data for public debt and current accounts, we interpolate between the IMF's 2019 values and 2020 forecasts, and take the September values. We use USD GDP data from the IMF for all countries in the sample, and similarly interpolate between the 2019 values and 2020 forecasts, taking the September values for the latest output.

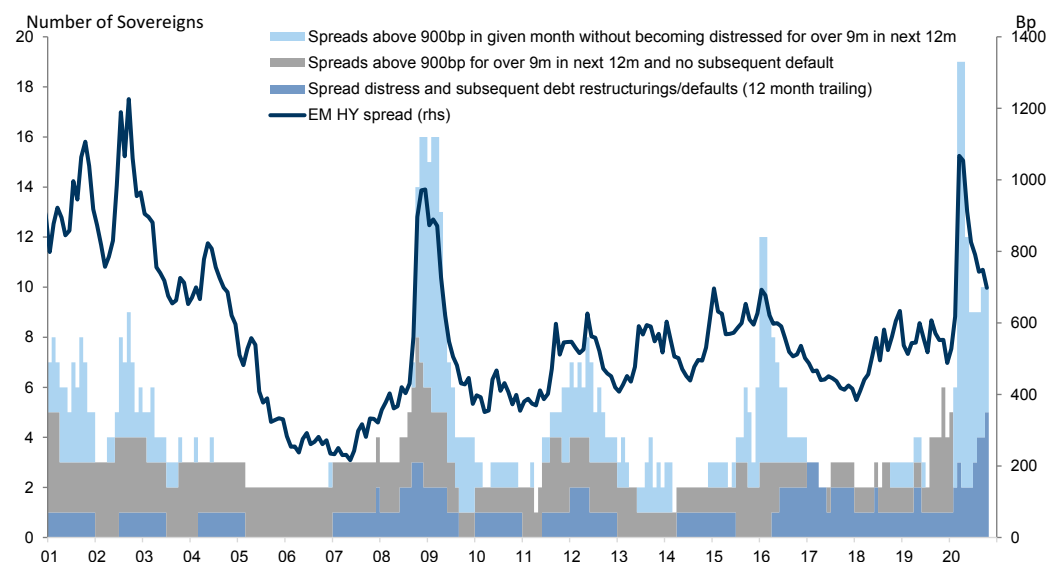
timely picture of the drivers of relatively long-term market spread distress, which could potentially result in a default.

Unsurprisingly, there is often an overlap between the countries that tend to have distressed spreads and the countries that default. Since 2010, a number of sovereigns have spreads at distressed levels for 20% of the time (or more), and have also defaulted. These include Argentina (defaulted in 2001, 2014 and 2019), Belize (defaulted in 2006-07, 2012-13 and 2017, and had a distressed debt event in 2020), Ecuador (defaulted in 2008-09 and 2020), Ukraine (defaulted in 2015) and Zambia (distressed debt event in 2020). Mozambique (defaulted in 2016-17) also stands out as distressed for a significant part of the past 10 years, including post-restructuring. Nevertheless, there are exceptions to this rule. For instance, the USD bonds of both Pakistan and Belarus have traded at distressed levels for prolonged periods of time (following the Global Financial Crisis and the Euro area sovereign crisis), and while both sovereigns have ended up in IMF programmes, they have not subsequently restructured their Eurobonds in the last 10 years.

For the purpose of our analysis, we define a sovereign as being “distressed” when its spread (measured through a weighted average of bonds outstanding) is above 900bp by month-end, which is roughly the 90th percentile of HY spreads since 2000. To capture this in a longer-term fashion, we look at when a sovereign has had spreads at distressed levels for over 9 months. That is, if a country is currently distressed and was distressed for more than 9 months in the following 12-month period, we label it as distressed.

As [Exhibit 18](#) displays, the number of sovereigns that have been distressed and/or defaulted in a given period is correlated with the overall index-level spread. For example, the GFC saw a rise in the number of countries with distressed spreads alongside sovereign defaults by Nicaragua (2007-08), Ecuador (2008-09) and Seychelles (2008-10). Similarly, at the peak of the crisis this year, the number of sovereigns with spreads at distressed levels reached a 20-year high, and credits such as Ecuador, Suriname, Belize and Zambia ended with debt events (while Lebanon defaulted in Q1 under conditions which were unrelated to Covid-19, the conditions following Covid-19 would likely have exacerbated the crisis here as well had it not already defaulted).

Exhibit 18: While sovereign defaults are fairly acyclical, the number of countries with spreads at distressed levels tends to be correlated with the index



As our dataset ends at end-September, we have not observed distress for over 9 months in 2020 yet

Source: Cruces and Trebesch (2014), Catao and Mano (2015), Moodys, Bloomberg, Datastream, Goldman Sachs Global Investment Research

Modeling distress: Dollarisation of deposits and FX valuations have been significant in capturing distress

We model distress in a similar fashion to debt restructurings, with a binary probability model, where the dependent variable – distress for over 9 months – is defined as described above. This Binary Logit model is estimated on an unbalanced panel of monthly frequency between 2005 and 2019 for all sovereigns that constitute the EMBI index, and where we have reliable data (72 in total). To improve the model fit, we condition the model to sovereigns which are rated sub-investment grade (HY), which constitutes 43 countries throughout the sample.

Similar to our default model, our explanatory variables include macroeconomic fundamentals such as the current account (%GDP), FX reserves (%GDP) and two measures of the debt stock – public debt (%GDP) and total external debt (%GDP). The current account (%GDP) is trimmed at extreme levels (if higher/lower than +15%/-10%), and then specified as the gap to the four-year moving average. As these fundamentals can be slow-moving, we also include two measures that would capture the macro-adjustments in a more timely manner: (i) the real effective exchange rate (REER) gap, CPI-based, specified as the difference between the 3- and 12-month moving average, 3 months prior to distress, and (ii) the change in dollarisation, captured by the ratio of local FX deposits to broad money. Finally, we control for the ratings below single B, through three different categories of dummies (B rated, B- rated and CCC rated or lower). For a clearer understanding of cause and effect, we lag all variables by 1 month (except for the REER as aforementioned).

The results are presented in [Exhibit 19](#), and show that the variables have the expected

sign, and are significant at the 1% level. More specifically, we find that **the probability of distress increases on a lower current account balance, lower FX reserves, higher debt stocks, an overvalued currency and a higher rate of dollarisation**, also when controlling for lower ratings.

We test the robustness of our preferred Model 1 to market variables, such as our Global Current Activity Indicator (to capture a higher frequency for external growth), US 10-year yields, changes in the oil price and the VIX. We find that Model 1 is generally robust to the inclusion of these variables, which also have the expected sign and significance (except for oil prices, which are not significant, and US 10-year yields which have a negative sign). Nevertheless, these variables are less robust, as seen with the Global CAI getting the wrong sign with the inclusion of the VIX.

While other cyclical variables such as growth and inflation also matter for the probability of distress, we have excluded these because of concerns around data quality, especially for frontier economies, where data frequency is lower (these typically do not have consensus data on a regular basis).

Exhibit 19: Modeling Distress

Dependent Variable: Distress Episode for over 9 months						
Explanatory variables (1 month prior to distress) ^a	(1)	(2)	(3)	(4)	(5)	(6)
Current Account (%GDP)	-14.8	-12.5	-11.5	-11.6	-11.6	-13.4
Difference to 4YMA	(2.79)***	(2.51)***	(2.91)***	(2.91)***	(2.92)***	(3.25)***
FX Reserves	-4.7	-4.7	-3.9	-3.9	-3.9	-6.3
(%GDP)	(0.77)***	(0.78)***	(0.8)***	(0.8)***	(0.8)***	(0.91)***
Public Debt	2.8	2.1	2.2	2.3	2.3	3.7
(%GDP)	(0.42)***	(0.36)***	(0.44)***	(0.44)***	(0.44)***	(0.49)***
External Debt	0.9	0.7	0.8	0.8	0.8	1.2
(%GDP)	(0.18)***	(0.18)***	(0.19)***	(0.19)***	(0.19)***	(0.2)***
REER ^a	0.04		0.03	0.03	0.03	0.04
Difference of 3MMA to 12MMA	(0.01)***		(0.02)**	(0.02)**	(0.02)**	(0.02)**
Dollarised deposits	1.1		0.9	0.9	0.9	0.9
Change in 12MMA	(0.26)***		(0.28)***	(0.28)***	(0.28)***	(0.29)***
B Rated	3.2	3.3	3.2	3.2	3.2	3.5
	(0.41)***	(0.41)***	(0.42)***	(0.42)***	(0.42)***	(0.43)***
B- Rated	3.7	3.7	3.4	3.5	3.5	3.6
	(0.41)***	(0.41)***	(0.41)***	(0.41)***	(0.41)***	(0.43)***
CCC or lower	4.1	4.1	4.1	4.1	4.1	4.3
	(0.44)***	(0.42)***	(0.44)***	(0.44)***	(0.44)***	(0.45)***
Global CAI			-0.12	-0.10	-0.10	0.22
			(0.04)***	(0.04)**	(0.04)**	(0.05)***
US 10y yields				-0.9	-0.8	
1st difference				(0.37)***	(0.41)*	
Brent price					-0.01	
% change					(0.01)	
VIX						0.1
						(0.01)***
Constant	Yes	Yes	Yes	Yes	Yes	Yes
McFadden R-squared	31%	27%	28%	28%	28%	34%
Distress Observations	208	233	182	182	182	182
Total Observations	4064	4428	3815	3815	3815	3815

^aAs stated in the text, the REER is lagged by 3 months rather than 1 month.

Source: IMF, World Bank, IFS, Bruegel, Federal Reserve Board, Financial Times, Wall Street Journal, Haver Analytics, Datastream, Bloomberg, Goldman Sachs Global Investment Research

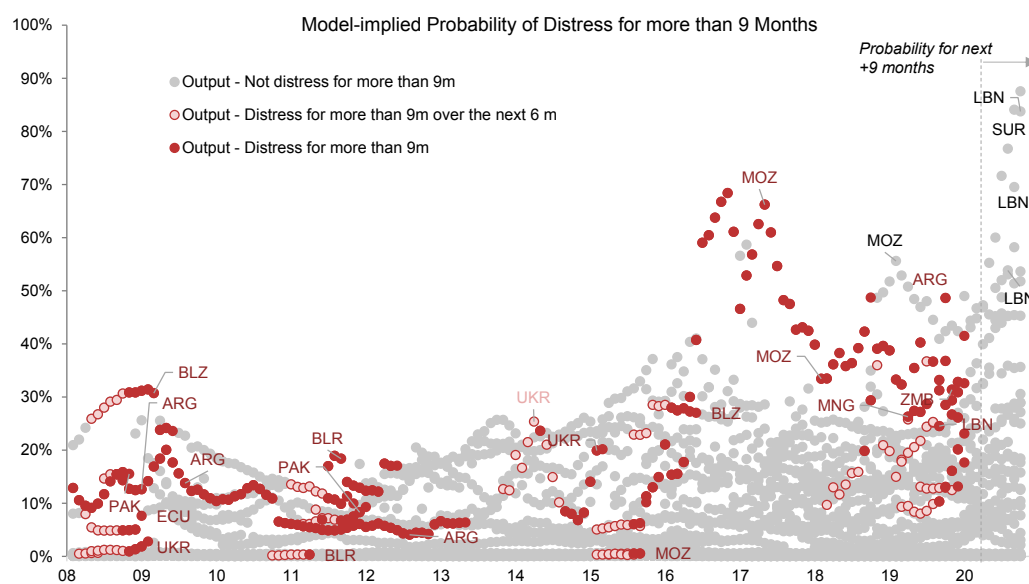
Model output: The implied probability of distress has risen across sovereigns in recent years

Model 1 is our preferred specification as it focuses on country-specific fundamentals which allows us to better track idiosyncratic developments. This model also has the second-highest R-squared, only below Model 6, which we find to have low out-of-sample predictive power due to the interaction with the VIX. Nevertheless, Model 1 poses a challenge for sovereigns that are in monetary unions (such as Senegal and Cote d'Ivoire), dollarised (such as Ecuador), or have capital controls currently in place

(like Argentina). In these instances, as well as when the data are missing, we refer to Model 2, where we exclude the REER and the dollarisation measures.

Exhibit 20 shows the model-implied output over time, colour-coded to indicate when sovereigns experienced long-term distress in that month (red dot) or some time in the next 6 months (pink dot), or when they did not (grey dot). Apart from a few specific country examples – such as Mozambique in 2017 and Argentina in 2019 – and the recent increases in 2020, the probabilities are range-bound below ~40%, having increased over time, partly driven by rising debt levels and also lower ratings in aggregate.

Exhibit 20: Apart from a few specific countries and the recent increases in 2020, the probabilities are range-bound below ~40%, having increased over time



Source: IMF, World Bank, IFS, Bruegel, Haver Analytics, Datastream, Bloomberg, Goldman Sachs Global Investment Research

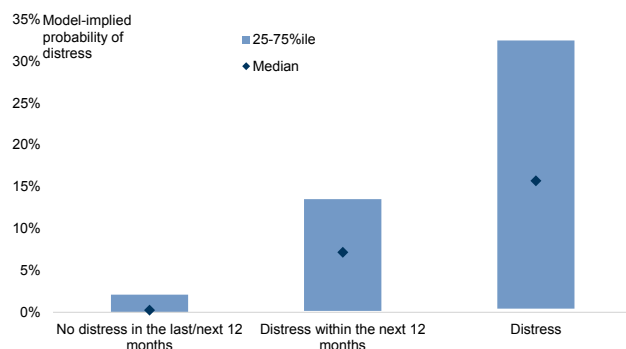
We also find information in the output on an aggregate level, as the median model-implied probability of long-term distress is 16% for credits that end up in distress, compared with a median probability of 0% for sovereigns that do not (Exhibit 21).

To see what the model tells us about 2020 (which is out-of-sample), Exhibit 22 displays the latest 3-month moving average of the probability of longer-term distress for countries with probabilities at or above the conditional median of ~16%, and compares this with the historical range of probabilities for each sovereign (i.e., the probability a year prior, the historical median, as well as the respective 9-95th percentile range, since either 2005 or the earliest data available).

Here, we find that the model-implied probability of distress is currently high for 15 sovereigns in the EM universe, of which 6 experienced debt events over the last 1-2 years. More specifically, the probability of distress currently stands at or above the 95th percentile of the historical range for all of these countries, and to a far greater extent in both Suriname (with an ~85% probability of long-term distress) and Lebanon (with a ~75% probability), consistent with the former having restructured part of its debt in July, and the latter having defaulted on its debt in Q1. Other sovereigns that stand out include

Zambia, which has already stated that it is planning to restructure and has missed coupon payments, and Angola, which renegotiated its liabilities with China and the official sector this year. For Argentina and Ecuador we use Model 2, given capital controls in the former and a dollarised economy in the latter (we switch to Model 2 in Argentina after October 2019). Here, both continue to screen high currently, and saw their probability of distress rise in the run-up to their default episodes, as with the other countries which experienced debt events over the past year (Exhibit 23).

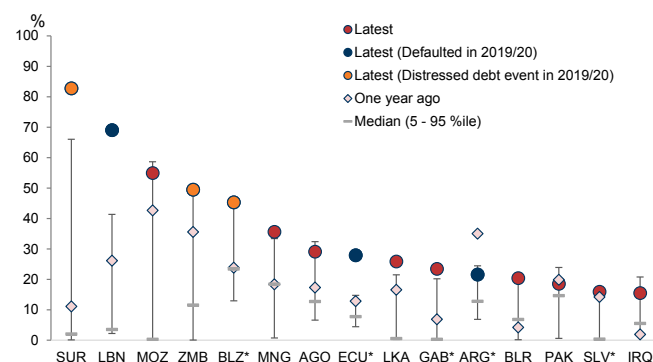
Exhibit 21: A model-implied probability of 16% tends to be associated with episodes of debt distress



Distress in this exhibit refers to our dependent variable of distress for over 9 months in a 12-month period

Source: IMF, World Bank, IFS, Bruegel, Haver Analytics, Datastream, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 22: A number of sovereigns currently have a high probability of distress



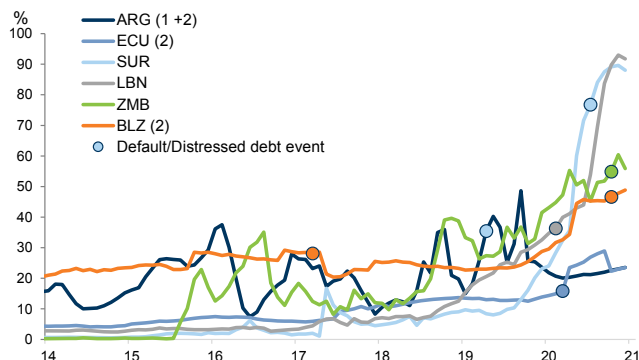
Latest values are the 3MMA for September 2020. *Belize, Ecuador, Gabon, Argentina and El Salvador's outputs are based on Model 2.

Source: IMF, World Bank, IFS, Bruegel, Haver Analytics, Datastream, Bloomberg, Goldman Sachs Global Investment Research

As a final step, we compare the signal from our Distress Probability model with that of the default model (here we use the percentile ranks of the model-implied probabilities). The assumption here is that while there are limitations to both frameworks, taken together they can help identify where some of the left-tail risks lie within EM HY credit.

Exhibit 24 portrays a scatter of the latest output of both models. Here, we limit the sample to the sovereigns with the latest probabilities in our distress model at or above the conditional median of ~16%, and compare this to the percentile rank of their model probability of default. This framework also points to recent defaulters like Ecuador and Lebanon as having a high probability of default and debt distress. Moreover, it points to a high probability of prolonged debt distress in Iraq, Sri Lanka, Angola and Gabon, which also have a relatively high probability of default, albeit below their historical medians for the three latter. Finally, the models point to a few sovereigns with a high probability of debt distress but lower probability of default; these include Belarus, Mozambique, El Salvador, Mongolia and Pakistan.

Exhibit 23: Countries that recently defaulted/restructured also saw their distress probability rise prior to the event

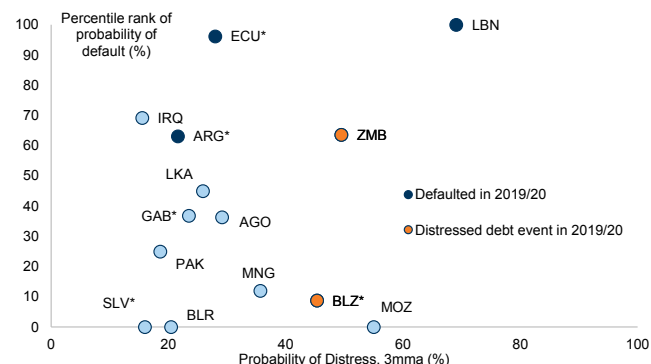


For ARG we show the output from Model 2 from October 2019

Source: IMF, World Bank, IFS, Bruegel, Haver Analytics, Datastream, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 24: The sovereigns that currently have a higher probability of default and debt distress are Iraq, Sri Lanka, Gabon and Angola

*For ARG, BLZ, ECU, GAB and SLV we use the probability of distress for over 9m from Model 2.



The y-axis shows the percentile rank of HY sovereign's probability of default for the next 12 months (sample typically covers 1990-2020). The x-axis shows the probability of spreads being at distressed levels for the next 12 months. Latest values are for September 2020. Exhibit does not include Suriname, given a short history for the model-implied probability of default (i.e., low information in the percentile rank).

Source: Cruces and Trebesch (2014), Catao and Mano (2015), Moodys, IMF, World Bank, IFS, Bruegel, Haver Analytics, Datastream, Bloomberg, Goldman Sachs Global Investment Research

Haircuts

While identifying the probability of a default and debt distress is an important first step in estimating potential investor losses, identifying the potential haircut (i.e., loss given default) on debt once a default has occurred is equally important. In broad terms, haircuts can be defined as creditor losses following a debt restructuring. To estimate potential haircuts, economists will often focus on the level of debt which is required to make debt sustainable, and compare this to the current debt outstanding. This type of analysis is dependent on assumptions on the sustainable interest rate and trend growth (among other things), which means results can vary significantly depending on the assumptions. As a result, we try to complement the standard approach with a top-down analysis, to identify which macro variables have been good predictors of haircuts in a cross-sectional framework.

For this, we use the historical haircuts of sovereigns over time from a combination of Cruces and Trebesch (2014), Moody's and our own estimates for more recent default episodes. More specifically, we focus on the definition of the net present value (NPV) of haircuts suggested by Struzenegger and Zettelmeyer (2008), which **compares the present value of new debt to the present value of old debt**, using a common discount rate for both, as we note that this is a better representation of the potential trade-off for creditors (similar to Cruces and Trebesch, 2014).² Similar to the default analysis, we adjust for haircuts of serial defaulters on bank loans which effectively resulted in exclusions from markets for several years. Here, we take the weighted

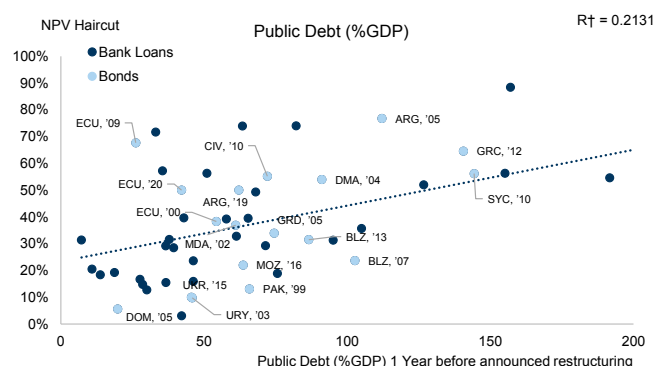
² Struzenegger and Zettelmeyer (2008) suggest using the yield on the new debt on the first day of trading as the discount rate.

average of the NPV haircuts over the period where issuers are excluded from the market (weighted by USD amount on defaulted debt).

Stylised facts: Higher debt and more managed FX regimes tend to be correlated with higher haircuts

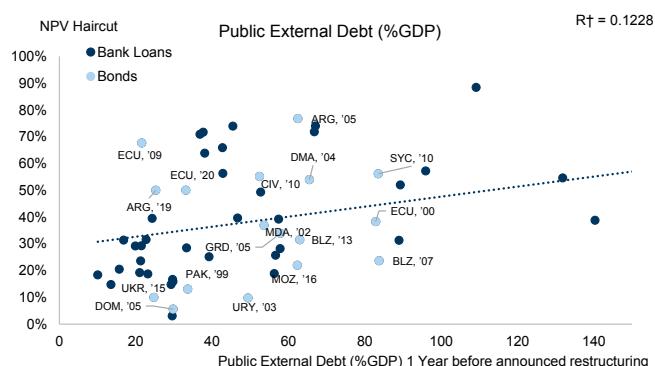
As described earlier, changes in NPV haircuts over time may partly be a function of the shift from sovereigns funding themselves through bank loans to bonds. Nevertheless, we do find a correlation between NPV haircuts and fundamentals in a cross-section, in particular with stock variables. For example, we find a positive correlation between NPV haircuts and sovereigns that had higher public debt relative to GDP ([Exhibit 25](#)), higher public external debt ([Exhibit 26](#)) and higher total external debt relative to gross national income ([Exhibit 27](#)) in the year before a debt restructuring was announced³. Moreover, we find a correlation with sovereigns' FX regimes (described above in the section on Sovereign Debt Restructurings), whereby more flexible exchange rate regimes tend to be correlated with lower NPV haircuts ([Exhibit 28](#)). To us, this is intuitive as it suggests that countries which are unable to use their exchange rate to adjust for external imbalances might have to impose a larger haircut on creditors.

Exhibit 25: Countries with higher public debt (%GDP) tend to see higher NPV haircuts



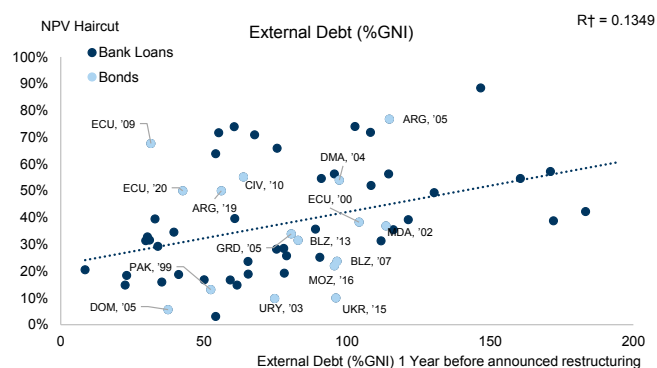
Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moodys, Goldman Sachs Global Investment Research

Exhibit 26: Similarly, countries with higher public external debt tend to see higher NPV haircuts ...

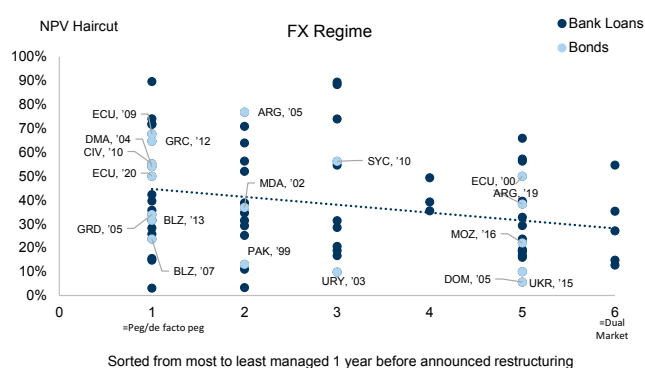


Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moodys, Goldman Sachs Global Investment Research

³ The year of default and announced debt restructuring may differ for some sovereigns. For example, for the Argentina default from 2001-14, we take 2001 as the year of default, and 2005 as the year of announced restructuring.

Exhibit 27: ... alongside countries with higher total external debt

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, Haver Analytics, Moodys, Goldman Sachs Global Investment Research

Exhibit 28: On average, countries with highly managed FX regimes have also had higher NPV haircuts

Source: Ilzetzki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), Moodys, Goldman Sachs Global Investment Research

Modeling haircuts: The size of external debt matters significantly in the cross-section

To identify the macro variables which matter for NPV haircuts across countries, we use a simple cross-sectional OLS regression (Exhibit 29). We start by looking at the four variables explained above, to test how they interact in a relative framework (Model 1). Interestingly, we find that only the total external debt (%GNI) is robust to the inclusion of other variables, which suggests that total external debt on its own may be one of the best predictors of relative NPV haircuts. This finding therefore challenges the standard practice of using the overall size of public debt to identify a sovereign's potential haircut, and suggests that contingent liabilities which are captured in total external debt may also play an important role.

Nevertheless, we find that a higher share of public external debt (relative to total external debt) does tend to increase the NPV haircut (Model 2), and that FX regimes have some predictive power for NPV haircuts when controlling for the size of a country's GDP (Model 3). The interesting finding here is that countries with higher GDP tend to have higher haircuts, which is not necessarily a function of GDP rising over time (as shown in Model 4, the year of default is not a strong predictor of NPV haircuts). Conversely, we do not find strong evidence that other macro variables, such as the output gap (measured as the level of GDP vs. 5-year trend), the REER gap, short-term external debt relative to FX reserves, trend GDP growth or inflation matter for NPV haircuts in the cross-section.

Exhibit 29: Modeling NPV Haircuts**Dependent variable: NPV Haircuts**

Explanatory variables (1 year prior to debt restructuring)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
External Debt (%GNI)	0.26 (0.09)**	0.18 (0.06)**	0.25 (0.07)***	0.24 (0.08)**	0.24 (0.07)**	0.23 (0.07)**	0.26 (0.07)***	0.23 (2.21)**
Public External Debt (%Total External)		0.32 (0.07)***	0.30 (0.09)**	0.28 (0.18) "	0.33 (0.1)**	0.31 (0.1)**	0.23 (0.1)**	0.38 (2.6)**
FX Regime (Higher is less Managed)	1.56 (1.49)		-2.74 (1.58)*	-2.78 (1.65)*	-3.16 (1.62)*	-2.78 (1.59)*	-2.20 (1.62) '	-3.62 (-1.48) "
Log GDP			2.81 (1.75) "	2.72 (1.99) '	2.97 (1.8) "	2.80 (1.75) "	2.84 (1.73) "	2.76 (1.17)
Public Debt (%GDP)	0.09 (0.13)							
Public External Debt (%GDP)	0.01 (0.19)							
Year of Default				0.00 (0.01)				
GDP vs. 5 year trend					-5.13 (7.79)			
Reer Gap (Positive is stronger than 5-year average)						-0.07 (0.09)		
Short-Term External Debt (%FX Reserves)							0.00 (0.01)	
Trend Real GDP Growth (%yoy)								-1.11 (-0.62)
Trend Inflation (%yoy)								0.00 (0.25)
Observations	45	49	49	49	48	49	48	35
Adjusted R Square	72%	78%	79%	78%	79%	79%	78%	75%
Standard Error	0.21	0.19	0.19	0.19	0.19	0.19	0.18	0.21
Constant	No	No	No	No	No	No	No	No

Standard errors in parenthesis. *** if p-value <1%, ** if p-value <5%, * if p-value <10%, " if p-value <15%, ' if p-value <20%

Here, we use the 'fine' category for FX regimes, and recode these to include 6 categories, such that dual markets and freely falling currencies are in the same category (6), and countries in currency regimes are the lowest (1).

Source: Ilzetzki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), World Bank, IMF, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

Generally, our results obtain a relatively good fit, with an adjusted R-squared of ~78-79% for most model specifications. However, it is important to note that this analysis is based on relatively few observations spanning several years, and will be sensitive to incoming data (i.e., may prove less robust over time). For the purpose of our exercise, which is to use this framework for future analysis as well, Model 3 is our preferred specification, as it has a slightly higher R-squared and contains more variables which will allow for additional differentiation in the cross-section.

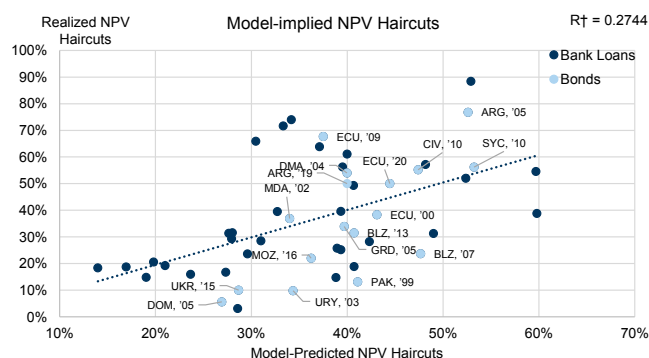
Model output: A cross-sectional model works relatively well, but misses the tails

To analyse the model output, we compare the model-implied NPV haircuts over time to the realised NPV haircuts. The results are shown in [Exhibit 30](#), which highlights bond restructurings over the last ~20 years. Here, we find a relatively good fit between the model-implied and realised output, although it is worth noting that a linear model such as this has worked less well in predicting the tails, such as the lower NPV haircuts seen in Ukraine in 2015 and Uruguay in 2003, or the higher NPV haircuts recorded in Argentina and Ecuador in 2005 and 2008-09 respectively.

We can also use this framework to infer potential NPV haircuts for sovereigns currently

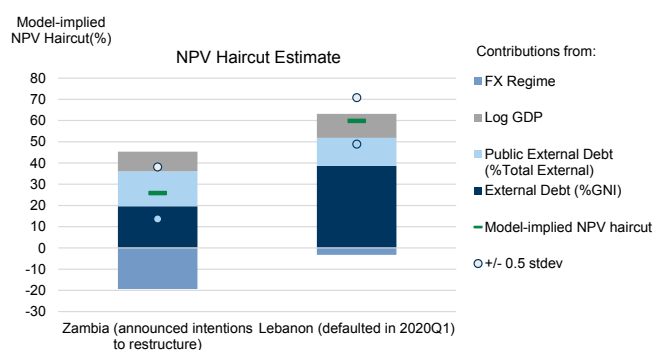
in default (such as Lebanon) or which have announced intentions to restructure their debt (such as Zambia, which also recently missed a coupon payment). Here, using last year's values for external debt (%GNI), the share of external debt that is public, the FX regime and logarithm of USD GDP, the model points to an NPV haircut of ~30% of Zambia, compared with ~60% for Lebanon, albeit with relatively large standard errors around these estimates ([Exhibit 31](#)). While the market is currently implying a larger NPV haircut for both sovereigns (inferred from USD bond cash prices), our results are nevertheless in line with the relative pricing (the market is implying a higher haircut for Lebanon), and our [economists' estimates](#) of a relatively high haircut for Lebanon. Moreover, it is worth highlighting to note that these results are likely to change, as they are based on the macro fundamentals in the year prior to an announced debt restructuring. Since neither have currently announced a final plan for their debt restructurings, changes in macro fundamentals between now and the announced debt restructuring could influence the model-implied results further.

Exhibit 30: We find a relatively good fit between the model-implied and realised NPV haircuts, albeit with a weaker fit in the tails



Source: Ilzetzki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), World Bank, IMF, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

Exhibit 31: This framework points to a relatively low haircut in Zambia compared with a much higher haircut in Lebanon



Source: Ilzetzki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), World Bank, IMF, Haver Analytics, Moody's, Goldman Sachs Global Investment Research

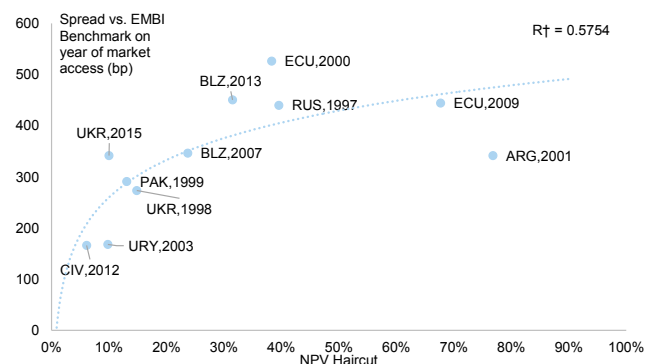
The costs of defaults and haircuts

While frameworks such as these can be useful in identifying expected investor losses, our results have relatively high standard errors and point to the difficulty in predicting sovereign defaults and haircuts with high certainty. In our view, however, this is to be expected, both because of the quality of the data, but also because sovereign defaults are often complex, and depend on a number of factors which are not necessarily captured by the macro fundamentals at the time. In some instances, sovereign defaults occur in relation to changes in political leadership (as seen with [Ecuador in 2008-09](#)), and may not always be a reflection of macro developments at the time (indeed, we found a relatively low probability of default for Ecuador in 2008-09; [Exhibit 16](#)). More often than not, however, we would expect the unforeseen costs of a sovereign default on external commercial debt to weigh on the decision to default.

While previous literature found that markets tend to 'forgive and forget' sovereigns with a history of default relatively quickly (i.e., Borensztein and Panizza, 2009), recent

literature has found evidence of the contrary. For example, Catao and Mano (2015) found that countries often pay a premium relative to their fundamentals once they regain market access after a default, and Meyer, Reinhart and Trebesch (2019) found that serial defaulters tend to have a higher yield. As for haircuts, Cruces and Trebesch (2013) found that countries with higher haircuts also tend to have wider spreads once they regain access to the international markets.

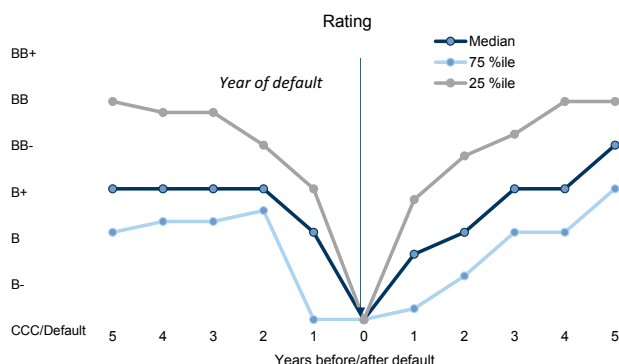
Exhibit 32: We find a positive correlation between NPV haircuts and the sovereign's spread relative to the benchmark on the year of market access



Source: Datastream, Cruces and Trebesch (2014), Moody's, Goldman Sachs Global Investment Research

Exhibit 33: It typically takes a number of years for ratings to recover to their 'post default' levels

Exhibit shows the median/modal rating from Moody's, S&P and Fitch for the LT External Debt



Source: Bloomberg, Cruces and Trebesch (2014), Catao and Mano (2015), Goldman Sachs Global Investment Research

While our analysis does not extend to the potential long-term costs for sovereigns that default, our data do show that countries with higher NPV haircuts may have had wider spreads in the year they regained access to international markets ([Exhibit 32](#)). Similar to Panizza, Sturzenegger and Zettelmeyer (2009), we also find that it takes several years for ratings to return to their "post-default" levels ([Exhibit 33](#)), which could be a function of fundamentals, or a function of an additional "default premium" imposed by rating agencies.

Implications for EM HY credit

Putting it all together, we highlight three high-level conclusions for EM HY credit from our analysis.

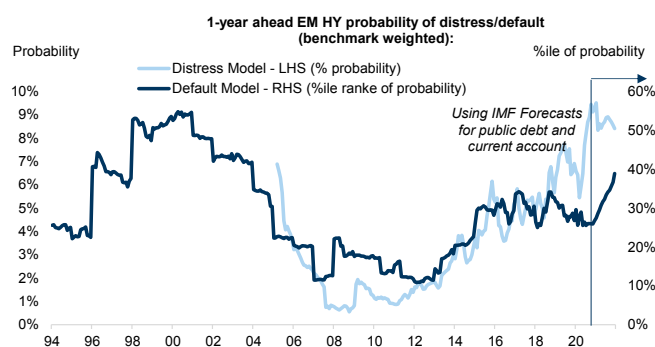
First, looking at the aggregate historical benchmark-weighted probability of debt default (using the percentile ranks of probabilities), we find that, in aggregate, the probability of default has increased over time and is on course to increase further in 2021 under the IMF's forecasts for still-high public debt levels and current account deficits ([Exhibit 34](#)). **However, while the probability of broad sovereign distress pricing has also increased through 2020, it appears to have peaked.** In part, this reflects the fact that the sharp deterioration in debt levels leaves a more longer-lasting footprint in our models of default probability relative to distress probability.

A different way to see this is to examine the number of countries for which the

probability of default and distress exceeds a certain threshold. IMF projections for 2021 point to a rise in the number of sovereigns with a probability of default greater than the 70th percentile of its history, although still in the 2-5 range that has prevailed since the GFC ([Exhibit 35](#)). The number of sovereigns with a probability of distress above 16% (which we found to be the threshold which correctly identifies most episodes of distress) has already risen sharply, but is projected to flatten out.

Interestingly, our measure of the expected default rate is higher than the realised default rate, which had been relatively constant in recent years prior to Covid-19. The divergence between the expected and realised investor losses in the sovereign space is intuitive to us, given the difficulty in assessing sovereign risk. For example, while fundamental developments may point to a default, the high costs related to a default, alongside the availability of official-sector support in times of crisis, could work to keep the realised losses low. This year was a prime example of this, where several sovereigns accessed [official sector support](#), and the official sector offered debt relief through measures such as the [G20 Debt Service Suspension Initiative](#) relief efforts. Other examples include [Angola](#), which restructured its short-term debt with China this year, but remained current on its Eurobonds.

Exhibit 34: The probability of distress and default has been rising for EM HY sovereigns ...



The LHS shows the weighted-average probability of spreads being at distressed levels for the next 12 months. The RHS shows the weighted average percentile rank of HY sovereign's probability of default (sample typically covers 1990-2020). Here, 40% means that the weighted-average EM HY sovereign's probability of default is at the 40th percentile of its history.

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, IFS, Bruegel, Haver Analytics, Bloomberg, Datastream, Moody's, Goldman Sachs Global Investment Research

Exhibit 35: ... although there has been a divergence between the number of sovereigns with a high probability of distress and high probability of default

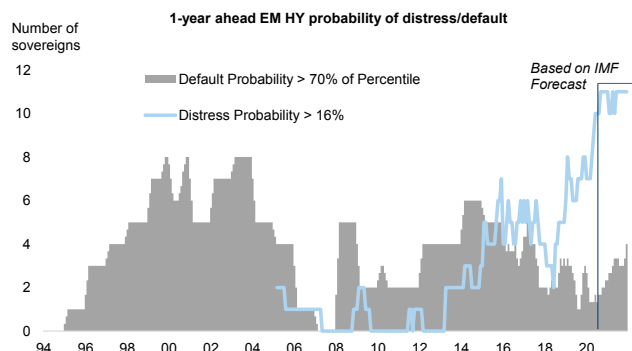


Exhibit shows the number of sovereigns where the probability of having spreads at distressed levels is higher than 16% (light blue line), and the number of sovereigns where the probability of default is greater than the 70th percentile of its history (grey shaded area).

Source: Cruces and Trebesch (2014), Catao and Mano (2015), IMF, World Bank, IFS, Bruegel, Haver Analytics, Bloomberg, Datastream, Moody's, Goldman Sachs Global Investment Research

Second, our results also suggest an increase in the predicted NPV haircuts in 2020, as a result of rising external debt in particular ([Exhibit 36](#)). In our view, however, the notion that expected losses overall (measured as the product of the probability of default and loss given default) are rising is broadly in line what can be observed in other fundamental markers of EM sovereign credit. For example, rising expected investor losses have been mirrored by rating downgrades for the average EM sovereign since 2012 ([Exhibit 37](#)), although this has been masked by compositional changes on an index level (such as the GCC index inclusion in 2019, which is mostly IG sovereigns).

Exhibit 36: The model-implied NPV haircut has risen slightly for the average EM HY sovereign

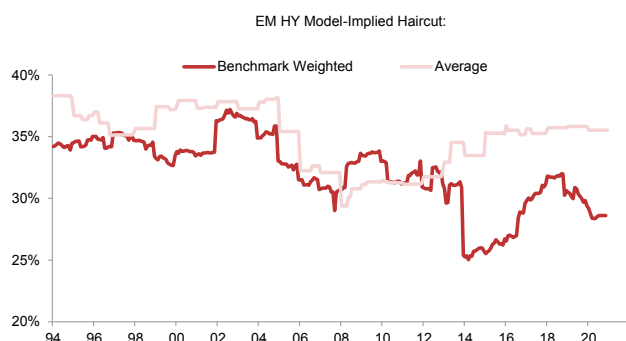
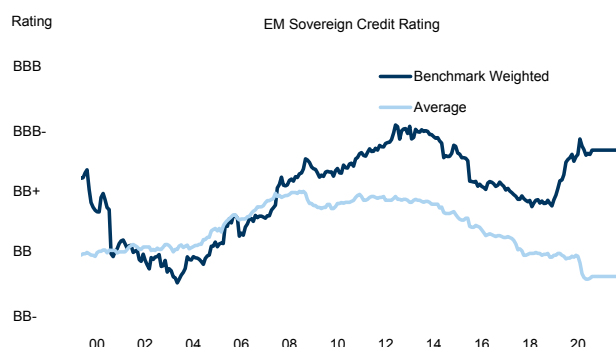


Exhibit shows the average and benchmark-weighted model-implied NPV haircut across EM HY sovereigns (not conditional on the default probability).

Source: Ilzetzki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), World Bank, IMF, Haver Analytics, Bloomberg, Datastream, Moody's, Goldman Sachs Global Investment Research

Exhibit 37: The average rating for EM sovereigns has been on the decline since 2012

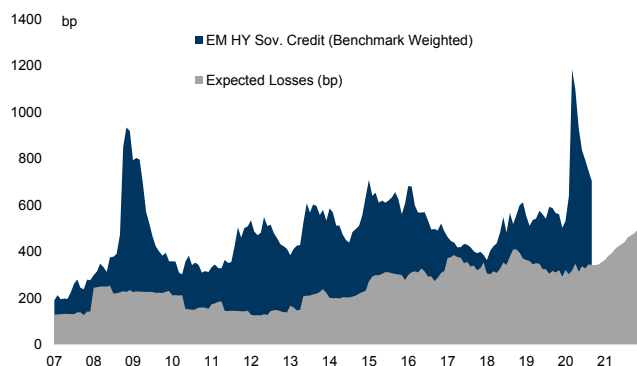


Source: Bloomberg, Goldman Sachs Global Investment Research

Third, while our model projects that expected losses will rise, higher EM HY credit spreads offer a generous compensation for this, in our view. We can translate the expected investor losses (we use the percentile rank of the probability of default times the modeled NPV haircuts) into basis points by running a simple regression of historical expected investor losses on EM HY sovereign credit spreads (on an index level). The results, shown in [Exhibit 38](#), indicate that the current EM HY spread level offers a generous cushion for the projected increase in expected losses.

[Exhibit 39](#) shows a similar picture. Here, we compare the risk premium (i.e., the difference between spreads and expected investor losses) that would have been inferred from the default probability and NPV haircut model, with the market-related risk premium that we obtained from our [EM sovereign credit model](#) (here the expected losses are a function of the macro fundamentals used to model spreads on a monthly frequency). Interestingly, the exhibit shows that despite using very different frameworks, the conclusion is similar: the risk premium in EM HY credit is currently at the higher end of the historical range (~90th percentile for both), and has been rising in recent years. Of course, if one includes the forward-looking estimates of expected investor losses based on IMF forecasts for public debt and current account developments in the default model (which is based on 3-year trends), the risk premium inferred from the current level of spreads falls significantly.

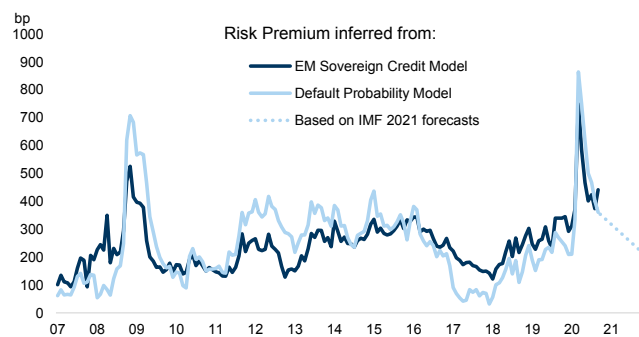
Exhibit 38: Higher spreads in EM HY have compensated for rising expected investor losses



Expected Losses = Probability of Default x Loss Given Default. Here, we use the percentile rank of the default probability model as a proxy for the probability of default, interacted with the model-implied NPV haircuts. We translate the expected losses from percentages to basis points through a simple OLS regression of EM HY spreads on Expected Losses over time.

Source: Ilzetki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), World Bank, IMF, Haver Analytics, Bloomberg, Datastream, Moody's, Goldman Sachs Global Investment Research

Exhibit 39: Various frameworks point to a relatively high risk premium for EM HY sovereign credit



Risk premium = Spreads - Expected Losses

Source: Ilzetki, Reinhart and Rogoff (2016), Cruces and Trebesch (2014), Catao and Mano (2015), World Bank, IMF, Haver Analytics, Bloomberg, Datastream, Moody's, Goldman Sachs Global Investment Research

Appendix

Data Sources

Variable	Source	Frequency
Defaults	2014 data update from Cruces, J., & Trebesch, C. (2013). Sovereign Defaults: The Price of Haircuts. <i>American Economic Journal: Macroeconomics</i> , 5, 85-117; Catao, L., & Mano, R. C. (2015). Default Premium. <i>IMF Working Paper</i> , Moody's, Goldman Sachs Global Investment Research	Annual
Spread Distress	Datastream, Bloomberg, Haver Analytics	Monthly
Haircuts	2014 data update from Cruces, J., & Trebesch, C. (2013). Sovereign Defaults: The Price of Haircuts. <i>American Economic Journal: Macroeconomics</i> , 5, 85-117; Catao, L., & Mano, R. C. (2015). Default Premium. <i>IMF Working Paper</i> , Moody's, Goldman Sachs Global Investment Research	Annual
Years since last default	2014 data update from Cruces, J., & Trebesch, C. (2013). Sovereign Defaults: The Price of Haircuts. <i>American Economic Journal: Macroeconomics</i> , 5, 85-117; Catao, L., & Mano, R. C. (2015). Default Premium. <i>IMF Working Paper</i> , Moody's, Goldman Sachs Global Investment Research	Annual
FX Reserves (% GDP)	IMF, IFS, Haver Analytics	Annual and Monthly
Current Account (% GDP)	IMF, respective countries' central banks, Haver Analytics	Annual and Quarterly
Short-term External Debt (%GDP)	World Bank, IIF, IMF, respective countries' central banks, Haver Analytics	Annual
Central Government Debt (% GDP)	World Bank, IMF, Haver Analytics	Annual
Total External Debt (%GDP)	World Bank, IMF, respective countries' central banks, Haver Analytics	Annual and Quarterly
Public External Debt (%GDP)	World Bank, IMF, Haver Analytics	Annual
Public Debt (%GDP)	IMF, respective countries' central banks, Haver Analytics	Annual
External Debt (%GNI)	World Bank, Haver Analytics	Annual
GDP per capita	IMF, Haver Analytics	Annual
GDP	IMF, Haver Analytics	Annual
Bayesian Corruption Index	Standaert, S. (2015). Divining the Level of Corruption: a Bayesian State Space Approach, <i>Journal of Comparative Economics</i> , 43 (3) 782-803.□	Annual
Inflation (% yoy)	IMF, Haver Analytics	Annual
Real GDP growth	IMF, IFS, Haver Analytics	Annual
Primary balance (%GDP)	IMF, Haver Analytics	Annual
REER	Bruegel, JP Morgan, Haver Analytics	Annual and Monthly
Dollarized deposits	Respective countries' central banks, Haver Analytics	Monthly
Rating	Moody's, S&P, Fitch	Monthly
Global CAI	Goldman Sachs Global Investment Research	Monthly
US 10y rates	Federal Reserve Board, Haver Analytics	Daily
Brent Crude oil price	Financial Times, Haver Analytics	Daily
VIX	Wall Street Journal, Haver Analytics	Daily
FX regime	Ilzetzi, E., Reinhart, C. H., & Rogoff, K. S. (2016). Exchange Arrangements Entering the 21st Century: Which Anchor Will Hold?	Annual

Bibliography

Aguiar, M., & Gopinath, G. (2006). Defaultable Debt, Interest Rates and the Current Account. *Journal of International Economics*, 69, 64-83.

- Asomuna, T., Niepelt, D., & Ranciere, R. (2017). Sovereign Bond Prices, Haircuts and Maturity. *NBER Working Paper Series*, 23864.
- Borensztein, E., & Panizza, U. (2009). The Costs of Sovereign Default. *IMF Staff Papers*, 56(4), 683-741.
- Catao, L., & Kapur, S. (2006). Volatility and the Debt-Intolerance Paradox. *IMF Staff Papers*, 53, 195-218.
- Catao, L., & Mano, R. C. (2015). Default Premium. *IMF Working Paper*.
- Cole, H., & Kehoe, T. (1998). Models of Sovereign Debt: Partial versus General Reputations. *International Economic Review*, 39, 55-70.
- Cruces, J., & Trebesch, C. (2013). Sovereign Defaults: The Price of Haircuts. *American Economic Journal: Macroeconomics*, 5, 85-117.
- Eaton, J., & Gersovitz, M. (1981). Debt with Potential Repudiation: Theoretical. *Review of Economic Studies*, 48, 289-309.
- Eichengreen, B., & Portes, R. (1996). Debt and default in the 1930s: Causes and Consequences. *European Economic Review*, 30.
- Esteves, R. P. (2007). Quis custodiet quem? Sovereign Debt and Bondholders'. *University of Oxford Department of Economics Discussion Paper*.
- Gelos, R. G., Sahay, R., & Sandleris, G. (2004). Sovereign borrowing by developing countries: What determines market access? *IMF Working Paper*, 83.
- Gennaioli, N., Martin, A., & Rossi, S. (2010). Sovereign Default, Domestic Banks and Financial Institutions. *CEPR Discussion Papers*, 7955.
- Ilzetzki, E., Reinhart, C. H., & Rogoff, K. S. (2016). Exchange Arrangements Entering the 21st Century: Which Anchor Will Hold?
- Manasse, P., & Roubini, N. (2005). "Rules of thumb" for sovereign debt crises. *IMF Working paper*.
- Meyer, J., Reinhart, C. M., & Trebesch, C. (2019). Sovereign Bonds since Waterloo. *NBER Working Paper*, 25543.
- Moody's. (2020, August). Sovereign Defaults Series: The causes of sovereign defaults.
- Panizza, U., Sturzenegger, F., & Zettelmeyer, J. (2009). The Economics and Law of Sovereign Debt and Default. *Journal of Economic Literature*, 47, 651-98.
- Paoli, B. D., Hoggarth, G., & Saporta, V. (2009). Output costs of sovereign crises: some empirical estimates. *Bank of England Working Paper*, 362.
- Reinhart, C., Rogoff, K., & Savastano, M. (2003). Debt Intolerance. *Brookings Papers on Economic Activity*, 1-70.
- Richmond, C., & Dias, D. A. (2009). Duration of Capital Market Exclusion: An Empirical Investigation.

- Rose, A. K. (2005). One Reason Countries Pay their Debts: Renegotiation and International Trade. *Journal of Development Economics*, 77, 189–206.
- Sandleris, G. (2008). Sovereign defaults: Information, investment and credit. *Journal of International Economics*, 76, 267-275.
- Standaert, S. (2015). Divining the Level of Corruption: a Bayesian State Space Approach, *Journal of Comparative Economics*, 43 (3) 782-803.
- Sturzenegger, F., & Zettelmeyer, J. (2006). *Debt Defaults and Lessons from a Decade of Crises*. Cambridge: MIT Press.
- Sturzenegger, F., & Zettelmeyer, J. (2008). Haircuts: Estimating Investor Losses in Sovereign Debt Restructurings. *Journal of International Money and*, 27, 780-805.
- Trebesch, C., Papaioannou, M. G., & Das, U. S. (2012). Sovereign Debt Restructurings 1950-2010 : Literature Survey, Data, and Stylized Facts. *IMF Working Papers*.

Disclosure Appendix

Reg AC

We, Sara Grut, Kamakshya Trivedi and Teresa Alves, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage.

Analyst compensation: Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with subject company, public appearances and trading securities held by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of the Global Investment Research Division of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Instruction 598 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Instruction 598, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** Goldman Sachs Canada Inc. is an affiliate of The Goldman Sachs Group Inc. and therefore is included in the company specific disclosures relating to Goldman Sachs (as defined above). Goldman Sachs Canada Inc. has approved of, and agreed to take responsibility for, this research report in Canada if and to the extent that Goldman Sachs Canada Inc. disseminates this research report to its clients. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 951-A, Rational House, Appasaheb Marathe Marg, Prabhadevi, Mumbai 400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan and Type II Financial Instruments Firms Association. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to

any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

The Global Investment Research Division of Goldman Sachs produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Ombudsman Goldman Sachs Brasil: 0800 727 5764 and / or ouvidoriagoldmansachs@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Ouvidoria Goldman Sachs Brasil: 0800 727 5764 e/ou ouvidoriagoldmansachs@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by either Goldman Sachs Canada Inc. or Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom and European Union.

European Union: Goldman Sachs International authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, has approved this research in connection with its distribution in the European Union and United Kingdom.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by our Global Investment Research Division. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees, will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and

https://www.fiadocumentation.org/fia/regulatory-disclosures_1/fia-uniform-futures-and-options-on-futures-risk-disclosures-booklet-pdf-version-2018.

Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by the Global Investment Research division of GS may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2020 Goldman Sachs.

No part of this material may be (i) copied, photocopied or duplicated in any form by any means or (ii) redistributed without the prior written

TOP^{of} MIND

AMERICAS FIRST



The Trump Administration kicked off 2026 with a new, more forceful foreign policy, driven by the so-called “Donroe Doctrine”. So, what could “America First” evolving to “Americas First” mean for economies, assets, and geopolitical risk? Trump’s former Special Envoy for Latin America, Mauricio Claver-Carone, and Johns Hopkins SAIS’ Hal Brands discuss the motives behind the Donroe Doctrine and agree it portends more action in the Western Hemisphere. GS’ Alberto Ramos then explores the implications for regional economies, and GS’ Neil Mehta and Daan Struyven dig into the implications for the asset at the center of recent events: oil. But with perhaps the most profound implication being the rise in geopolitical risk, we ask the Centre for Liberal Strategies’ Ivan Krastev what’s ahead for US-Europe relations

(a rocky period, but not a divorce) and GS’ Christian Mueller-Glissmann how to protect portfolios (diversify, don’t time), with GS’ Kamakshya Trivedi also seeing value in a surprisingly resilient corner of the market: EM assets.



The prospect of a fundamental shift in US policy is more real now than at any point in the last 70 years.

- Hal Brands

President Trump has long believed that the US can’t be the preeminent global power if it’s not the preeminent regional power.

- Mauricio Claver-Carone

The [US-Europe] relationship looks to be headed for a rocky period, but not a divorce.

- Ivan Krastev



WHAT’S INSIDE

INTERVIEWS WITH:

Hal Brands, Professor, Johns Hopkins School of Advanced International Studies, Senior Fellow, AEI

Mauricio Claver-Carone, Managing Partner, LARA Fund, Former US Special Envoy for Latin America and Senior Director for Western Hemisphere Affairs at the NSC in the Trump Administration

Ivan Krastev, Chair, Centre for Liberal Strategies in Sofia

VENEZUELA: WHAT’S NEXT?

Alberto Ramos, GS LatAm Economics Research

Q&A ON IMPLICATIONS FOR VENEZUELAN OIL

Neil Mehta, GS Natural Resources Equity Research, and Daan Struyven, GS Commodities Research

WHEN GEOPOLITICS MEETS PORTFOLIOS

Christian Mueller-Glissmann, GS Asset Allocation Research

EM ASSETS: CONTINUED RESILIENCE

Kamakshya Trivedi, GS FX and EM Research

Q&A ON DEFENSE SECTOR GEOPOLITICAL RISK

Noah Poponak and Sam Burgess, GS Equity Research ...AND MORE

Allison Nathan | allison.nathan@gs.com

Jenny Grimberg | jenny.grimberg@gs.com

Ashley Rhodes | ashley.rhodes@gs.com

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S. This report is intended for distribution to GS institutional clients only.

The Goldman Sachs Group, Inc.

Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

- We [now expect](#) two 25bp cuts in Jun and Sept (vs. Mar and Jun previously) to a terminal rate range of 3-3.25%.
- We [recently lowered](#) our 12m US recession probability to 20% (vs. 30%) partly owing to early signs of labor market stabilization.

Datapoints/trends we're focused on

- US growth; we [expect](#) somewhat stronger growth of 2.5% in 2026 (Q4/Q4) as the tariff drag gives way to a fiscal boost.
- US government shutdown, the [odds](#) of which have spiked.
- DoJ investigation into Fed Chair Powell, which we [don't expect](#) to impact Fed policy for now, but we [find](#) that a less independent Fed would put upward pressure on inflation.
- [AI job apocalypse](#) concerns, which we [think](#) are overblown.

Technological change: a key driver of job growth

Employment by new and pre-existing occupations, millions



Source: Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

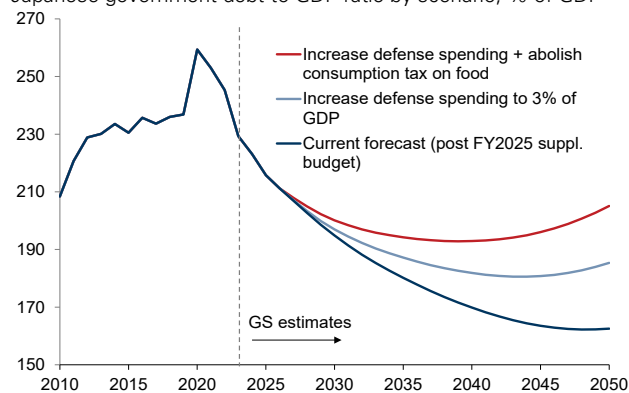
- No major changes in views.

Datapoints/trends we're focused on

- Japanese growth; we [expect](#) still-steady growth of 0.8% in 2026 (yoy) led by domestic demand, but see external demand slowing slightly amid higher Japan-China tensions.
- BoJ policy; we [expect](#) the BoJ to hike at a semiannual pace, with the next hike in July, to a policy rate of 1.5% by mid-2027, though the timing of the next hike is uncertain.
- Japanese fiscal fears, which have risen on the back of the [looming Lower House election](#) (February 8), with all political parties advocating for a [consumption tax reduction](#).

Japan: a potential undermining of fiscal health ahead

Japanese government debt-to-GDP ratio by scenario, % of GDP



Source: Ministry of Finance, Cabinet Office, Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

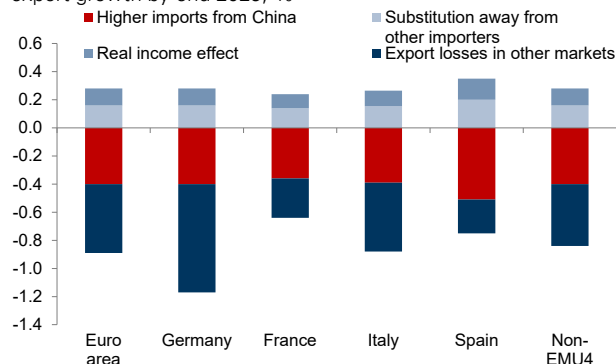
- No major changes in views.

Datapoints/trends we're focused on

- EA growth; we [expect](#) real GDP growth of 1.2% in 2026 (yoy) reflecting real income gains and a fading fiscal drag but growing [headwinds](#) from increased competition with China.
- UK economy, for which we [expect](#) another mixed year in 2026 characterized by trend-like growth and a further rise in unemployment but much lower inflation and more BoE cuts.
- Trade policy; we [see](#) only limited downside to European growth and limited upside to inflation if the US-EU trade deal implementation remains stalled.

Europe: a meaningful drag from China Shock 2.0

Estimated country-level real GDP impact of higher Chinese export growth by end-2029, %



Source: Haver Analytics, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

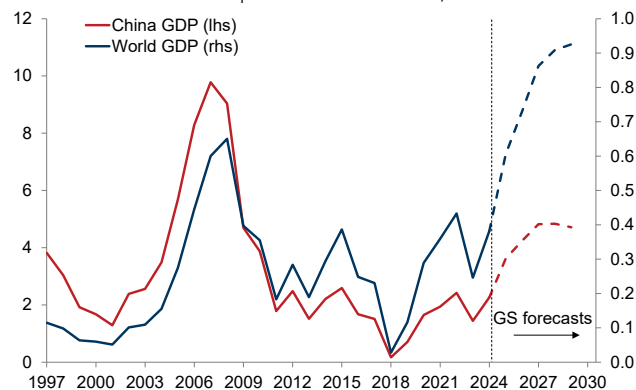
- No major changes in views.

Datapoints/trends we're focused on

- China's current account surplus, which we [expect](#) to rise further over the next several years to almost 1% of global GDP, the [biggest of any country in recorded history](#), owing to China's manufacturing strength and weak domestic demand.
- China policy easing; we [expect](#) more easing this year, including two 10bp rate cuts, which should support our above-consensus growth and inflation forecasts.
- EM policy normalization, which we [expect](#) to continue amid further inflation normalization, including in [Brazil](#) and [CEEMEA](#).

China current account surplus: rising to record levels

China current account surplus as a share of GDP, %



Source: IMF, Haver Analytics, Goldman Sachs GIR.

Americas First

The Trump Administration kicked off 2026 with a new, more forceful foreign policy, driven by the so-called “Donroe Doctrine”—a corollary to the Monroe Doctrine that aims to expand US influence and control in the Western Hemisphere (see pg. 8). What the evolution of “America First” to “Americas First”—and potentially beyond—means for economies, assets, and geopolitical risk more broadly is Top of Mind.

We first turn to Mauricio Claver-Carone, President Trump’s former Special Envoy for Latin America and Senior Director for Western Hemisphere Affairs at the NSC, and Johns Hopkins School of Advanced International Studies’ Hal Brands for more insight into the Donroe Doctrine and the Trump Administration’s potential next moves.

Claver-Carone explains that the Donroe Doctrine reflects Trump’s long-held belief that “the US can’t be the preeminent global power if it’s not the preeminent regional power” (see pg. 9). This belief is driving Trump to reassert US dominance in Latin America with the overarching goal of strengthening US security—namely, national security and, importantly, energy security, which Claver-Carone says are closely linked. He sees no contradiction between this and Trump’s “America First” policy, arguing that “Americas First” is its natural extension.

Brands, for his part, sees two motivating factors behind the Donroe Doctrine, which he characterizes as “a reversion to an earlier style of statecraft”: the US’ desire to consolidate its position in the Western Hemisphere amid a new era of great power rivalry, and Trump’s desire to increase America’s control of hemispheric resources (think oil and territory).

But Claver-Carone and Brands agree: the Donroe Doctrine portends more action in the Western Hemisphere ahead. Claver-Carone expects the Trump Administration to continue pursuing “practical partnerships” with Latin American countries. And Brands underscores the importance of the Greenland issue (see pg. 14), suggesting that it likely won’t be easily resolved, though he also argues that “we shouldn’t view the Donroe Doctrine as a constraint on action elsewhere” (e.g. Iran).

So, what could this all mean for regional economies? Alberto Ramos, GS Head of LatAm Economics Research, doesn’t expect major economic spillovers from the developments in Venezuela to the rest of Latin America given their minimal linkages owing to Venezuela’s strikingly prolonged and deep depression and hyperinflation. He sees similarly limited political implications as voters in several Latin American countries (Peru, Brazil, and Colombia) are set to head to the polls later this year.

And what about exposed assets? Ramos recounts the many uncertainties facing Venezuela’s defaulted bonds amid potential debt restructuring. Neil Mehta, GS Head of North American Natural Resources Equity Research, and Daan Struyven, Co-head of Global Commodities Research, then dig into the global asset at the center of the recent developments—oil. Struyven expects the Administration’s plan to revitalize Venezuela’s

beleaguered oil sector to result in a modest increase in Venezuelan oil production and a modestly negative impact on oil prices over the next several years, with more significant oil price impacts over the longer run as production ramps up further (see pg. 11 for a map of Venezuelan oil assets). And Mehta sees select US majors, refiners, and global service companies well-positioned to benefit from this opportunity, though he finds that it is already largely priced in.

But the most profound and potentially lasting implication of the recent developments is the rise in geopolitical risk (see pg. 18). We speak with the Centre for Liberal Strategies’ Ivan Krastev about the future of the US-Europe relationship, which he describes as at a “turning point but not a breaking point” given Europe’s security, economic, and technological dependence on the US as well as American political realities that may constrain US policy. But he expects a rocky period in the relationship ahead given that trust has been broken, not just because of Greenland, but also because of US actions vis-à-vis Ukraine and the US’ broader treatment of Europe.

And Brands sees even broader geopolitical implications, arguing that US designs on Greenland may test the global territorial status quo that has prevailed post WWII. He warns that the US deciding to play a fundamentally different role in the world than it has in the past 70 years could lead to significant changes in everything from nuclear proliferation, to the security of seaborne transit and democracy, to Dollar dominance.

So, what does this elevated geopolitical risk environment mean for global investors? Christian Mueller-Glissmann, GS Head of Asset Allocation Research, offers guidance on how to protect portfolios against this risk. He argues that rather than trying to time geopolitical shocks, investors’ first line of defense should be diversification. He sees value in several diversification strategies: turning to bonds and other safe havens like the Swiss Franc, increasing allocations to assets that could benefit from geopolitical shocks (like gold and the defense sector, which GS defense analysts Noah Poponak and Sam Burgess believe has not yet fully priced in elevated geopolitical risk), limiting direct and indirect exposures, and hedging with options.

Kamakshya Trivedi, GS Chief FX & EM Strategist, also sees value in a surprisingly resilient corner of the market: EM assets, which have held up despite elevated geopolitical risk, which typically weighs on EM relative performance. He expects this resilience to continue this year, with EM assets’ role as diversifiers further increasing their value proposition. In this context, Trivedi believes frontier assets are also worth a look.

Allison Nathan, Editor

Email: allison.nathan@gs.com
Tel: 212-357-7504
Goldman Sachs & Co. LLC



Interview with Hal Brands

Hal Brands is Henry A. Kissinger Distinguished Professor of Global Affairs at the Johns Hopkins School of Advanced International Studies and Senior Fellow at the American Enterprise Institute. Below, he discusses the US' renewed assertiveness in the Western Hemisphere, which he argues could have major repercussions for the global order.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: In 2024, you argued that a second Trump presidency would feature a “revitalized Monroe Doctrine”. What factors are motivating the US’ more forceful posture in the Western Hemisphere?

Hal Brands: I see two motivating factors—one structural, one personal. The structural factor is the US’ historic

pattern of attempting to consolidate its position in the Western Hemisphere when the rest of the world is falling apart. During both World Wars and the Cold War, the US worked hard to try to ensure its dominance in the Western Hemisphere. So, it’s not surprising that the US is now doing the same as the world enters a new era of great power rivalry.

The personal factor revolves around President Trump’s long-held interest in renewing the Monroe Doctrine to increase the US’ control of hemispheric resources, like oil, and territory, specifically from Panama, Canada, and Greenland. The President is also intensely focused on tangible threats to American sovereignty and security rooted in hemispheric issues like drug trafficking and migration. Putting all this together, it’s not surprising that this Administration has focused on reconsolidating US primacy in the Western Hemisphere.

What is a bit more surprising is how energetic the Administration has been in doing so. The Venezuela intervention has received substantial attention, but it’s just the capstone of a much bigger and broader campaign that has featured diplomatic pressure on Colombia and Panama, economic support to allies like the Milei government in Argentina, partnerships with El Salvador’s Bukele government and other ideological fellow travelers, lethal strikes on drug boats, and an array of other military, diplomatic, and economic initiatives. In many ways, this has been the centerpiece of Trump’s foreign policy so far.

Allison Nathan: So, it’s right to think that oil was a key motivation for the Venezuela actions?

Hal Brands: Resource control is undoubtedly at the forefront of the President’s calculus. He has long associated it with national power, often remarking that the primary mistake in the American invasion of Iraq was not “taking the oil”. In describing US aims in Venezuela since the intervention, Trump has said very little about democracy and good governance, and even relatively little about drug trafficking. He talks a lot about the US exerting greater control over the disposition and sale of Venezuelan oil and the desire for greater openness of the Venezuelan oil industry to the US oil majors. Others in the Administration may value different things in this Trump Corollary to the Monroe Doctrine, but for Trump himself, the

primary issues are essentially about economic gain and a neo-mercantilist approach to international economics.

Allison Nathan: Is that what differentiates the “Donroe Doctrine” from the Monroe Doctrine?

Hal Brands: The Monroe Doctrine’s meaning has evolved over time. It was initially about preventing the reimposition of colonial control by European powers. During the 20th century, it became about keeping fascist and communist regimes out of the hemisphere. In some ways, the “Donroe Doctrine” is a throwback—it is a reversion to an earlier style of statecraft that is much less apologetic about the desire to control the resources of weaker countries and much more willing to talk about outright territorial acquisition and other ideas that went out of style with the end of the age of imperialism. It’s sometimes said that Trump is trying to revive a 19th century style of foreign policy. I agree that he would likely feel quite at home if he were presiding over some of the US’ interventions in Latin America in the late 19th and early 20th centuries. Indeed, the tactics he has used in Venezuela—overthrowing a government and exerting control of the country’s trade revenues—aren’t that different from what the US did in places like the Dominican Republic well over a century ago.

Allison Nathan: With that context, why do you think the Administration chose to leave the rest of the Maduro regime in place?

Hal Brands: Trump is seeking maximum leverage and gain with minimum long-term investment of resources and political capital. He burst onto the domestic political scene 10 years ago by arguing that the US had made terrible mistakes by undertaking nation-building missions in Afghanistan and Iraq. And flying Maria Corina Machado or Edmundo Gonzalez into Caracas in the hours after Maduro’s apprehension probably would have incited a lot of instability by putting them in conflict with the remnants of the Maduro regime. So, the mechanism Trump has settled on is a 21st century version of gunboat diplomacy, where the US will work through the existing authorities while threatening them with punishment if they don’t do the Administration’s bidding.

Allison Nathan: What do these developments mean for other countries in the region, many of which are slated to hold elections this year?

Hal Brands: They serve notice to any vulnerable anti-American regimes playing strategic footsie with China and Russia that they will likely face greater US pressure. Cuba might be the best example. That said, I’m skeptical that the US would undertake a large-scale military intervention in Cuba because a collapse of that regime might produce a flood of refugees headed for the US. More broadly, countries from Colombia to Mexico will likely feel pressure to act with greater urgency on

issues related to narcotrafficking, which we're already seeing evidence of in the discussions the US and Mexico appear to be having about a potential role for the US military in going after cartels operating in Mexico, as well as some of Trump's comments about Colombia.

The domestic political implications of these shifts will vary given the array of political cultures and levels of susceptibility to US interests throughout the region. In Brazil, the political culture to some degree has long been interwoven with resistance to US influence in Latin America, even in periods when conservative military governments in Brazil worked very closely with the US. So, I wouldn't be shocked if US efforts backfire to some degree in the next Brazilian election. But it could go the other way there and/or in other countries, where the US either has more influence or where a homegrown swing to the right or in the direction of neo-populist politics occurs.

Allison Nathan: What lessons do you think China and Russia are taking from the recent US actions, and how are they shifting their calculus in the region?

Hal Brands: Beijing and Moscow have run up hard against the reality of US hard power in the Western Hemisphere. Whether it's the US Navy and Coast Guard seizing shadow fleet tankers, the US forcibly deposing Maduro, or the Pentagon shooting up drug boats in the Caribbean, neither China nor Russia can do much to contest the application of American military power in the Western Hemisphere. That said, China's influence in the region in particular is multifaceted and deeply embedded in everything from trade relationships to the diffusion of technology to physical and digital infrastructure, all of which will persist to some degree.

It's important to note that China's influence is not simply imposed on Latin American countries, but a response to their needs. Big gaps in physical and digital infrastructure exist in the region, which China is well-placed to address. So, Beijing will play the long game and continue to attempt to win influence in the region through economic, political, and, in some cases, low profile security relationships that will focus more on internal security than traditional military functions. If anything, the recent developments likely mark the start of an even more intense phase of competition for influence in the region.

Allison Nathan: How does Greenland fit into Trump's plan, and what might that suggest about his next steps there?

Hal Brands: Greenland is both a hemispheric and global issue. It's a hemispheric issue in the sense that the Trump Administration views consolidating US control of Greenland as part of the larger effort to revive the Monroe Doctrine. And they can point to historic examples of the US increasing its influence in Greenland as part of a strategy of hemispheric defense, as President Roosevelt did in the run-up to US entry into WWII.

It's also a global issue, for two reasons. First, Greenland powerfully affects the fate of NATO and the transatlantic community; it's hard to overstate the degree of outright alarm that Canada and many European countries feel about US designs on the territory. Second, Greenland is the best test of whether the US will seek changes to the global territorial status quo. That's an explosive issue because China and Russia are attempting to do much the same thing in the South China

Sea/Himalayas and Ukraine, respectively. So, a world in which the US also aims to redraw borders through coercion or even force is a world in which the three most powerful countries are all violently or coercively disrupting the territorial status quo. That's something we haven't seen since the 1930s and could be deeply corrosive to the post-1945 global order.

Allison Nathan: So, recent US actions have helped further the ambitions of China and Russia in some respects?

Hal Brands: Both countries were undoubtedly unhappy to see Maduro go, especially at the hands of ruthlessly efficient US military power. But recent developments aren't entirely bad news for Beijing and Moscow. A world in which international law and norms matter less, and great powers are free to do as they like, especially in their own backyards, is a world in which Chinese and Russian leaders would feel comfortable. In fact, that's exactly the world they aim to bring about. And if an intensification of US military activity in the Western Hemisphere pulls resources away from other regions, as was the case when Trump redeployed an aircraft carrier covering the Middle East and Europe from the Mediterranean to the Caribbean, that's even better news for them.

Allison Nathan: How does Iran fit in here?

Hal Brands: Iran is a key reason why we should be skeptical of the idea that the world is rapidly transitioning into a spheres of influence arrangement. Trump wants an American sphere of influence in the Western Hemisphere, but he also likes to maintain unhindered freedom of action globally when he views it as advantageous to him. This was clear in the recently-released national security strategy. This Administration has also conducted two major military interventions in the Middle East and threatened others. We shouldn't view the Donroe Doctrine as a constraint on action elsewhere.

Allison Nathan: Amid all this, what geopolitical risks should companies and investors be most focused on?

Hal Brands: The mega risk is the potential for the US to decide to play a fundamentally different role in the world. We have lived for so long in the post-1945 international order in which US power was wielded to secure strong prohibitions on forcible conquest, freedom of navigation to support and propel the global economy, and protection of human rights and the spread of democracy, that it's difficult to conceive of a world without that leadership. Such a world would likely see more countries pursuing nuclear weapons, greater disorder on the high seas that would impact global trade and finance, a challenge to Dollar dominance, and a global retreat of democratic values.

30 years ago, I would have said there was little chance of such a shift. 15 years ago, I would have said signs of retrenchment-minded inclinations in the US foreign policy debate had grown. We now have a US president who often expresses a desire for fundamental change in US foreign policy and the way the US engages in the global economy, even if the policies don't always match the rhetoric. So, the prospect of a fundamental shift in US policy is more real now than at any point in the last 70 years. The stakes are high, and the choices the US makes in the coming years will not only determine its own role on the world stage, but also the trajectory of the international order.

Interview with Mauricio Claver-Carone

Mauricio Claver-Carone is Managing Partner of the Latin America Real Assets Opportunity Fund. He served as US Special Envoy for Latin America and Senior Director for Western Hemisphere Affairs at the National Security Council in the Trump Administration. Below, he argues that President Trump views US dominance in the Western Hemisphere as essential for global preeminence, and that “Americas First” is a natural extension of “America First”.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Jenny Grimberg: You played a key role in foreign policy in the Western Hemisphere during both Trump Administrations. What's behind President Trump's focus on the region, and do you sense a shift in that focus from his first term?

Mauricio Claver-Carone: President Trump has long believed that the US

can't be the preeminent global power if it's not the preeminent regional power. That is the “Donroe Doctrine”. And contrary to widespread perception, this reflects the President's own thinking, not his advisors' influence. President Trump is a developer from New York—he thinks in terms of building neighborhoods. He has traveled throughout the region and knows it well. And he believes that no region in the world impacts Americans' daily lives more than Latin America and the Caribbean from a security, economic, and social perspective.

I've seen firsthand many times how focused the President has been on this, and on the US' waning influence and positioning in the region over the last two decades, especially vis-à-vis China. During periods of conflict in Afghanistan and Syria in his first term, President Trump would incredulously remark on how eagerly and easily the US sent troops to those countries yet was too scared to have any type of presence in its own hemisphere. And I remember at one point during the intensifying crisis that gripped Venezuela in 2019, the President being disturbed to learn that the US Navy had no ships in the vicinity of the Caribbean because they were all deployed to the Persian Gulf, South Asia, and the Mediterranean.

That said, given his team was distracted by the Middle East and Asia at the time, President Trump's first term was largely about facilitating a pivot from China in the region. The Administration focused on setting frameworks to, mostly through soft power, execute this pivot, which led to the creation of the America Crece Initiative to catalyze private sector investment in energy and infrastructure projects across Latin America and the Caribbean.

The second term, by contrast, is about the implementation of President Trump's vision for the region. He made that vision extraordinarily clear during his second inauguration as he harkened back to President McKinley and the golden era of American influence, projected not only through policy—the Monroe Doctrine and, later, the Roosevelt Corollary to it—but also through brick-and-mortar initiatives like the Panama Canal. And this time around, his team is aligned with that vision. Secretary of State and National Security Advisor Marco Rubio understands and cares about the region and President Trump's vision for it. Deputy Chief of Staff for Policy and Homeland

Security Advisor Stephen Miller is also focused on the region owing to his holistic approach on immigration and his view that immigration would be less of a problem for the US if Latin American countries were thriving. And, in contrast to the secretary of defense in the first Administration, who didn't believe that the Department of Defense should be involved in counternarcotics, Secretary of Defense Pete Hegseth is very involved.

Jenny Grimberg: What does the President hope to achieve by reasserting US influence in Latin America?

Mauricio Claver-Carone: The Trump Administration's actions in Latin America—from the Venezuela operation to the relationship it created with Mexico to the historic support it lent Argentina during its economic crisis last year—are based first and foremost on US security. This runs the gamut from national security in terms of halting narcotic and migrant flows and reducing terrorism risk to energy security. The Achilles heel of Latin America and the Caribbean in particular is energy. Fidel Castro always understood that, which is why he helped install Hugo Chavez and viewed Venezuela as his golden crown. And Caribbean countries' pressing need for energy led to the formation of PetroCaribe, an initiative launched by Chavez to supply Venezuelan oil to these countries on favorable financial terms, which allowed Venezuela to essentially control the Caribbean for many years.

Energy security is closely intertwined with national security, as high energy costs have been a key driver of migration throughout Central America. So, energy security is incredibly important, which is consistent with President Trump's desire to increase oil production in the Western Hemisphere. The US is an energy powerhouse in its own right, and many Latin American countries could be too with proper management. US influence can play a crucial role here. Guyana currently produces more oil than Venezuela, largely because the Guyanese government has worked hand-in-hand with US companies to develop the right policies from the get-go and leave behind the nationalist ideological strains that were embedded into the psyche of the regional population during the 20th century revolutions.

Jenny Grimberg: Does expanding US influence in Latin America conflict with the “America First” pillar of President Trump's Make America Great Again (MAGA) platform?

Mauricio Claver-Carone: Not at all. “Americas First” is the natural extension of America First as President Trump works to counter the notion of every DC think tank expert that the 21st century would be the Chinese century. The President was adamant during his inauguration that it would instead be

another American century, and potentially the greatest American century. And contrary to what many believe, the President's Latin America strategy is not about boots on the ground or nation-building. It is about practical partnerships between nations as the US shifts from a policy of non-interventionism—which served neither the local population nor the US well by leading to the rise of Chavez and China—to non-imperial expansionism, a term coined to explain the strategy of deepening US influence in the region through partnerships.

These are not political partnerships—the US is currently working with the right-leaning governments of Argentina, El Salvador, Costa Rica, and Chile as well as the left-leaning governments of Mexico and Venezuela—but practical, business-oriented relationships that benefit not only the US but also these countries, whose populations desire security and economic opportunities. The best example of that is Argentina. Many people, including myself, worried that the Trump Administration's financial support ahead of last year's midterm elections could spark a backlash against President Milei's Conservative party given that strains of Peronism still exist in Argentina. But the election results showed that most of the party's support came from the poor, who are fed up with ideologies that neither pay the bills nor put food on the table. So, the region has undergone a dynamic social shift.

Jenny Grimberg: How does the Trump Administration's decision to depose Maduro fit into that?

Mauricio Claver-Carone: Let me first clarify that the Administration didn't "depose" Maduro. This was a law enforcement operation with military support to capture an individual indicted for narcoterrorism in the US. Now, this was a particularly complicated operation because Maduro was de facto running a nation-state with a military, which showed the world that nobody is outside of US reach and sent a message of US influence, ability, and capability. But the operation also underscored the Trump Administration's preference for partnership. President Trump gave Maduro plenty of opportunities to work together; he expressly rejected every offer, which led the Administration to execute a carefully thought-out operation to capture him. That sends an important message: when President Trump offers you a deal or an opportunity, you take it.

The Trump Administration's subsequent actions in Venezuela are also a testament to its partnership preference. The US undoubtedly had the capabilities for a broader operation, but assessed that the other people in Venezuela controlling the institutions of power would want to work with the US. And so far, that assessment has proven right. In the days following the operation, the US reached a deal to purchase 30-50 million barrels of Venezuelan oil and several million barrels moved shortly after. Venezuela is also increasingly and consistently releasing political prisoners. And the US is working with the Venezuelan authorities to put the country on a path toward becoming a more stable and secure presence in the region with an open, participative process. Much remains uncertain, but I'm optimistic.

Jenny Grimberg: The million-dollar question is what President Trump will do next in Latin America and the broader Western Hemisphere. What's your view?

Mauricio Claver-Carone: Further efforts at forming and consolidating partnerships with countries in the region are likely. Upcoming elections in Brazil and Colombia will be important. Conservative candidates recently won Honduras' and Chile's presidential elections, and if right-wing candidates also prevail in Colombia and Brazil, it would result in a historic ideological consolidation for the region. But again, this isn't about ideology but whether the new leaders will choose the US first and foremost as their partner. That would give the US a historic opportunity to modernize its business relationships and commercial partnerships across Latin America.

That could start with Mexico and the upcoming USMCA renegotiation, which seems to be the path the President is heading down. But President Trump's intense focus on trade is reactive. He feels that the trade deals of the 1990s/early 2000s were all flawed, forcing him to focus on fixing them. As those issues resolve, the focus of US partnerships with regional allies could shift to making deals that ensure a US strategic presence in critical sectors like energy and infrastructure. The hope is that by the end of President Trump's term, nobody will ever again talk about China's expanding influence in the region, because it will be unquestionably clear that the US is the partner of choice for Latin America and the Caribbean.

Jenny Grimberg: As an investor in the region, what advice do you have for investors attempting to navigate Latin America's evolving geopolitical landscape?

Mauricio Claver-Carone: Investors looking to monetize the "Donroe Doctrine trade" by focusing more on Latin America should look beyond the big-ticket deals in Mexico and Brazil, which have often been walloped by currency issues, to smaller countries. The deals are smaller, but the returns are bigger as a percentage, and these countries are among the region's fastest-growing economies, offering great opportunities for investors. The secret sauce is local partners.

I would also advise against relying on multilateral development banks. As a former president of the Inter-American Development Bank, I've seen firsthand that these institutions aren't focused on finding or creating the best deals. Investors should instead focus on US domestic finance institutions like the International Development Finance Corporation (DFC), which partners with the private sector to mobilize capital for strategic investments around the world and recently secured a six-year reauthorization, to gain exposure to long-term deals in strategic sectors.

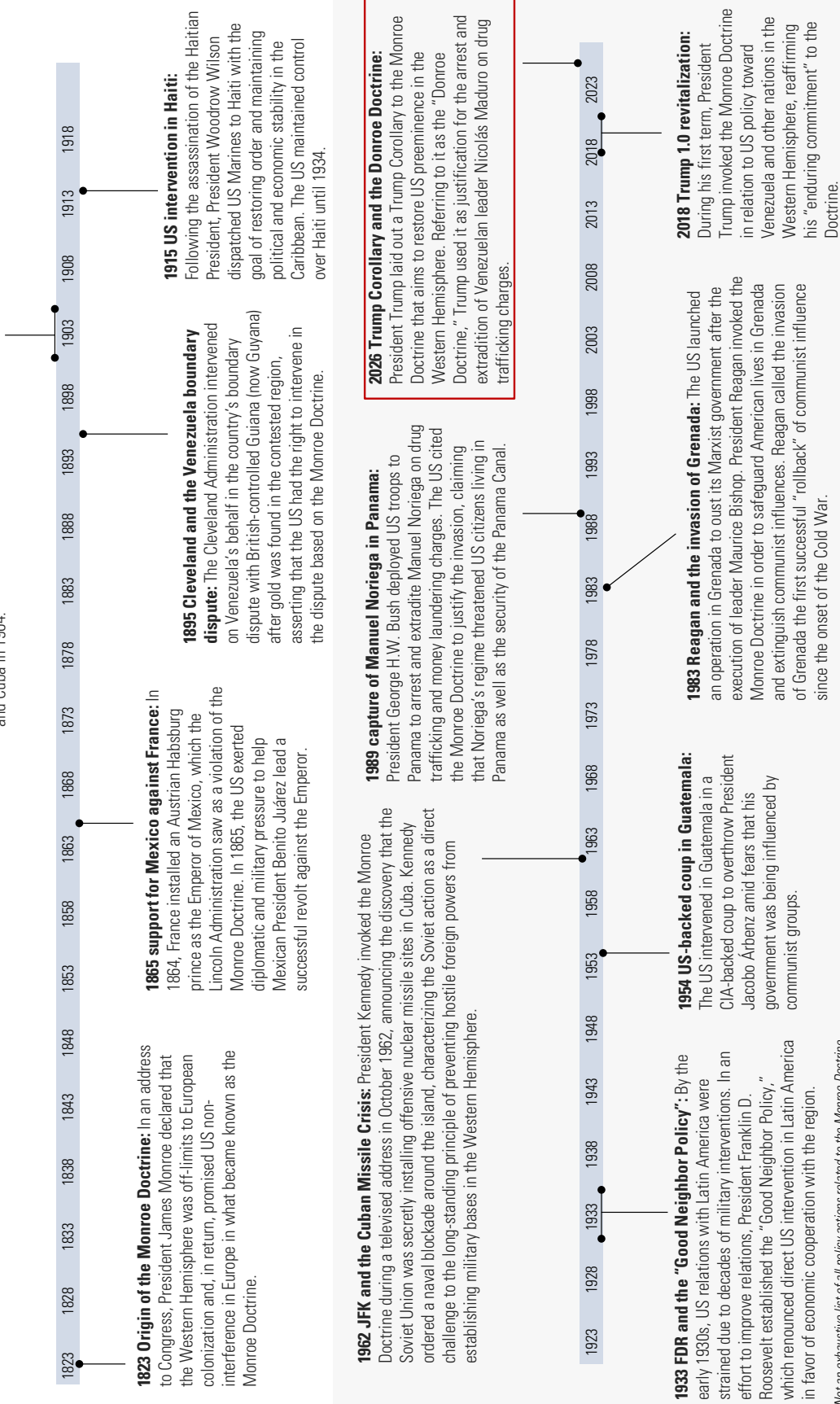
I also see significant short and medium-term investment opportunities in the energy and infrastructure space. The timing is right for these investments because the US government is focused on and geopolitically invested in Latin America and wants to encourage private investment in the region, and that focus isn't going away given President Trump's "Americas First" philosophy.

The Monroe Doctrine through the years

INVOKING THE MONROE DOCTRINE

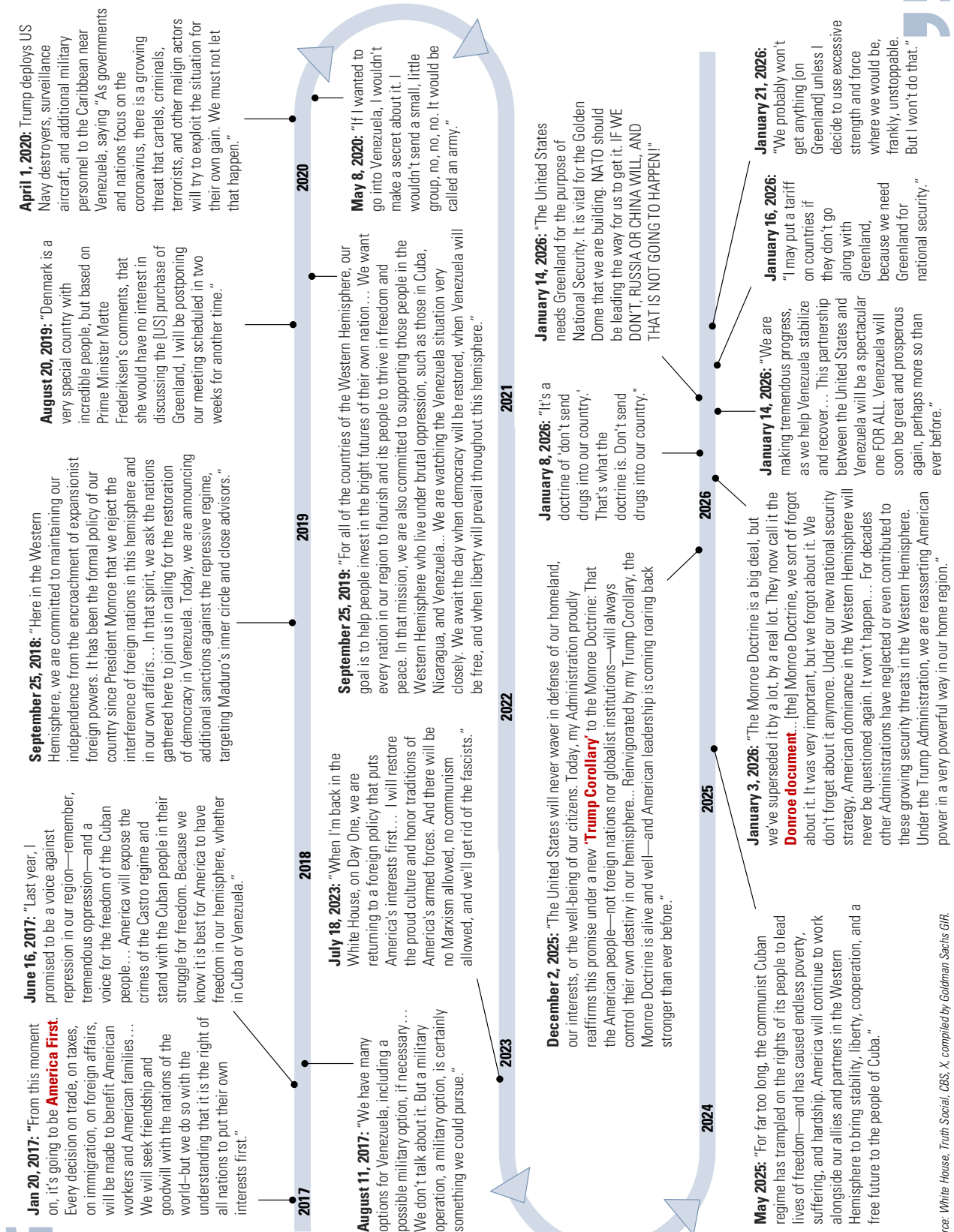
The Monroe Doctrine, first articulated by President James Monroe in 1823, declared that European powers should not interfere in the affairs of the newly independent nations of the Americas, a stance motivated by concerns over potential European re-colonization. Since its inception, US presidents have frequently invoked the Doctrine to justify various foreign policy actions across the Western Hemisphere.

1902-1904 Roosevelt Corollary: In 1902, a fleet of British, German, and Italian gunboats set up a blockade of Venezuela's ports after the nation defaulted on its debts to European lenders. President Theodore Roosevelt, as part of his foreign policy to "speak softly, but carry a big stick," issued a corollary to the Monroe Doctrine that gave the US "international police power" over the Western Hemisphere, justifying US intervention in Latin American affairs to prevent European influence. Roosevelt used this to justify US intervention in Venezuela, as well as Panama in 1903 and Cuba in 1904.



*Not an exhaustive list of all policy actions related to the Monroe Doctrine.
Source: US Department of State, National Archives, Office of the Historian.*

The Donroe Doctrine, in Trump's words



Source: White House, Truth Social, CBS, X, compiled by Goldman Sachs GIR.

Venezuela: what's next?

Alberto Ramos explores what developments in Venezuela could mean for the country's economy and the broader LatAm region

Following the US operation in Venezuela, the situation on the ground remains very fluid and the outlook complex and uncertain. The Trump Administration is now working with the remnants of the Maduro regime to stabilize the Venezuelan economy, focusing first and foremost on the battered oil sector, and eventually hold open elections.

An economic collapse of historic proportions

Venezuela has been trapped in a seemingly never-ending economic depression and hyperinflationary spiral. Large macro and financial imbalances have spawned an economic collapse of historic proportions. In fact, Venezuela has experienced one of the worst peacetime economic downfalls in modern history, with real GDP contracting far more than in the US during the Great Depression (-28%), Spain during the Spanish Civil War (-28%), and the 2009 Greece crisis (-27%). Per the IMF, in Dollar terms, Venezuelan GDP has shrunk by ~80% since 2012, with seven consecutive years of negative real GDP growth over 2014-2020 for an astonishing cumulative decline of 74.3%.

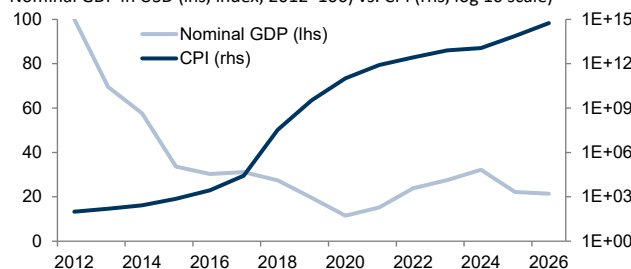
Quite telling and illustrative of the major economic collapse of the last 20 years is the dramatic shift within Venezuela's oil sector—the country's key source of foreign exchange. Oil production fell from 2.7mb/d in 2015 to 0.93mb/d in 2025, resulting in an oil-export hard-currency crunch that imposed a severe and debilitating constraint on imports. As a result, Venezuela's economy has contracted to ~\$80bn today, roughly the size of Uruguay and less than a quarter the size of Chile.

Hyperbolic inflation immiserizes

Annual inflation over the decade leading up to 2025 was an astronomical 14,544% on average—reflecting years of hyperinflation (inflation was ~130,000% in 2018). Over the same period, Venezuela's currency (the Bolivar) lost ~98% of its value. The economic hardship has helped generate a mass exodus of people over the last 16 years. According to the UNHCR, nearly 8mn people fled the country, close to a quarter of the population and one of the world's largest displacements, with most settling in nearby countries, particularly Colombia.

Venezuela has experienced hyperinflation and depression

Nominal GDP in USD (lhs, index, 2012=100) vs. CPI (rhs, log 10 scale)



Source: IMF, Goldman Sachs GIR.

Likely limited economic and financial spillovers...

It therefore comes as no surprise that the economic and financial linkages between Venezuela and the rest of LatAm have become minor. Accordingly, we do not anticipate major economic spillovers from the situation in Venezuela to the rest

of LatAm. However, if the situation on the ground were to deteriorate into widespread lawlessness and violence, we could see renewed migration flows, likely affecting neighboring countries first and foremost. Beyond migration, the situation in Venezuela could indirectly impact the region through their impact on key commodity prices, particularly oil and gold.

...and political implications

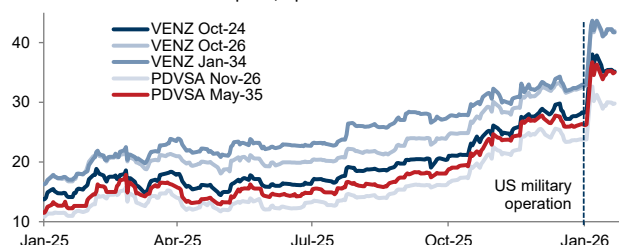
The political implications for the rest of LatAm also seem limited. The reactions of regional governments were mixed and clearly divided along political/ideological lines, but even those critical of the situation were cautious and contained in their responses. However, US actions in Venezuela could stress its relationship with Mexico, Colombia, and Brazil, especially if the US expands its focus to the regime in Cuba and/or conducts military strikes on drug cartels inside Mexico or Colombia. At this juncture, we do not expect the ongoing developments in Venezuela to impact upcoming elections in Brazil (general election in October), Colombia (congressional elections in March, presidential in May), or Peru (general elections in April).

Debt restructuring key to watch from here

Venezuela defaulted on international bonds in late 2017 due to the compounding effects of the deep economic crisis and sanctions. Defaulted debt is estimated at north of \$150bn, including around \$60bn (face value) in defaulted bonds (sovereign and state oil company PDVSA), other PDVSA obligations (e.g., suppliers' debt), bilateral loans (e.g., with China, Russia), and multilateral debt, with significant unpaid interest as well as legal claims from expropriation cases compounding the total since the 2017 default.

Venezuela bond prices rallied following the US military operation

Venezuela and PDVSA bond price, bp



Source: Bloomberg, Goldman Sachs GIR.

Venezuela bond prices rallied in 4Q25 on expectations that rising US pressure and military buildup would eventually lead to regime change. And the initial market reaction to the January 3 US airstrikes was positive. Indeed, defaulted Venezuelan bond prices rallied 8-9pts (including PDVSA) on January 5 on expectations of a brighter outlook for Venezuela's economy and oil sector as well as the potential for debt restructuring.

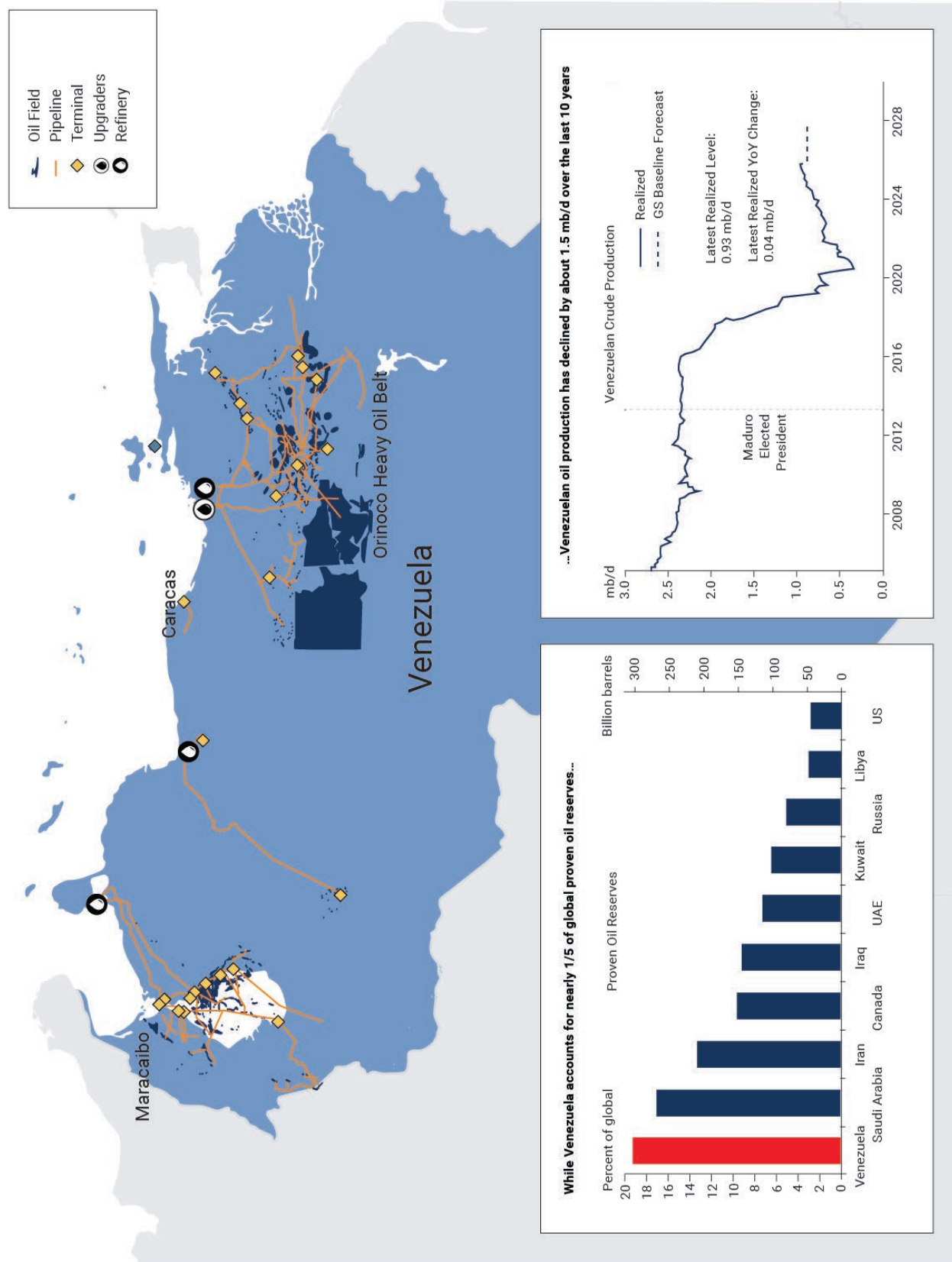
However, bond prices have since given back some of these gains due to the complex and still highly uncertain economic and political picture, the perception that debt restructuring may not be a near-term priority, President Trump's executive order protecting Venezuelan oil revenue held in US Treasury accounts, and bond valuations commensurate with lower-bound recovery levels. The market debate has now shifted to "what type of restructuring" and when and with whom creditors will eventually be negotiating.

Alberto Ramos, Head of LatAm Economics Research

Email: alberto.ramos@gs.com
Tel: 212-357-5768

Goldman Sachs & Co. LLC

Venezuela oil infrastructure, mapped out



*We use OPEC data on proven crude oil reserves as of 2024 for all the countries above except Canada, where we use Energy Institute proven reserves estimates as of 2020 (latest available).
 Sources: OPEC, Energy Institute, secondary sources, S&P Global Energy, Goldman Sachs GIR.
 Special thanks to GS Commodities Research team for charts.

Q&A on implications for Venezuelan oil

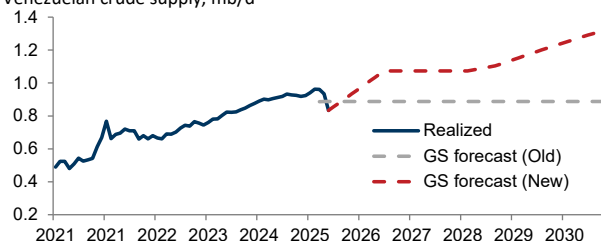
Neil Mehta and Daan Struyven answer key questions about the implications of US plans to revitalize Venezuela's oil sector¹

Q: What's the size of the opportunity for the oil industry of the US' plan to revitalize Venezuela's oil sector?

A: Venezuela accounts for roughly one-fifth of the world's proven oil reserves², yet its production is constrained at ~800-900kb/d, which is less than ~1% of global output. The Trump Administration has [called for](#) ~\$100bn of investment into the country over the next decade to rebuild infrastructure and bring production back to peak levels of ~3mb/d. We believe gains of a few hundred thousand barrels per day are achievable in the near term. Specifically, we expect Venezuelan crude production to rise by 0.2mb/d over the next 12m (from 0.8-0.9mb/d in Dec 2025) as an anticipated easing of sanctions and modest investment in existing assets help ease some operational bottlenecks. We expect production to rise by an additional 0.2mb/d by 2030 (to 1.3mb/d). But a complete restoration of the country's output to ~3mb/d would likely extend into the next decade and require significant capital investment. Some estimates suggest an initial ~\$5-10bn of capital investment over the next 2-3 years could yield an additional ~500kb/d in Venezuela, but the subsequent ~1mb/d would require an estimated ~\$70-80bn. Upstream production expansions would also require a reactivation of damaged upgraders that process heavy crude, extension of pipeline infrastructure, investments in the power grid, and continued political stability.

We expect a modest increase in Venezuelan crude supply

Venezuelan crude supply, mb/d



Source: Secondary sources, Goldman Sachs GIR.

Q: What types of energy companies are expected to benefit from this opportunity?

A: Across the energy complex, US major Chevron, Gulf Coast refiner Valero, and international-levered service company SLB are well-positioned to profit from additional Venezuela production. Specifically, Chevron is uniquely positioned as the only US major oil company to retain a foothold in Venezuela, leveraging its longstanding involvement to expand production to over ~200kb/d, although this still comprises only ~1-2% of the company's CFFO. Chevron's continued engagement in Venezuela sets it apart from many of its international counterparts that have withdrawn from the country. Chevron now [sees](#) a path to double its liftings from Venezuela in the very near term and increase its production by 50% (or ~100kb/d) over the next 18-24 months.

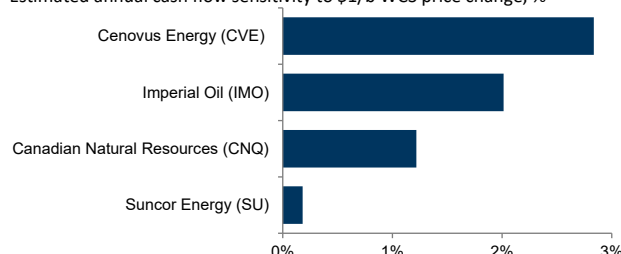
Gulf Coast refiners, including Valero, are also incremental beneficiaries as their facilities are designed to process the heavy, sour crude that Venezuela produces, and wider light-heavy crude differentials should support refining earnings. International service companies are also poised to benefit from increased activity in Venezuela, such as Halliburton and Weatherford, which have historically operated in the country. Lastly, with SLB's position as the only operating oilfield service company currently in the country, an increase in activity would drive potential upside over the long term.

Q: Conversely, what types of companies could be harmed by the revitalization of the Venezuelan oil industry?

A: On a relative basis, Canadian oil companies could be negatively impacted by higher oil production from Venezuela. In particular, Venezuelan barrels could displace Canadian heavy crude on the Gulf Coast and result in wider WTI-WCS differentials, which would reduce realized prices for Canadian producers. This includes companies like Canadian Natural, Cenovus, and Imperial Oil, which are major producers of heavy crude oil. Cenovus and Imperial are the most exposed to widening light-heavy differentials given lower relative protection on heavy crude.

Venezuelan barrels could displace Canadian heavy crude on the Gulf Coast and result in wider WTI-WCS differentials

Estimated annual cash flow sensitivity to \$1/b WCS price change, %



Source: Goldman Sachs GIR.

Q: How have US oil companies responded to these potential opportunities/risks so far?

A: Among the US Majors, Chevron has already positioned itself to take advantage of the new opportunities in Venezuela, including doubling its liftings from the country immediately and raising its production by 50% (or ~100kb/d) over the next 18-24 months. By contrast, ConocoPhillips and Exxon remain focused on recovering previous debt following the nationalization of their assets. Exxon has also publicly discussed its view that Venezuela is difficult to invest in without significant changes to the legal and commercial environment. Among oilfield services companies, SLB has discussed its ability to scale quickly as needed in the country, as it currently has operations on the ground. Halliburton and Weatherford have also commented on their ability and interest in allocating resources to Venezuela if activity and production ramp up.

Q: How much of the upside/downside of these developments is already priced into stocks?

A: Energy equities reacted sharply on the back of the initial Venezuela headlines, including the stocks of US majors, US

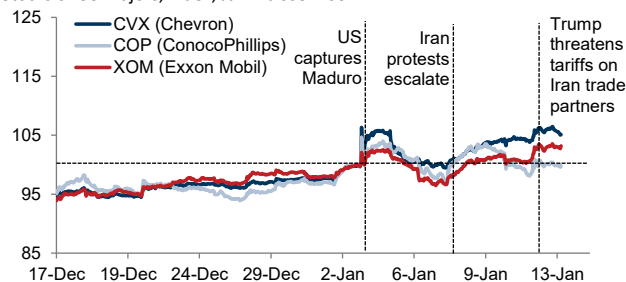
¹ The research views in this section are those of the analysts who cover the relevant areas. Equity research views are expressed by Neil Mehta, an Equity Research Analyst covering Americas Integrated Oil & Refiners. Commodities views are expressed by Daan Struyven, a Macro Research Analyst covering global commodities.

² According to OPEC data.

refiners, international service companies, and Canadian oil producers. While we believe the latest developments are now largely reflected in the equities, we continue to highlight strong fundamentals across the energy complex.

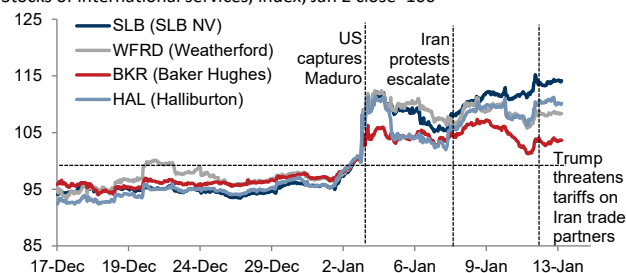
Energy equities reacted sharply on the back of the initial Venezuela headlines, including across US Majors...

Stocks of US Majors, index, Jan 2 close=100



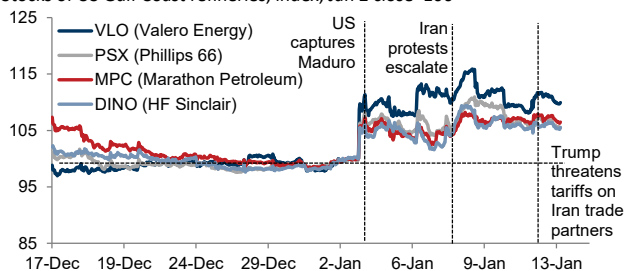
...international services...

Stocks of international services, index, Jan 2 close=100



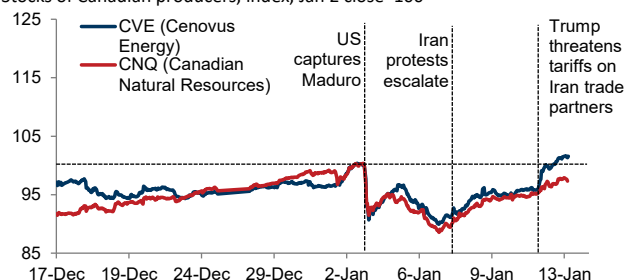
...US Gulf Coast refineries...

Stocks of US Gulf Coast refineries, index, Jan 2 close=100



...and Canadian producers

Stocks of Canadian producers, index, Jan 2 close=100



Source: Bloomberg, Goldman Sachs GIR.

Q: What, if anything, are investors underappreciating about the potential impacts of the Venezuela developments for US oil companies?

A: Given the state of Venezuela's energy infrastructure—which has deteriorated amid underinvestment—investors may be underestimating the challenges associated with sustainably ramping Venezuelan volumes. We also believe the scale of

investment required to increase their Venezuelan presence as well as the political and legal risks of doing so could deter companies from making significant new capital commitments in the country, which is a major reason we expect an only modest increase in Venezuela's production by 2030.

Q: So, how should investors be positioned?

A: For investors seeking exposure to Venezuelan upside optionality, we see value in Chevron, Valero, and SLB. While we believe the latest developments in Venezuela are mostly priced into these stocks, we remain constructive on their long-term fundamentals and believe they are well-positioned to benefit from any additional positive developments in the Venezuelan oil industry.

Q: What are you watching most closely ahead to gauge how this could all play out for the US oil industry?

A: Despite the recent developments in Venezuela, the outlook for the US oil industry should remain largely unchanged from our base case scenario, with three primary forces continuing to drive the industry's trajectory. First, after rising from 5-6mb/d in early 2000s to 14mb/d of crude today, the industry is now entering a period of slowing growth and maturity. Accordingly, US producers will likely focus on harvesting cash flow domestically, investing in international exploration, and driving consolidation through M&A.

Second, the larger companies have structural advantages of scale and the ability to execute an international development plan. That leaves the US majors, as well as international services companies, well-positioned for growth. This is reflected in our preference for ConocoPhillips, Chevron, Suncor, FTI, and SLB. Third, we believe that refiners are structurally well-positioned on a multi-year basis as crack spreads are likely to remain above historical averages and global oil demand is expected to outpace new capacity additions between now and 2030. Within this theme, our preferences include Valero³.

Q: What could the Venezuela developments mean for oil prices?

A: We estimate that the 0.2mb/d increase in Venezuelan oil production we expect over the next 12m will have a \$1 impact on Brent/WTI prices this year (baseline forecast of \$56/\$52/bbl average 2026 prices vs. \$57/\$53/bbl in a scenario of flat production). But the impact on long-run oil prices will likely be more significant. We estimate that the cumulative 0.4mb/d increase in Venezuelan production we expect by 2030 will have a \$2 impact on long-term oil prices. This, together with higher expected US and Russian production, led us to lower our long-run Brent/WTI price forecasts to \$75/\$71/bbl earlier this month (2030-35, from \$80/\$76/bbl). We estimate that a larger recovery in Venezuelan production to 2mb/d by 2030 would pose \$3 of downside to these revised forecasts.

Neil Mehta, Head of North American Natural Resources Equity Research

Email: neil.mehta@gs.com
Tel: 212-357-4042

Goldman Sachs & Co. LLC

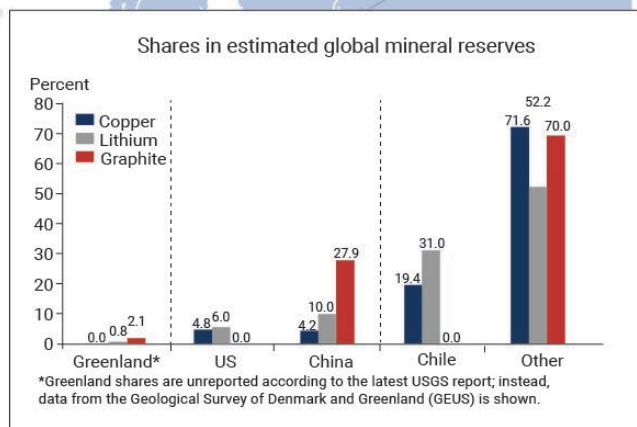
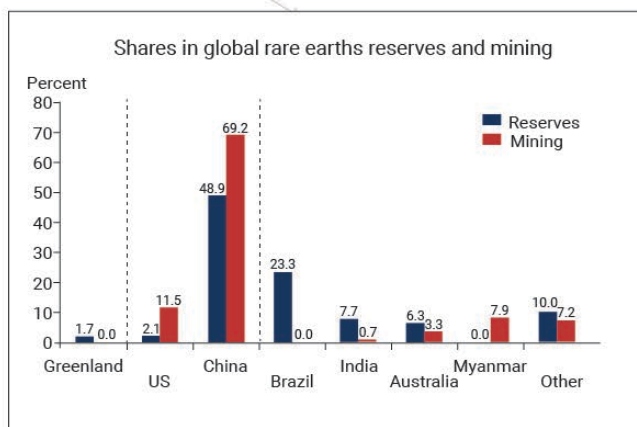
Daan Struyven, Co-head of Global Commodities Research

Email: daan.struyven@gs.com
Tel: 212-357-4172

Goldman Sachs & Co. LLC

³ For more on key single stock ideas, see our 10 Ideas for 2026 notes [here](#) and [here](#), and see our 2026 Energy, CleanTech & Utilities Conference landing page [here](#).

Greenland's strategic value, mapped out



Source: Bloomberg, USGS, GEUS, Goldman Sachs GIR.

Special thanks to GS Commodities Research team for charts.

Interview with Ivan Krastev

Ivan Krastev is Chair of the Centre for Liberal Strategies in Sofia, Bulgaria. Below, he argues that US-Europe relations have reached a turning point, but not a breaking point.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Do the Greenland developments mark a turning point in US-Europe relations?

Ivan Krastev: They mark a turning point but not a breaking point because alliances don't exist on paper, but in the minds of the public and policymakers. And the Greenland story, combined with the Trump

Administration's recently-released national security strategy and Venezuela actions, have substantially shifted how Europe views transatlantic relations. I was part of a recent European Council on Foreign Relations group that conducted global polling revealing that only 16% of Europeans see the US as an ally, while 20% view it as a rival, with most others seeing it as only a necessary partner. This marks a big shift from the past.

The Greenland issue has been especially difficult for Europeans to digest because the Danes are ready to cooperate with the US on almost anything it wanted from a security and even economic perspective. And the Danish government is one of the most pro-Atlantic governments in the European Union. So, Europeans are shockingly no longer sure about three things: one, the extent to which the Trump Administration believes that its relationships with Russia and Europe would be more difficult in NATO's absence; two, that one of the objectives of US trade policies is not the European Union's destruction; and, three, that the EU would be able to count on the US in times of major crisis, which was a key part of the magic of NATO.

Allison Nathan: So, why isn't this a breaking point?

Ivan Krastev: First of all, it's one thing to understand that your marriage is not working in the same way as before, but it's completely different to imagine what divorced life would look like. Europe is too dependent on the US not only for its security, but also technologically and economically, to prefer the divorced life today. And the stakes are too high to find out what such a life would look like, with the very survival of the European project on the line. Remember that the European Union was largely an American project. In the moment of crisis, Americans played a critical role in securing European unity.

Second, some Europeans believe that Trump's actions don't reflect the consensus among American society or the elite, and point to the walking back of the potential use of military force in Greenland as evidence of the constraints Trump faces. So, Europeans believe that when they stand against Trump on this issue, they are not standing against America. Third, while Trump has strong allies within some European countries on issues like immigration, they have not sided with him on Greenland because no European leader can convince their voters that the US getting its way in Greenland is in Europe's best interest. It's noteworthy that the European Parliament stalled ratification of the US-EU trade deal, and all MPs, from the mainstream parties to the nationalistic right, applauded the decision to stand behind Denmark and Greenland.

Now, this is largely about territory for both sides. But while Trump views land from the lens of a real estate manager, European countries, especially small ones, view land as a critically important part of their national sovereignty, and the idea that small countries don't matter is not good news for anyone in Europe. So, European leaders agree that caving on Greenland would be disastrous for Europe and its credibility. In that light, this issue has strengthened European unity. And Europe's weakness paradoxically means that it cannot *not* retaliate when it comes to this red line, because its weakness would be exposed. My sense is that Trump initially miscalculated on this point but now seems to understand it.

Allison Nathan: How could US-Europe relations evolve?

Ivan Krastev: The relationship looks to be headed for a rocky period, but not a divorce. For Europe, while there is much talk about finding new allies, aligning with others won't be a quick or easy process. The European Parliament essentially blocking the EU-Mercosur trade deal shows that too many competing interests exist among EU countries. And I'm skeptical that Europe and China will suddenly have a smoother relationship just because the US-Europe marriage has hit hard times. Instead, Europe will be focused on showing the US that Europe matters, not only because it can be helpful to the US, but also because it can be harmful. One action Europe can take to show that it is strong and serious about consolidating its economic power is making major progress on capital market integration.

Europe will also look inward in terms of focusing on building up its defenses, but here too it will encounter difficulties. Europe is struggling against time, and the blame doesn't lie with Trump. 15 years ago, then US Secretary of Defense Robert Gates warned Europe that the US would not pay for Europe's defense if it wasn't willing to increase its defense budgets, which Europeans chose to ignore, and now time is not on its side. But Europe's biggest problem is that budgets don't fight wars—people do. And Europe culturally is not prepared for the more hostile world it now finds itself in. That said, some European countries are taking their defense more seriously, and I expect closer cooperation among those countries, especially those closest to the Ukrainian crisis, as well as Germany and France in its capacity as the only European nuclear power. The Greenland issue has also brought the UK closer to the EU on defense, which will be critically important in bolstering European defense capabilities.

For the US, midterm elections loom large, which I suspect will continue to serve as a constraint on the Administration's actions. So, both the US and Europe will likely look for ways to de-escalate tensions and compromise, as has already begun. But the trust between the US and Europe has been broken, not just because of Greenland but also because of Ukraine in terms of US policy related to the conflict, and the broader US treatment of Europe. So, I don't expect the relationship to go back to the way it was even if, as we've discussed, the current moment doesn't prove to be a breaking point.

When geopolitics meets portfolios

Christian Mueller-Glissmann discusses how to navigate and hedge against geopolitical risk

Geopolitical risk has returned to the forefront following the recent developments around Venezuela, Iran, and Greenland, and is likely to remain there for some time as the US continues to reshape its geopolitical and economic relationships with the world. This has raised questions about how investors can protect themselves from geopolitical shocks, which can have material impacts on global growth, inflation, and sentiment, especially as the linkages between economies and markets have become increasingly complex. Rather than try to time geopolitical shocks, which can be difficult, we think robust portfolio construction and diversification should be investors' first line of defense.

Geopolitical shocks: rhyming, but seldom repeating

While geopolitical shocks are a key risk for portfolios, they tend to differ every time and are particularly difficult to position for as the timing and potential economic and market impacts are hard to anticipate. Geopolitical tensions can also linger for a long time before they escalate. To further complicate matters, geopolitical shocks often start out as local shocks but can extend globally and have lasting knock-on effects.

The broader macro backdrop also affects how markets react to geopolitical shocks. We find an imperfect relationship between the news-based Geopolitical Risk Index and equity volatility, with an increase in geopolitical tensions often leading to higher—but not extreme—volatility. Geopolitical risks alone have seldom pushed the VIX above 40. But the combination of such risks and an already-weak macro backdrop has often sent volatility sharply higher. A more resilient US economy and a Fed able to buffer negative growth shocks may help markets digest geopolitical shocks better today, though low volatility, elevated equity valuations, and bullish investor sentiment to start the year increase the risk of a volatility spike.

The individual asset impacts of geopolitical shocks can also vary. Historically, large geopolitical shocks like the Middle East wars have driven oil prices sharply higher, often resulting in a worsening global growth/inflation mix. By contrast, oil prices have remained in a downtrend amid the recent geopolitical developments due to excess supply, with the Dollar also weakening as investors have become more concerned about US assets.

Finally, the inherent unpredictability of geopolitical events, including the possibility of a reversal of the shock, creates material market timing risk. Equities on average have recovered and risen after geopolitical shocks. In some cases, uncertainty can decline rapidly as more information becomes available or a policy pivot occurs, which can drive sharp equity recoveries, as happened around "Liberation Day" last April, which showcased the difficulty of timing equity moves.

Protecting portfolios: diversify, don't time

Given these challenges, investors' first line of defense against geopolitical shocks should be robust portfolio construction and diversification. In a balanced multi-asset portfolio, bonds can provide a buffer for risky assets against unanticipated growth

shocks. However, bonds become a less effective hedge if the geopolitical shock increases inflation risk. US policy uncertainty and global fiscal risks have also weighed on bonds, with recent concerns about potential fiscal expansion in Japan increasing the upward pressure on long-dated bond yields globally. Long-dated bonds may provide a smaller buffer for risky assets going forward if fiscal risks persist.

Investors can also turn to other global safe haven assets like the Swiss Franc, which has become a more effective "risk-off" hedge vs. the Euro in periods of rising geopolitical risk. But it's important to keep in mind that while safe haven assets tend to outperform during risk-off episodes, they usually have lower returns through the cycle. So, as with bonds, investors need to consider the tradeoffs between risk reduction and returns.

Increased allocations to assets that could benefit from an escalating shock can also buffer geopolitical risk. Commodities like oil and gold have often been useful diversifiers in periods of geopolitical uncertainty, though their hedge value can vary depending on the drivers (if there are oil supply disruptions, for example) and again they tend to have relatively low risk-adjusted returns over longer horizons. That said, gold and precious metals more broadly have recently helped buffer both geopolitical and fiscal risks, with higher central bank demand further boosting their risk-adjusted returns. Defense stocks are another potential beneficiary of geopolitical shocks (see pg. 20).

Investors can also manage geopolitical risk by avoiding or reducing direct and indirect exposures. This could include avoiding the country/region facing elevated geopolitical risk or large energy consumers (in the event of sharply higher oil prices) as well as reducing indirect exposure from trade linkages. With global investor portfolios currently dominated by US assets due to their large weights in equity and bond benchmarks, investors could also increase home bias in portfolios to reduce FX risk, though the potential impacts of large shifts out of US assets on portfolio returns and risk should be carefully considered.

Finally, the fact that geopolitical shocks can reverse quickly suggests that options can be a valuable hedge. Hedging tail risk with options can undoubtedly be expensive, making prolonged exposure to risk management strategies that protect portfolios systematically unfeasible. That said, realized and implied volatility has been very low coming into the year, increasing the value of tactical portfolio overlay hedges.

So, for investors looking for ways to protect their portfolios from geopolitical shocks, the message is: diversify, don't time. Specifically, we recommend more regional, sector, and style diversification relative to global benchmarks and like safe assets such as gold and selectively hedging Dollar risk. With the volatility reset, we also like shorter-dated rates receivers as a hedge for major growth shocks, credit payer spreads/VIX call spreads to hedge corrections, and longer-dated calls to maintain overweight equity allocations despite the potential for more geopolitical shocks.

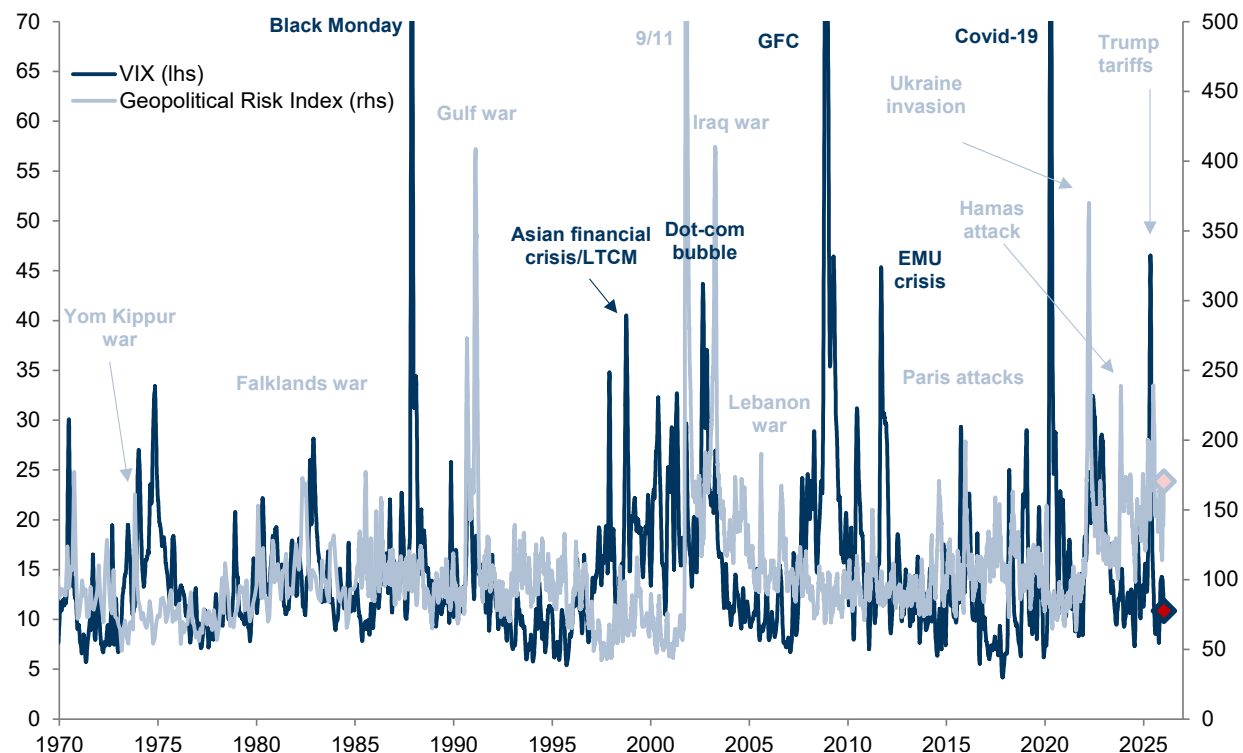
Christian Mueller-Glissmann, Head of Asset Allocation Research

Email: christian.mueller-glissmann@gs.com
Tel: 44-20-7774-1714

Goldman Sachs International

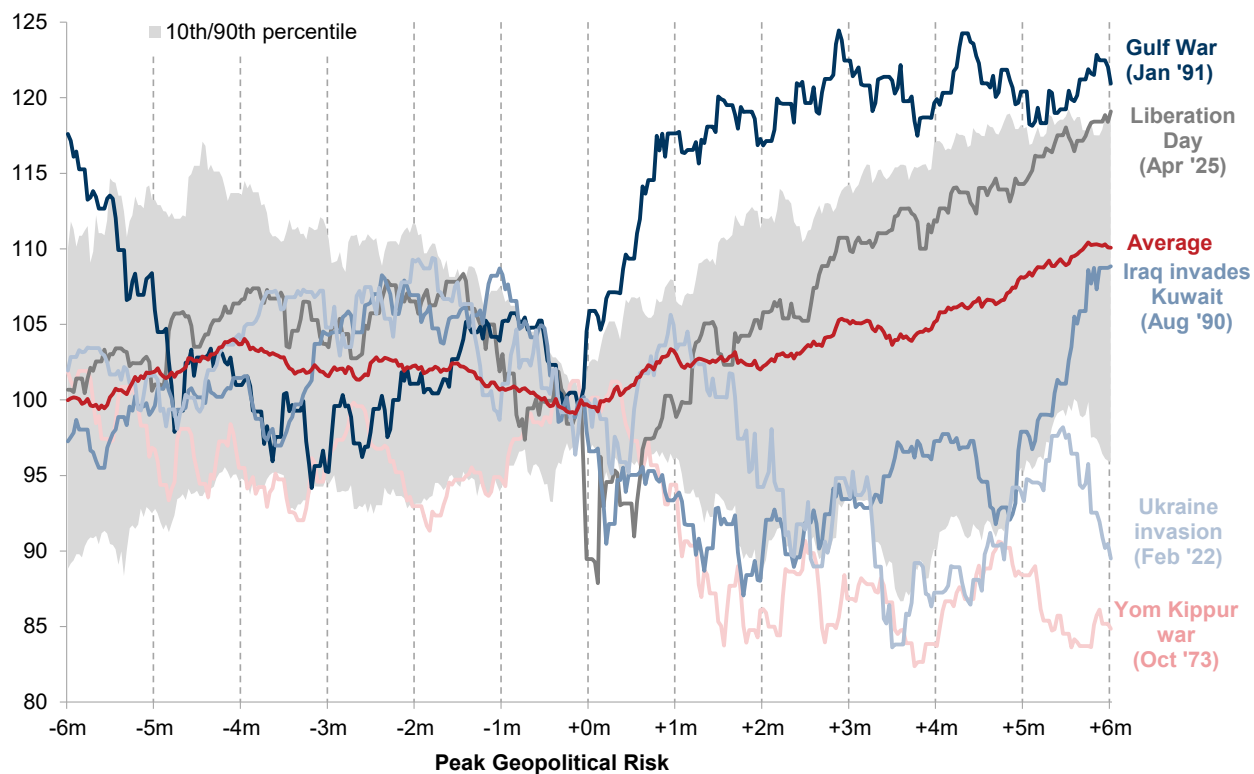
The relationship between geopolitical risk and equity volatility is mixed

VIX Index (lhs) vs. Geopolitical Risk Index (rhs), 1-month average



Timing equities around geopolitical shocks has been very difficult

Equity response to geopolitical shocks since 1970 (15 geopolitical shocks included)



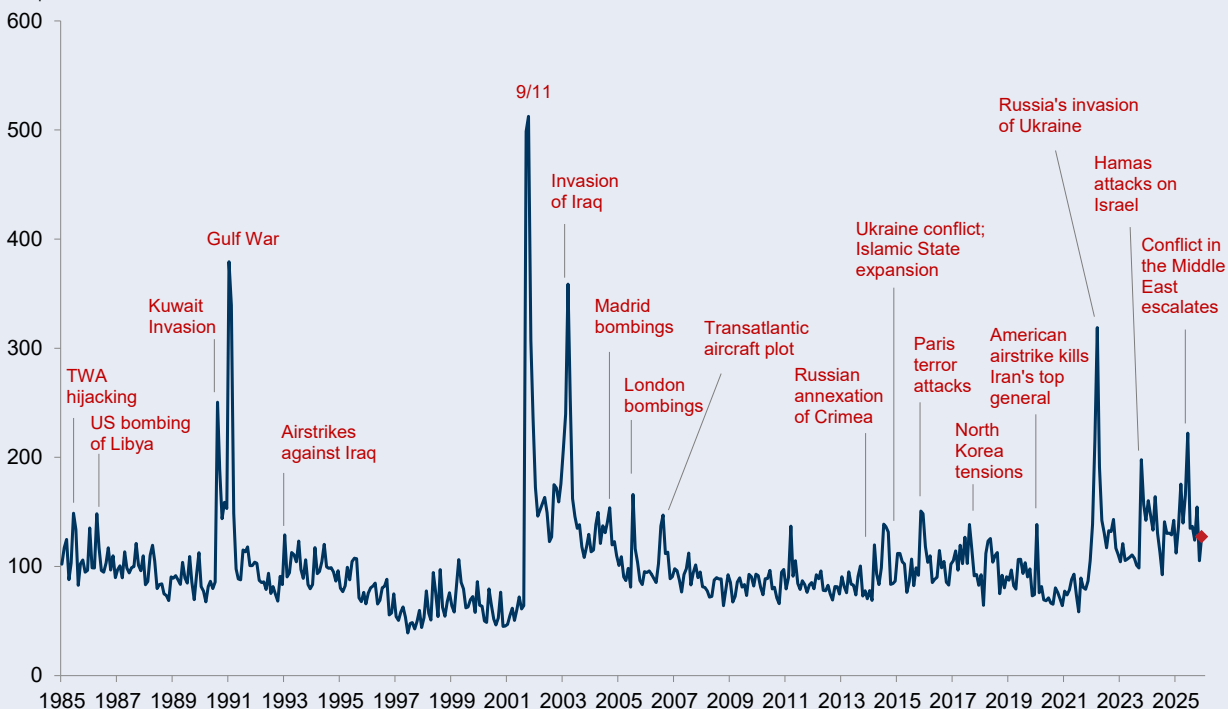
Source: Datastream, Bloomberg, Dario Caldara and Matteo Iacoviello, Goldman Sachs GIR.

Geopolitical risk through the years

Geopolitical tensions, which take many different forms, are difficult to measure. One proxy for assessing the geopolitical environment is the news-based Geopolitical Risk Index developed by economists from the Federal Reserve Board.

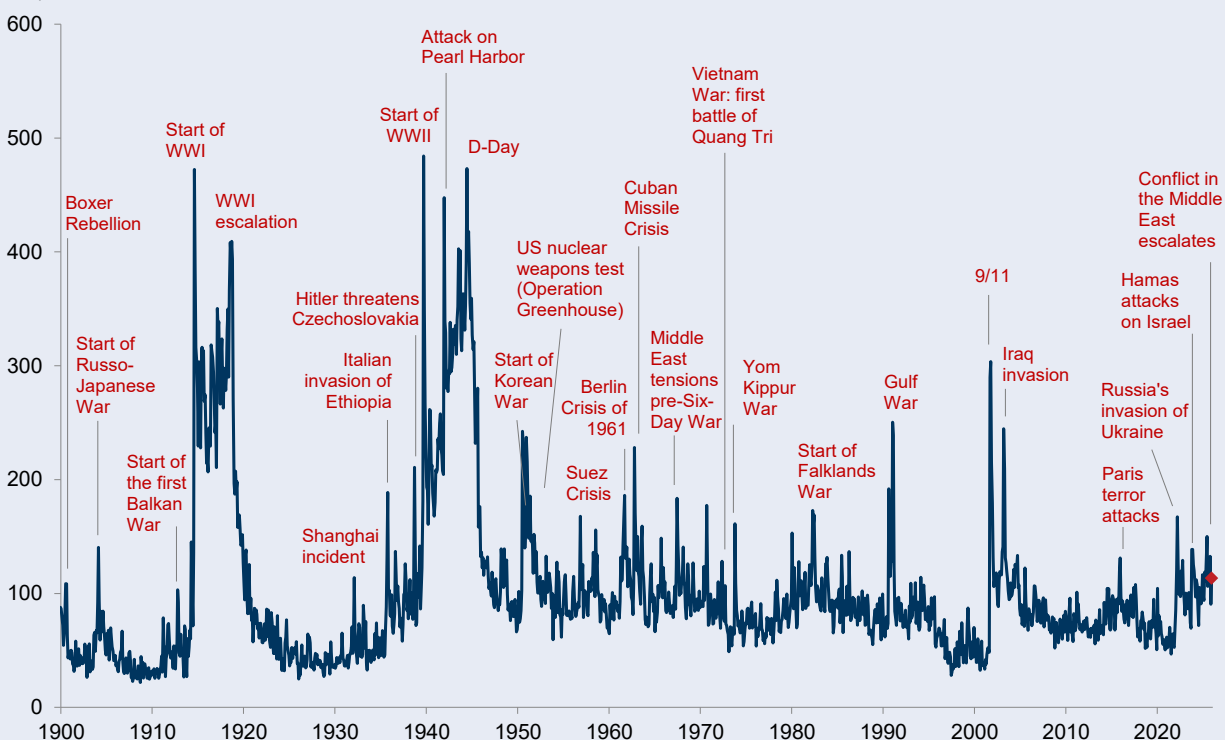
1985-Present

Geopolitical Risk Index, 1900-2019=100



1900-Present

Geopolitical Risk Index, 1985-2019=100



Note: The index from 1985 on counts the number of articles in 11 US, UK, and Canadian newspapers mentioning phrases related to geopolitical tensions. The index from 1900 on performs the same analysis using the archives of three newspapers: the New York Times, the Wall Street Journal, and the Financial Times. The choice of newspapers for both indices implies a measure of geopolitical risk as covered by the Anglo-Saxon press. See [here](#) for more information.

Source: [Dario Caldara and Matteo Iacoviello](#), Federal Reserve Board, Goldman Sachs GIR.

EM assets: continued resilience

Kamakshya Trivedi discusses the outlook for EM assets amid elevated geopolitical risk

Geopolitical risks have dominated headlines in 2026, from developments in Venezuela and Iran to Greenland. Despite that, EM equities have started the year strong and have comfortably outpaced DM equity markets, EM currencies are among the top performers year-to-date, and EM rates and spreads have held on. We think this is indicative of the [resilient upside](#) that we expect out of EM this year. Indeed, despite heightened geopolitical risks and several upcoming elections, our [global macro outlook](#) of sturdy growth, low inflation, and further Fed cuts is a [friendly one for EM assets](#), even if less intensely so than in 2025. Better local EM fundamentals and still-low global investor allocations should also support returns in 2026.

Limited spillovers from Venezuela

Venezuelan developments—while highly pertinent from a geopolitical standpoint—have had limited spillovers in part because of limited exposure in EM investor portfolios. Political shifts can create significant volatility in local markets—as seen in Argentina last year and the political ructions in Türkiye before that—but their relatively small share of aggregate EM portfolios typically limits any impact. Equally, the improvement in EM fundamentals in recent years—solid growth, further inflation normalization, and scope for rate cuts as well as contained fiscal and current account positions—has helped insulate the broader EM complex from political and geopolitical shocks.

A positive returns story for EM assets

We continue to expect positive returns for EM assets this year amid a friendly global macro backdrop. In particular, we expect [EM equities](#) to outperform DM equities again, though we look for more moderate returns of ~15% (vs. last year's record 30% gains) driven primarily by solid earnings growth (19%/12% in 2026/2027) spread across the EM geographies. In EM FX, after the year of carry in 2025, we think that [cyclical exposure](#)—via both equity and commodity price linkages—will be as important as carry levels in 2026 for relative spot returns. This includes ZAR, CLP, and KRW, which have high betas to cyclical pricing and to the appreciation we expect in the CNY anchor and remain among the more undervalued currencies vs. USD. In EM fixed income, the scope for deeper cutting cycles in EM high-yielders (like Hungary and Brazil) should provide the best opportunities in [local rates](#). We remain defensive on [EM sovereign credit](#), with total returns likely to be largely a function of yield given our expectation of a modest widening in very tight spreads.

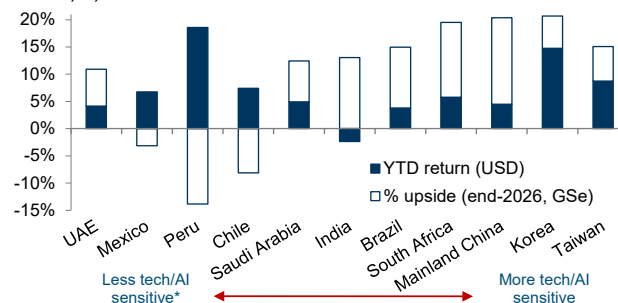
EM assets as diversifiers

EM assets can also provide an attractive opportunity for investors given their role as diversifiers. In particular, EM assets can help balance AI exposure amid growing investor concerns about valuations. For example, within EM, while we continue to favor the tech-sensitive equity markets of Korea, Taiwan (strong semiconductor earnings), and Mainland China (AI-driven growth and external market share gains), we prefer rotating some portfolio weight into the more domestic-oriented equity markets of South Africa (where a cyclical growth recovery and easing rates should help laggard domestic sectors catch up to the miners), India (with its mass-consumption revival), and

Brazil (where rate cuts should drive further gains).

EM has seen a broad-based rally this year

Returns, %, USD



*MSCI weight in tech/hardware, semis, internet, software, media/entertainment. Source: FactSet, Goldman Sachs GIR.

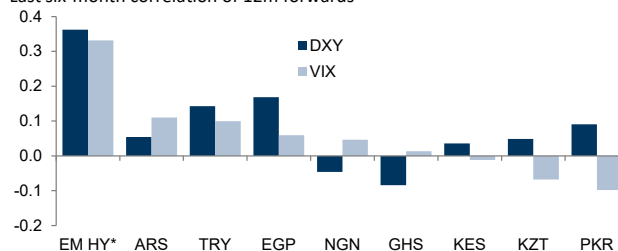
Uncorrelated risks and returns on the EM Frontier

Frontier markets also provide a source of returns that are relatively uncorrelated to the broader macro backdrop (although they come with their own risks, as Venezuela shows). Amid renewed geopolitical risks and the potential for higher volatility this year, we think that Frontier currencies continue to represent an attractive investment proposition in a more range-bound Dollar environment owing to elevated carry levels together with low correlation to Dollar strength and global macro volatility relative to other EM counterparts.

In particular, given still-healthy [GSDEER model valuations](#) and our [economists' constructive external outlook](#), we favor carry longs in CEEMEA, including EGP, NGN, and KES, where risks are skewed toward appreciation given historically high FX reserves. Specifically, a still-cautious central bank in Egypt, which could keep real rates elevated for longer than we previously expected, led us to recently extend the target on our short USD/EGP trade recommendation (opened last [July](#)). And in Türkiye, we expect TRY monthly total returns to remain around 1-1.5% as a slight reduction in carry is offset by a decline in the pace of nominal depreciation amid slowing year-on-year inflation. The main risk to this constructive view is elevated positioning in still-relatively illiquid FX markets that could reverse sharply on a negative shock. Finally, we [expect](#) ARS to weaken along wider FX flotation bands given Argentine policymakers' objective to rapidly accumulate international reserves this year from critically low levels.

Frontier currencies are less correlated with USD and global vol

Last six-month correlation of 12m forwards



Note: We take log changes of all series before computing correlations.

*Median of BRL, COP, MXN, and ZAR.

Source: Goldman Sachs GIR.

Kamakshya Trivedi, Chief FX & EM Strategist

Email: kamakshya.trivedi@gs.com

Tel: 44-20-7051-4005

Goldman Sachs International

Q&A on defense sector geopolitical risk

We ask our equity analysts about the potential implications of elevated geopolitical risk for the defense sector.

US Defense

Noah Poponak, GS Equity Research

Q: Is the upside for defense stocks on the back of elevated geopolitical risk already fully priced in?

A: A lot of upside is priced in some stocks, but it is not fully priced across the sector. The US Defense sector has outperformed year-to-date on the back of geopolitical developments and statements from the White House regarding potentially much higher defense budgets. The Administration currently appears to want to further grow the US defense budget, but efforts are also underway to shift more financial and technology development risk onto the industrial base, creating a somewhat mixed picture for defense firms' financials. Product category prioritizations are also driving different growth profiles across verticals and individual companies in the sector. And with the US defense budget already at an all-time high and questions remaining about the appropriate level of US government spending, the annual level of defense spending will continue to be debated notwithstanding the clear focus on national security. Valuations at the sector level are on average above their historical midpoint but below their highs, with enough dispersion at the company level to still find value.

Q: Where does value remain in the sector?

A: We see value in stocks leveraged to product and vertical prioritizations that have growth and margin upside potential (rather than facing more financial risk as the Administration and Pentagon seek to shift more of the investment burden onto firms) and trade at relatively attractive valuations. We are Buy-rated on Huntington Ingalls in Shipbuilding as the Navy has clearly become high priority and the company has been awarded funding to improve its labor base, which will flow to margins. We are also Buy-rated on L3Harris, which is levered to the growth areas of missile & missile defense, space, and C4ISR (Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance). CACI International also has off-the-radar high exposure to growth areas within defense and trades at an attractive valuation. We also see value in Defense Tech companies, leveraged to growth areas like AeroVironment and Kratos in drones, Leonardo DRS, which sells to the US Navy, and Beta Technologies in EVTOL (Electric Vertical Take-off and Landing) (these companies are all covered by Anthony Valentini). By contrast, we believe Lockheed and Northrop are worse positioned among traditional US large-cap primes as these companies are taking on more upfront financial and contractual terms risk as the Pentagon takes on less.

Q: What key catalysts are you watching to realize the upside?

A: The White House recently discussed the possibility of a much higher US defense budget. The government typically makes its official budget request for the next fiscal year in the spring, which then goes through the congressional armed services committees through the summer, and should be finalized by the fall. Where that budget ultimately lands will go a long way toward informing future growth. We're also watching the moving pieces under the hood to continue to understand the verticals and products in higher demand. New Administration frameworks have been incrementally informing major changes in the Pentagon's defense procurement process, capital requirements, and risk sharing processes. And Defense budgets have been rising in other places, especially Europe, which will impact defense companies in those regions as well as US companies given that other countries procure military technologies from the US.

European Defense

Sam Burgess, GS Equity Research

Q: Is the upside for defense stocks on the back of elevated geopolitical risk already fully priced in?

A: No. While geopolitical tensions, alongside an increasingly fractious NATO alliance, have driven a sharp re-rating in European defense equities—with the GS EU Defense basket (GSSBDEFE) up over 20% year-to-date—we still see value in the sector. On growth-adjusted metrics, European defense valuations remain materially cheaper than US peers, with the discount currently at one of the highest points over the past three years. As a result, we would push back on the idea of a European defense bubble—the sector has clearly re-rated, but valuations are not uniformly stretched.

Q: Where does value remain in the sector?

A: We think the context demands selective positioning rather than a blanket sector buy. We find that priced-in long-term earnings growth still looks relatively undemanding for Rheinmetall and BAE Systems. We see Rheinmetall as the most directly leveraged to Europe's rearmament given its exposure to capability gaps (land systems, munitions) and its position as a beneficiary of Germany's defense funding uplift. And we see BAE Systems as a quality, cash-generative defense prime, with diversified, long-duration programs and meaningful US exposure positioning it well for a potential increase in US defense spending.

Q: What key catalysts are you watching to realize the upside?

A: As global geopolitical developments continue to emerge, we are watching for catalysts that reinforce defense spending as a necessary "insurance" outlay rather than a discretionary budget item for European policymakers. We are also keeping an eye on European defense budgets as well as disbursements from the ~€150bn European Security Action for Europe (SAFE) fund, a financial instrument providing loans to EU member states for defense procurement, which should drive meaningful orders for European primes. And we will be monitoring European primes' ability to expand capacity and meet demand in the new environment of increased military spending.

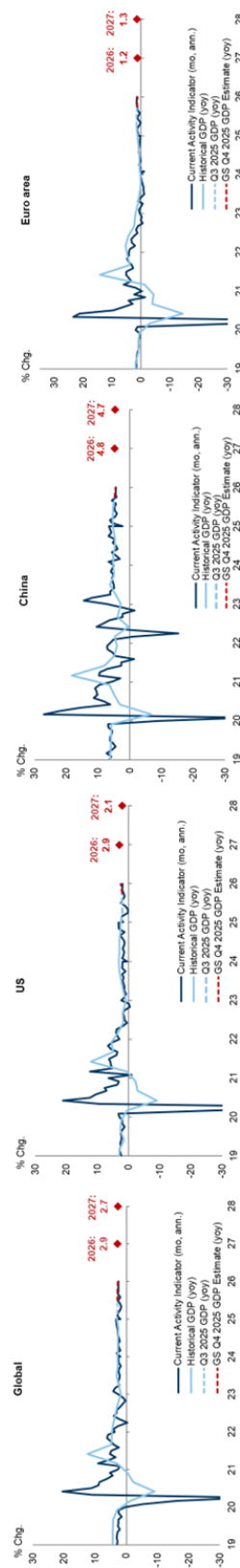
Summary of our key forecasts

Watching

- **Globally**, we expect sturdy and above-consensus real GDP growth of 2.9% yoy in 2026 amid fading headwinds from US tariffs and rising real income growth. We expect global core inflation to decline to 2.1% by the end of 2026 as the tariff-driven boost fades and shelter and wage inflation fall further.
- **In the US**, we expect real GDP growth to rise to an above-consensus 2.5% on a Q4/Q4 basis in 2026 as tax cuts provide a tailwind to disposable incomes and financial conditions ease, but uncertainty around trade policy presents a potential risk. We expect core PCE inflation to decline to 2.1% yoy by the end of 2026 as tariff pass-through fades while underlying wage and shelter inflation trends improve. We expect the unemployment rate to end 2026 at 4.5%.
- **We expect the Fed** to deliver two 25bp cuts this year in June and September for a terminal rate range of 3-3.25%.
- **In the Euro area**, we expect real GDP growth of 1.2% yoy in 2026 reflecting real income gains and a fading fiscal policy drag but growing headwinds from more intense export competition with China. We expect core inflation to decline to 1.8% by end-2026 amid lower energy prices, a stronger Euro, and the potential for a negative inflation effect from Chinese trade rerouting.
- **We expect the ECB** to remain on hold for the foreseeable future given our and the ECB's forecast of better growth ahead and target-consistent inflation over the medium term.
- **In China**, we expect above-consensus real GDP growth of 4.8% yoy in 2026 given resilient export growth, continued government policy easing, and a decreasing drag from the ongoing property market downturn, although sluggish domestic demand should remain a headwind. On the inflation front, China's deflation process will likely remain gradual, but CPI/PPI inflation should rise to 0.6%/-0.7% yoy this year owing to government policy efforts and a low base.
- **WATCH US POLICY, GEOPOLITICAL DEVELOPMENTS, AND FISCAL DYNAMICS.** Uncertainty about US policy remains elevated, especially around Greenland, tariffs, and a possible US government shutdown. Geopolitical developments more broadly pose risks worth watching, especially as the situation in Venezuela develops, tension in the Middle East continues, a potential Russia-Ukraine peace deal remains uncertain, and US-China relations remain fragile. And fiscal pressures—including in the US, UK, France, and Japan—remain important to watch.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.
Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI, please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics										Markets						Equities																								
GDP growth (%)		2026			2027			Interest rates 10Yr (%)			FX			Last			E2026			E2027			Returns (%)			12m	YTD	E2026 P/E												
		GS (Q4/Q4)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	Last	E2026	E2027	FX	Last	3m	12m	S&P 500	GS	Cons.	GS	Cons.	Q1	Q2	Q3	Q4															
Global		2.9	--	2.9	2.6	2.7	2.6	US	4.24	4.20	4.25	EUR/\$	1.18	1.18	1.25	Price	7,800					S&P 500	9.9	1.0	22.6x															
US		2.5	2.1	2.9	2.0	2.1	2.3	Germany	2.88	3.25	3.00	GBP/\$	1.36	1.33	1.36	EPS	\$305	\$311	\$336	\$356	MX40J	12.0	5.7	15.2x																
China		4.7	4.7	4.8	4.9	4.7	4.5	Japan	2.26	2.00	2.00	\$/¥	158	157	152	Growth	12%	15%	10%	15%	Topix	7.4	6.5	18.1x																
Euro area		1.4	1.4	1.2	1.4	1.3	1.2	UK	4.42	4.00	4.00	\$/£	6.94	6.95	6.85						STOXX 600	2.7	2.8	15.4x																
Policy rates (%)		2026			2027			Commodities			Last			3m			12m			Credit (bp)			Last			2026			4Q26			Consumer			2027			Wage Tracker 2025 (%)		
		GS	Mkt.		GS	Mkt.		Crude Oil, Brent (\$/bbl)	66	57	55	USD	IG	72	85	90	US	2.4	4.5	1.9	4.5	4.1	3.9	3.7	3.4															
US		3.13	3.23		3.13	3.33		Nat Gas, NYMEX (\$/mmBtu)	5.28	3.75	4.00	USD	IG	72	85	90																								
Euro area		2.00	1.93		2.00	2.20		Nat Gas, TTF (EUR/MWh)	39.47	38	30	HY	256	295	315		Euro area	1.8	6.4	1.8	6.3	--	--	--	--															
China		1.20	1.34		1.10	--		Copper (\$/mt)	13,049	12,665	10,870	EUR	IG	85	95	99	China	0.6	--	0.9	--	--	--	--	--															
Japan		1.00	1.22		1.50	1.60		Gold (\$/troy oz)	4,946	4,805	5,445	HY	257	298	315																									
Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/disclosure.html																																								
Market pricing as of January 23, 2026																																								

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.